MS-16J5/MS-1795

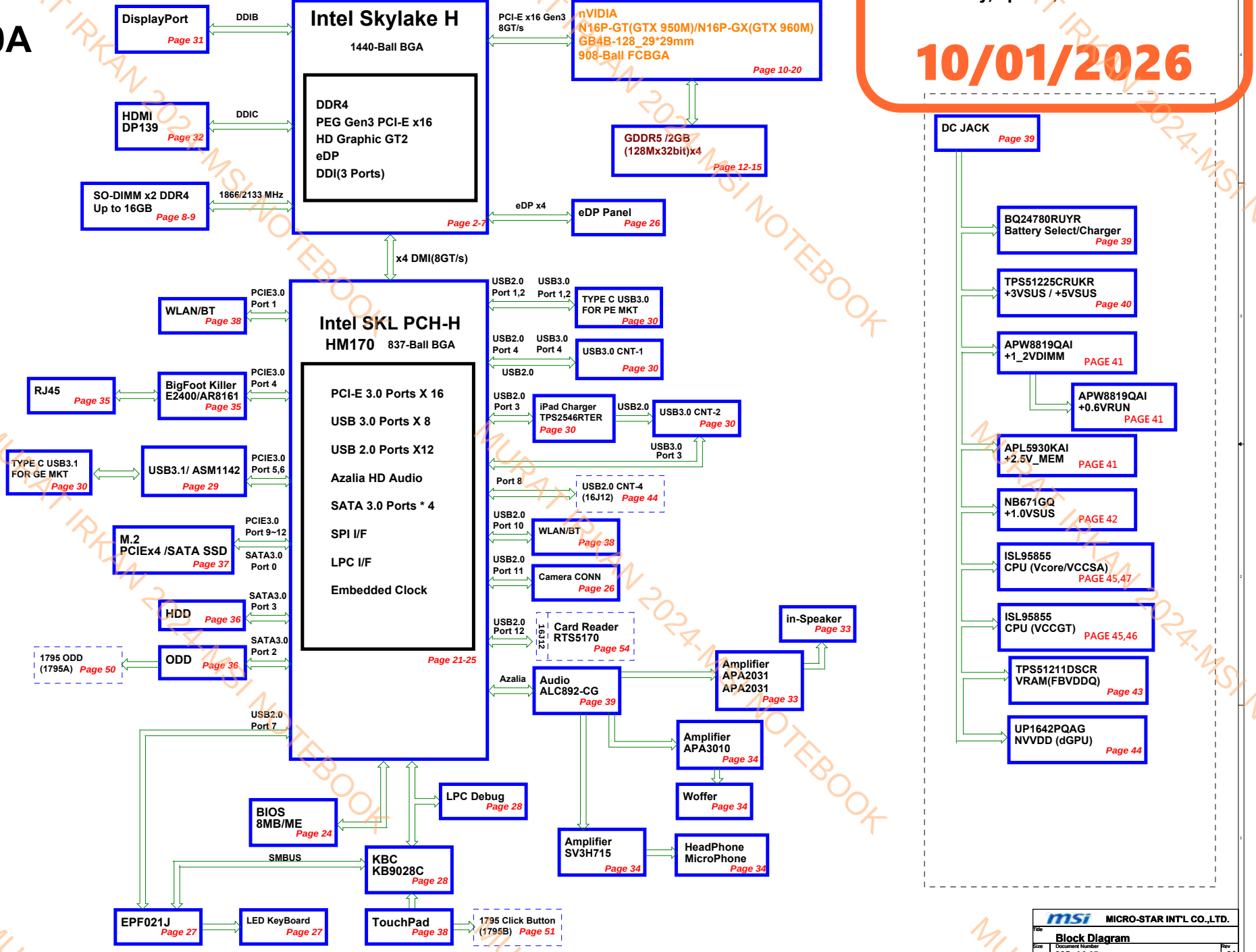
Ver:0A

Intel Skylake Mobile

MSI NOTEBOOK INTEL 1440 PIN

Monday, April 20, 2015

10/01/2026

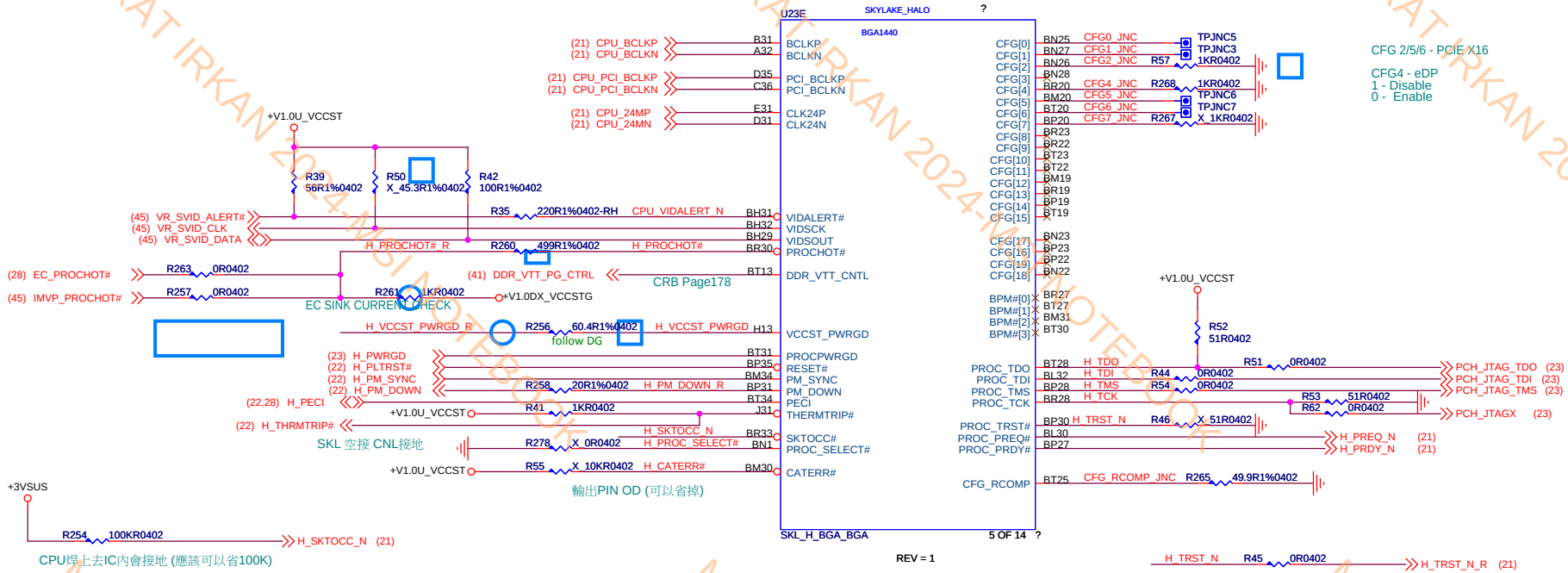


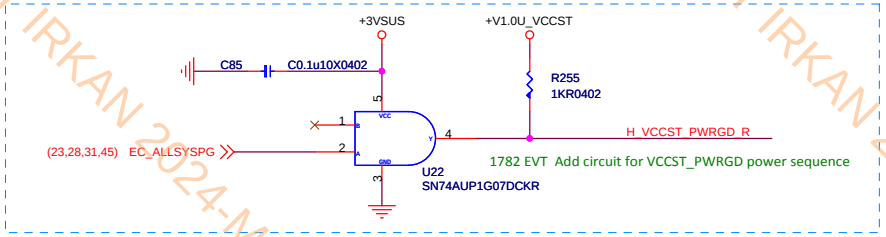
msi MICRO-STAR INT'L CO.,LTD.

Title: Block Diagram

Size | Document Number: MS-16J5 | Rev: 0A

Date: | Sheet 1 of 55





MSR Privacy Bit Feature	
CFG3	1 = Debug capability is determined by IA32_Debug_Interface_MSR (0xC80) bit[0] setting 0 = IA32_Debug_Interface_MSR (0xC80) bit[0] default setting overridden

eDP Enable	
CFG4	1 = Disabled 0 = Enabled

PEG DEFER TRAINING	
CFG7	1: (Default) PEG Train immediately following xxRESETB de assertion 0: PEG Wait for BIOS for training

PCIE Express * Static X16 Lane Numbering Reversal	
CFG2	CFG[2]: PCI Express* Static x16 Lane Numbering Reversal. 1 = Normal operation 0 = Lane numbers reversed.

PCI Express* Bifurcation	
CFG[6:5]	00 = 1 x8, 2 x4 PCI Express* 01 = reserved 10 = 2 x8 PCI Express* 11 = 1 x16 PCI Express*

Intel 54492 Page 37,121

MICRO-STAR INT'L CO.,LTD.			
Title: Skylake(HOST)			
Size: A3	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 2 of 55		

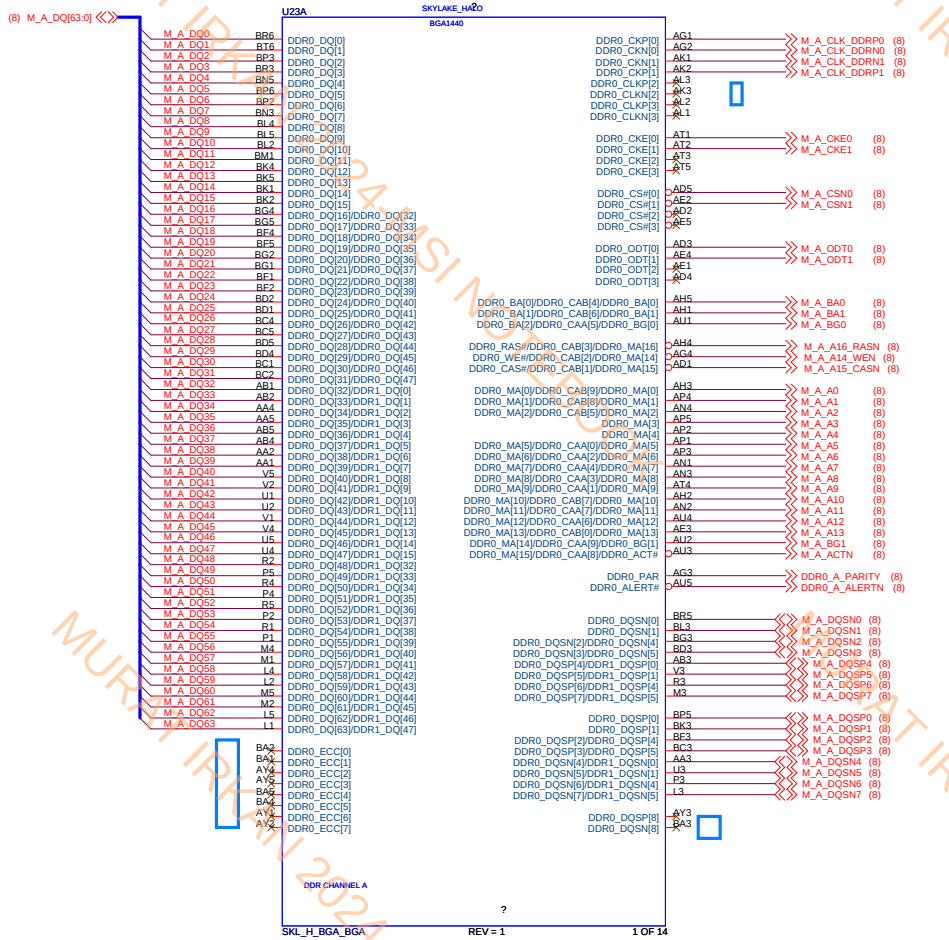
MURAT IRKAN 2024-MSI NOTEBOOK

DDR4 KANAL A-3600 MHZ
10/01/2026

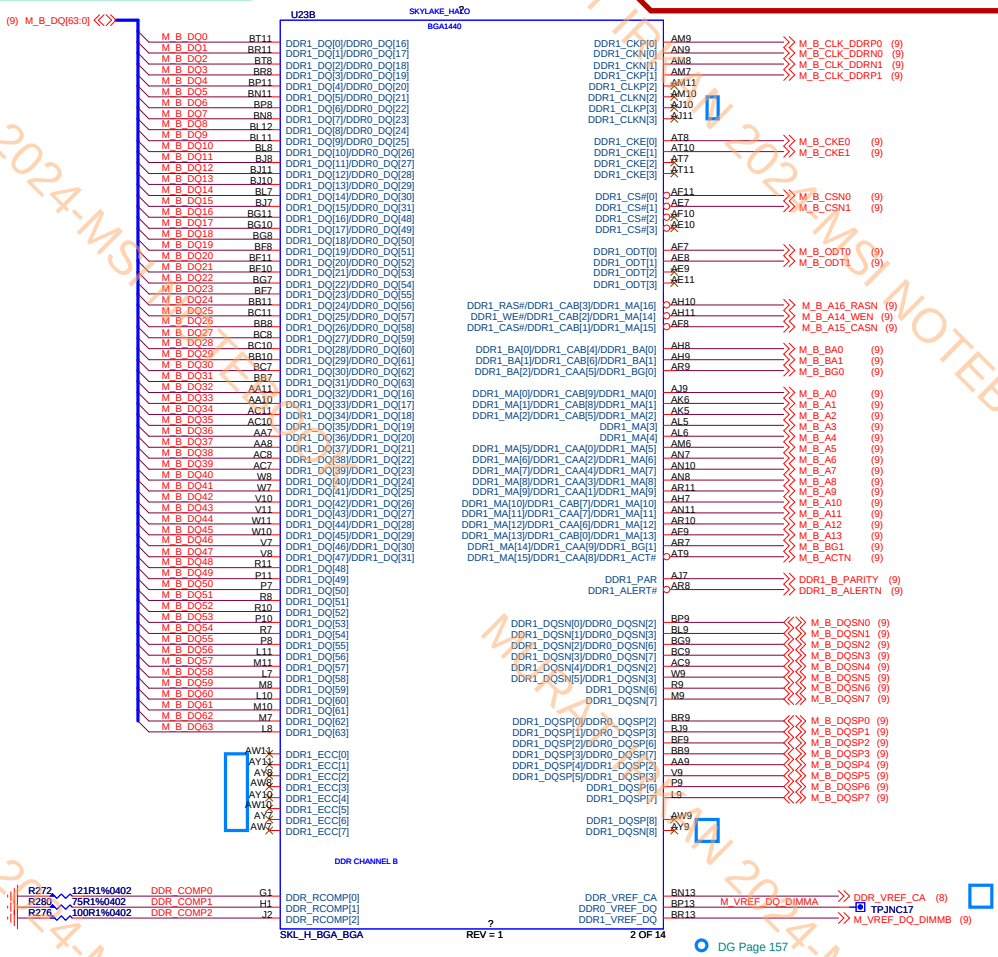
MSI INTEL SOKET 1440 PIN NOTEBOOK SİSTEMİ
10/01/2026

DDR4 KANAL B 3600MHz
10/01/2026

DDR Channel A



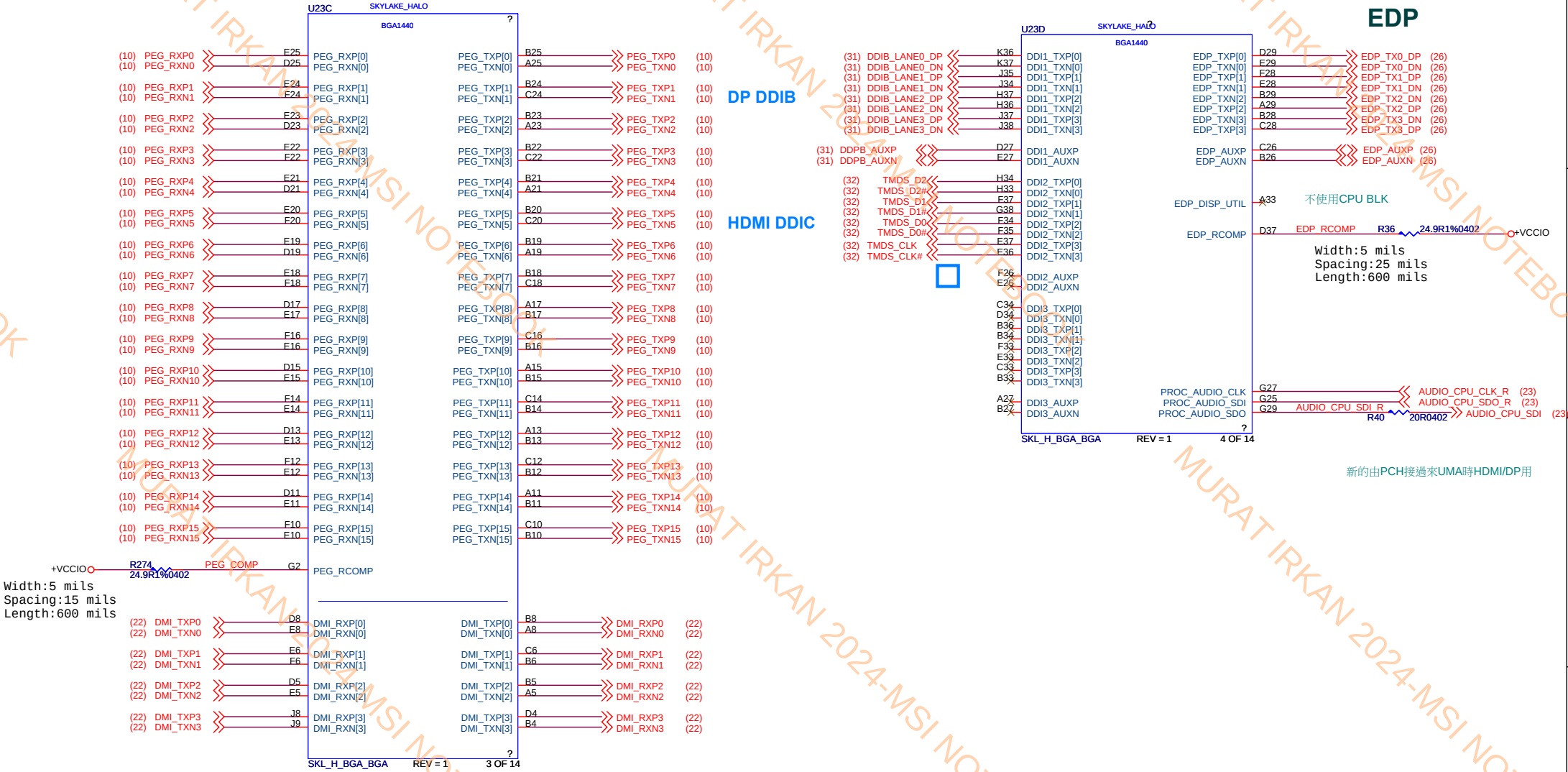
DDR Channel B



DG Page 157

msi	MICRO-STAR INT'L CO.,LTD.		
Title	Skylake(DDR4)		
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 3 of 55		

MURAT IRKAN 2024-MSI NOTEBOOK

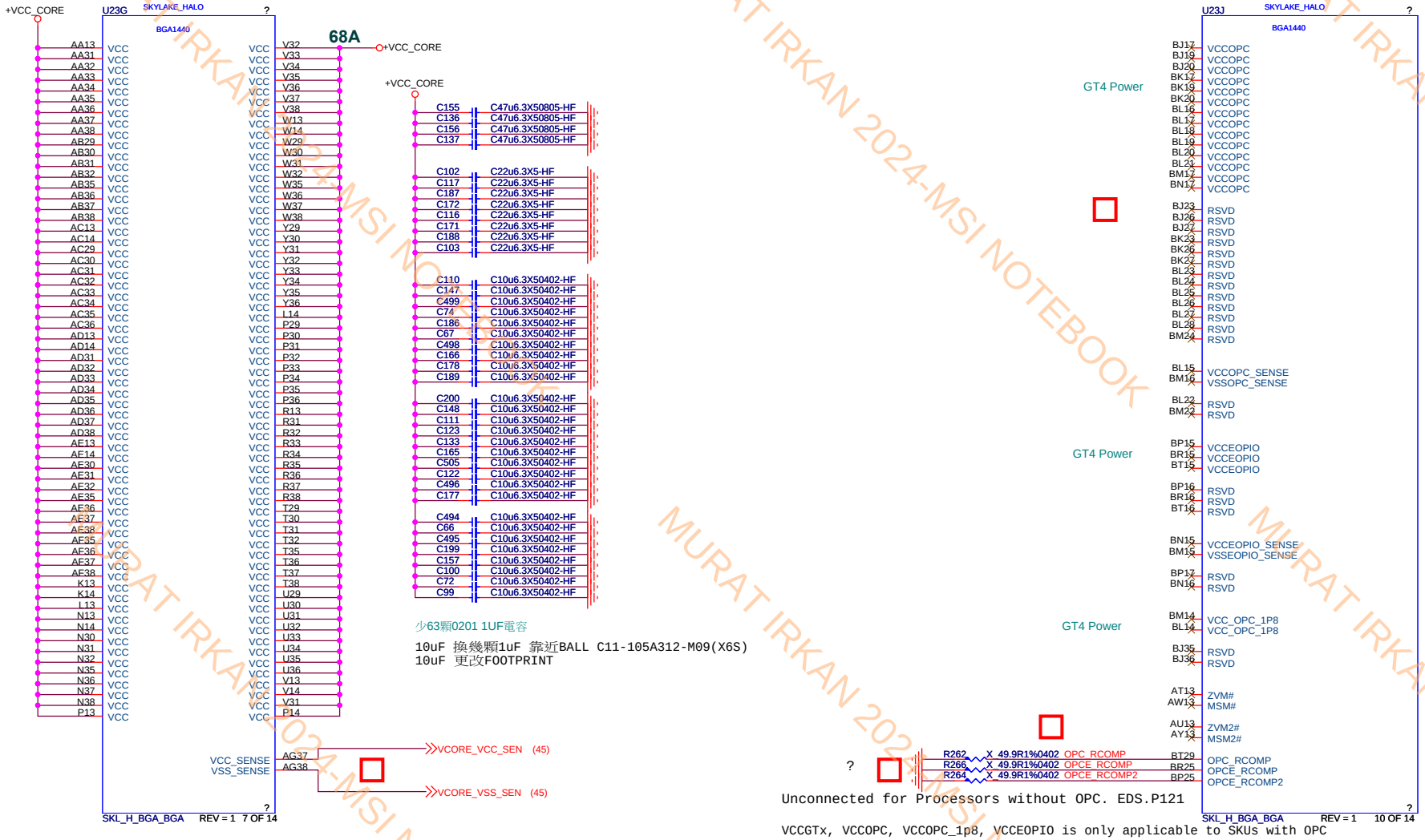


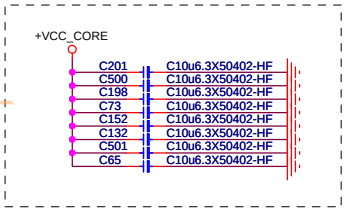
MSI INTEL 1440 PIN NOTEBOOK

10/01/2026

MICRO-STAR INT'L CO.,LTD.		
Title: Skylake(DMI/Display)		
Size: B	Document Number: MS-16J5	Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 4 of 55	

MURAT IRKAN 2024-MSI NOTEBOOK





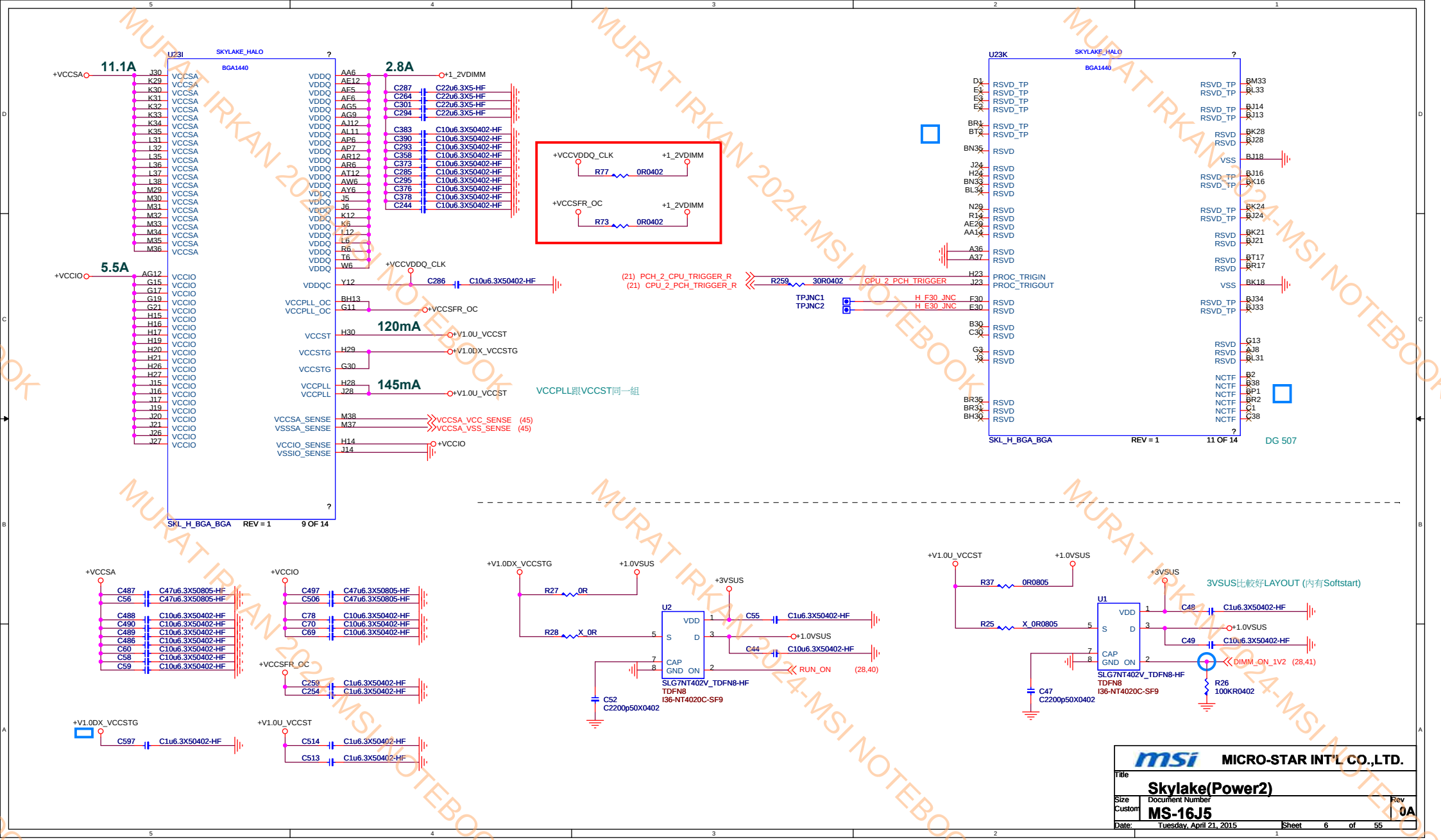
MICRO-STAR INT'L CO.,LTD.

Title: Skylake(Power1)

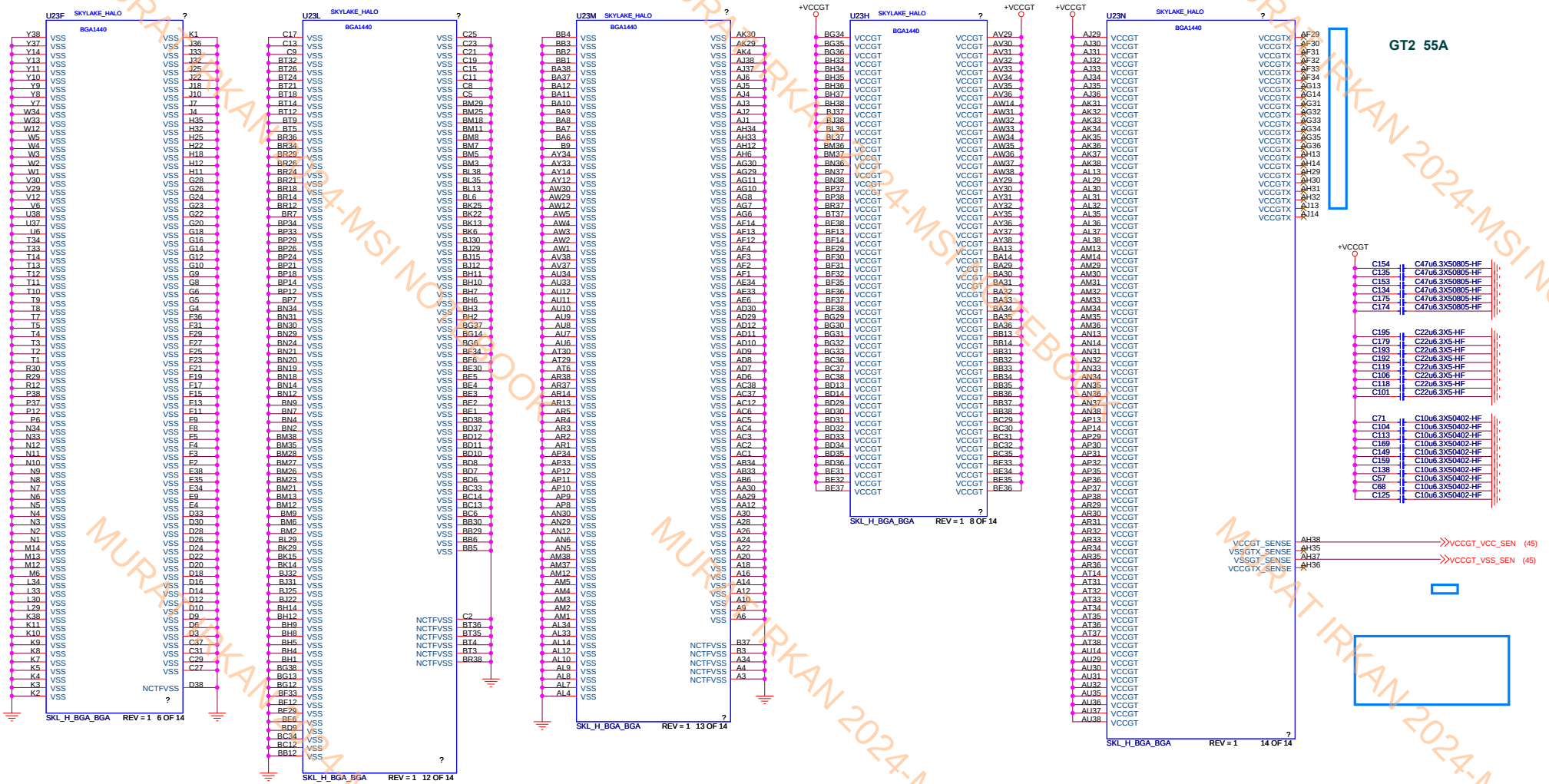
Size: A3 | Document Number: MS-16J5 | Rev: 0A

Date: Tuesday, April 21, 2015 | Sheet 5 of 55

MURAT IRKAN 2024-MSI NOTEBOOK



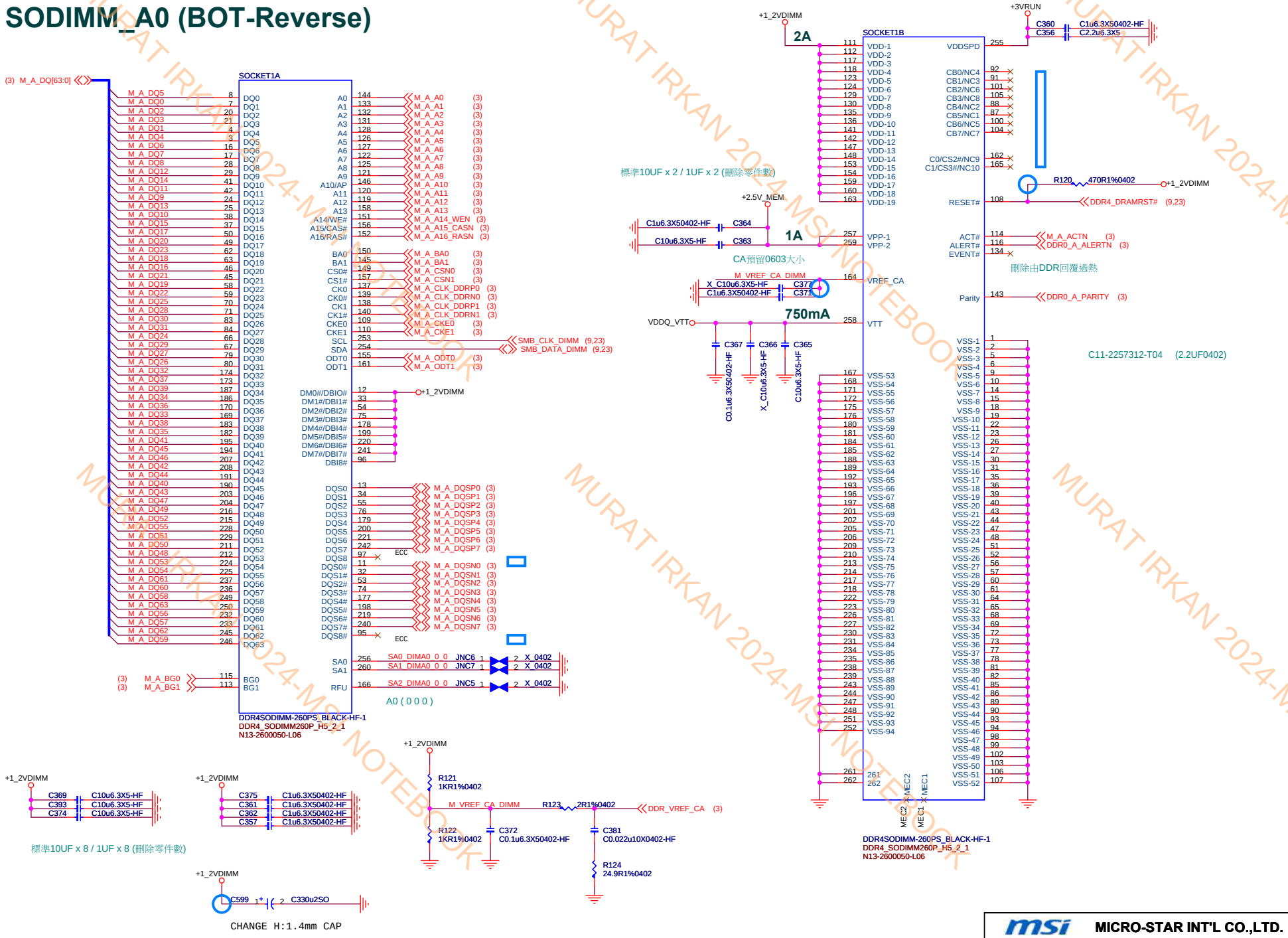
MURAT IRKAN 2024-MSI NOTEBOOK
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MICRO-STAR INT'L CO.,LTD.			
Title: Skylake(Power3)			
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 7 of 55		

MURAT IRKAN 2024-MSI NOTEBOOK

SODIMM_A0 (BOT-Reverse)

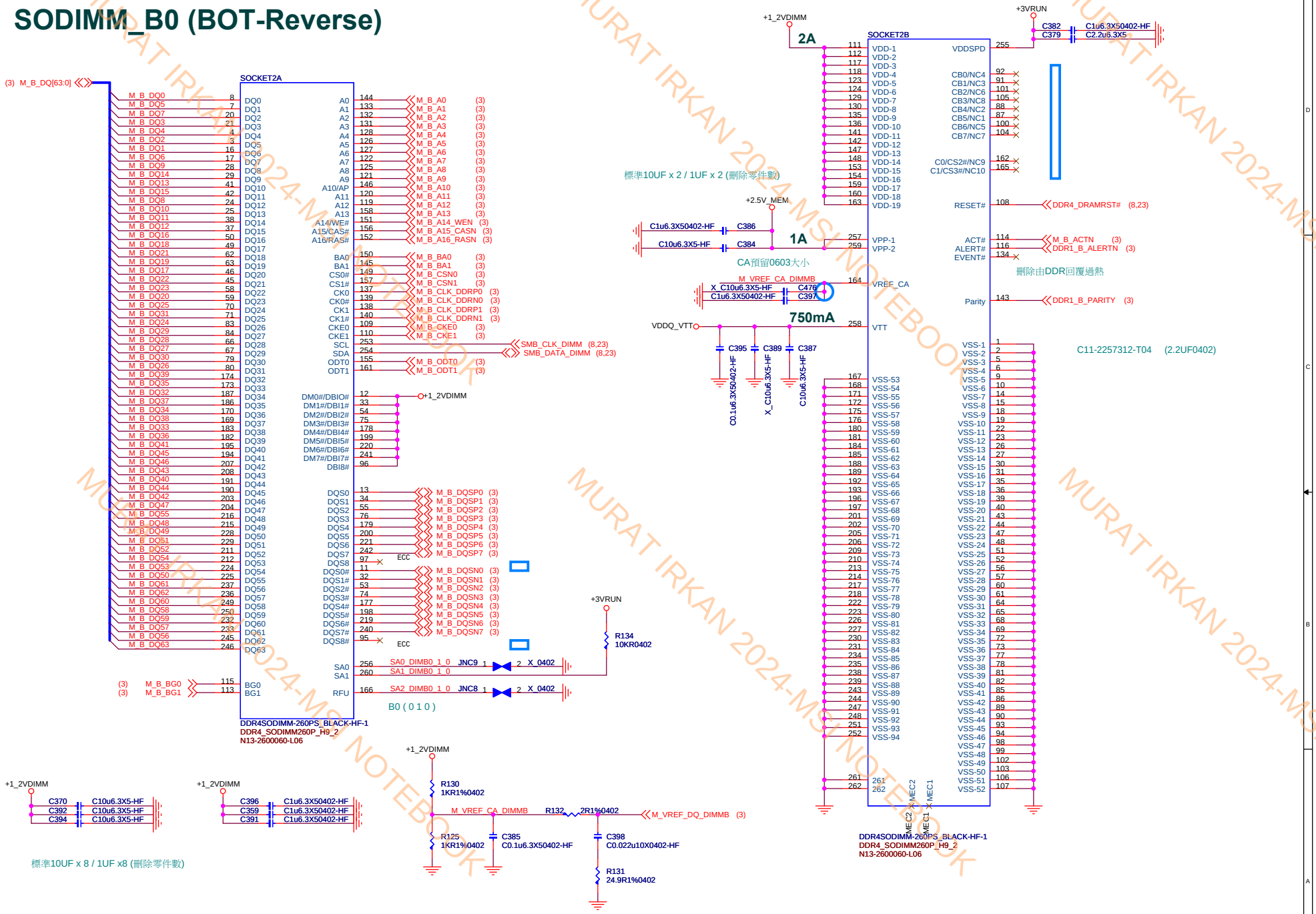


MURAT IRKAN 2024-MSI NOTEBOOK

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MICRO-STAR INT'L CO.,LTD.			
Title: DDR4_SODIMM_A0			
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 8 of 55		

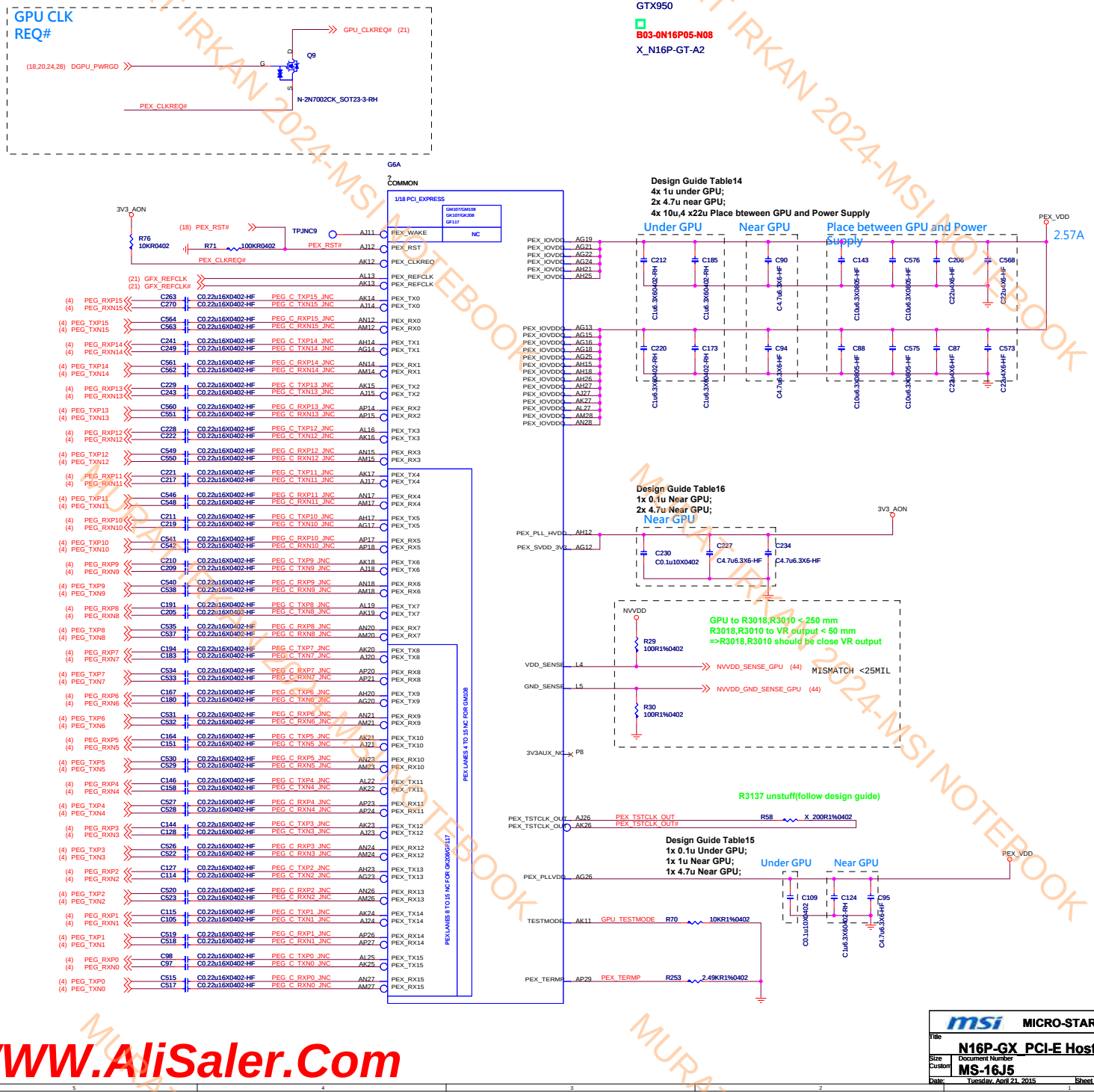
SODIMM_B0 (BOT-Reverse)



MICRO-STAR INT'L CO.,LTD.		
Title: DDR4_SODIMM_B0		
Size: Custom	Document Number: MS-16J5	Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 9 of 55	

MURAT IRKAN 2024-MSI NOTEBOOK

N16P-GX(PCI-Express Gen3 x16 Interface)



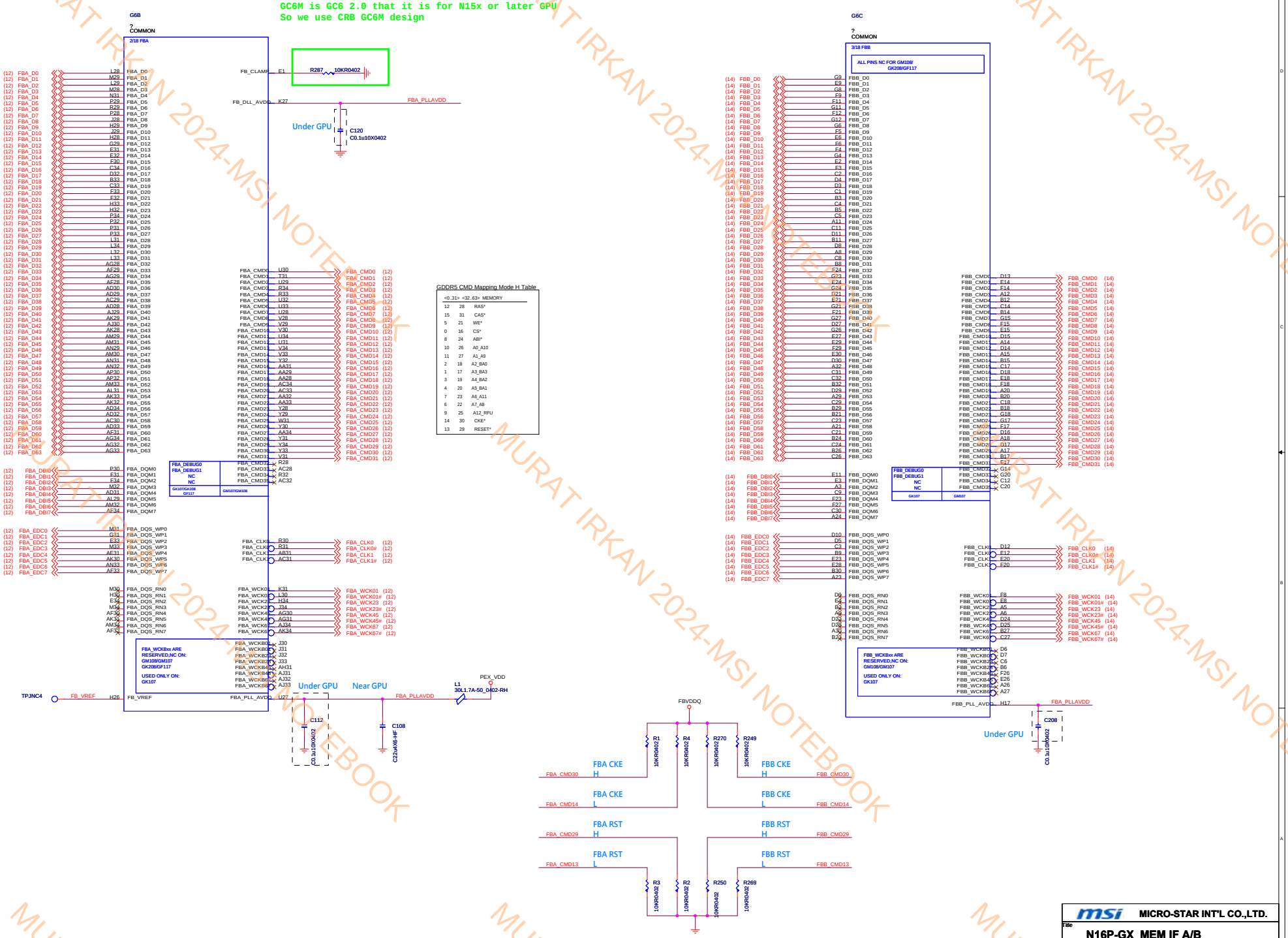
MURAT IRKAN 2024-MSI NOTEBOOK

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msi	MICRO-STAR INT'L CO.,LTD.	
Title	N16P-GX_PCI-E Host	
Size: Custom	Document Number: MS-16J5	Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 10 of 55	

N16P-GX(Frame Buffer Interface)

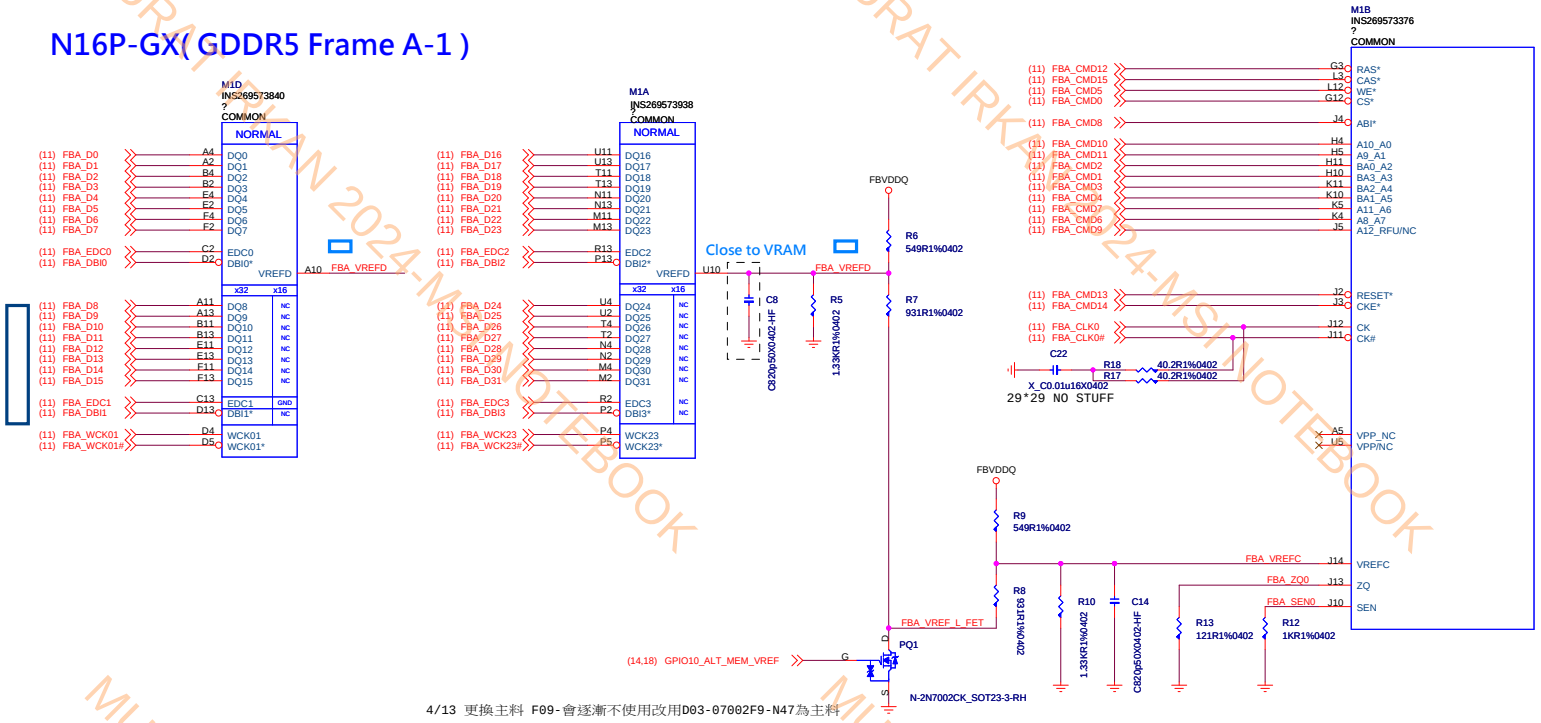
GC6M is GC6 2.0 that it is for N15x or later GPU
So we use CRB GC6M design



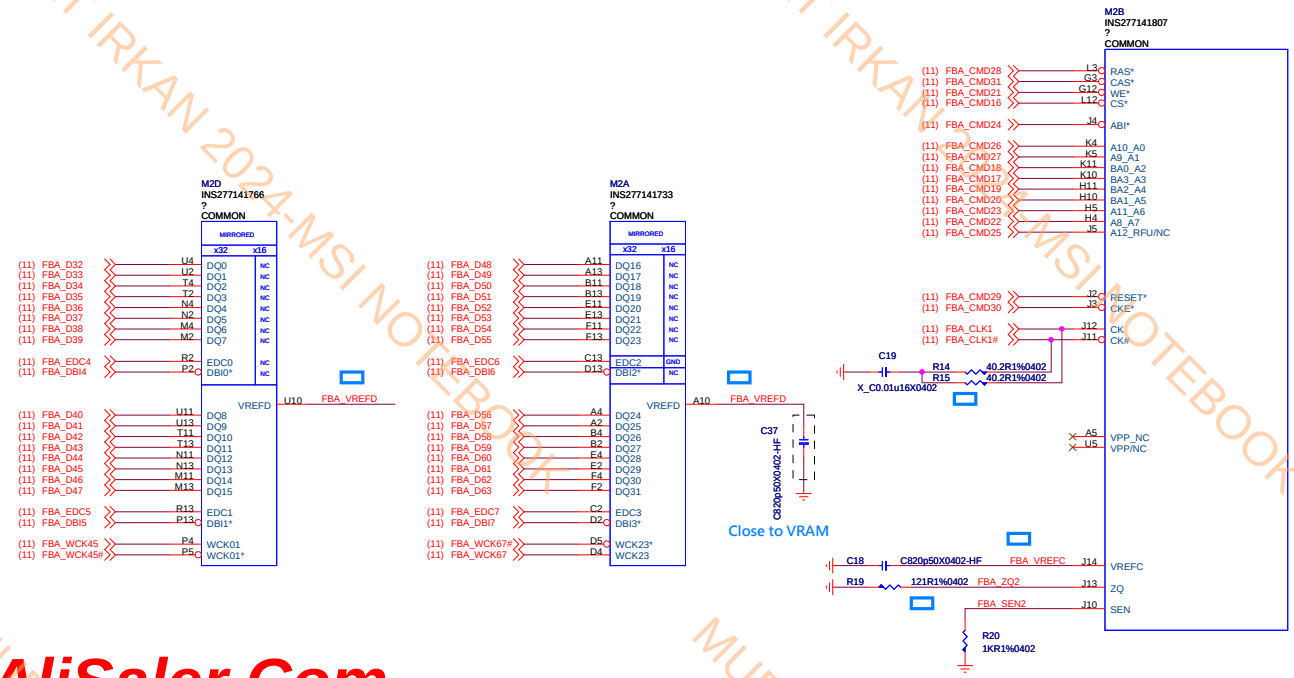
MICRO-STAR INT'L CO.,LTD.			
Title: N16P-GX MEM IF A/B			
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 11 of 55		

MURAT IRKAN 2024-MSI NOTEBOOK

N16P-GX(GDDR5 Frame A-1)



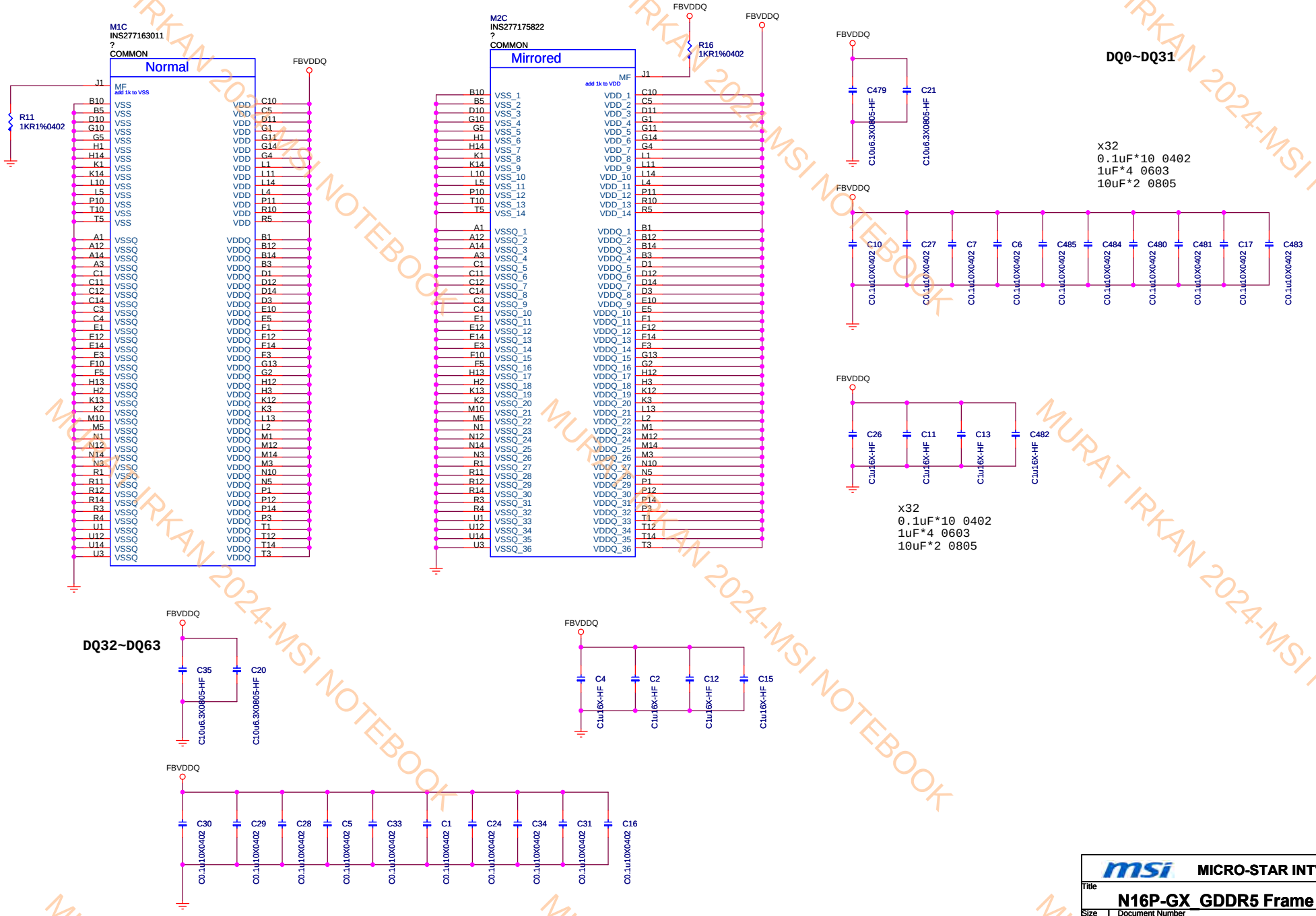
4/13 更換主料 F09-會逐漸不使用改用D03-07002F9-N47為主料



MICRO-STAR INT'L CO.,LTD.			
Title: N16P-GX_GDDR5 Frame A-1			
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 12 of 55		

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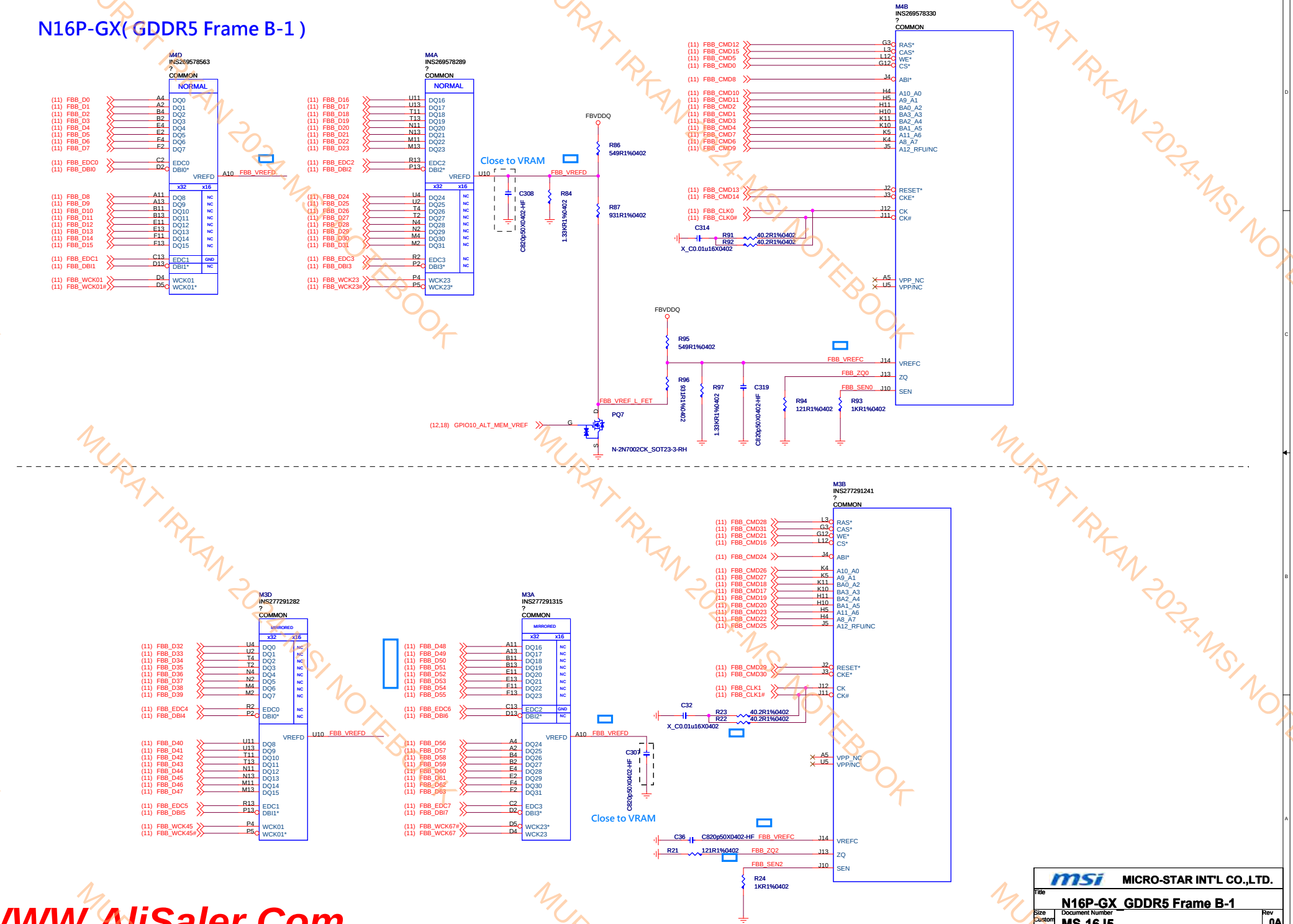
N16P-GX(GDDR5 Frame A-2)



MURAT IRKAN 2024-MSI NOTEBOOK

MICRO-STAR INT'L CO.,LTD.		
Title: N16P-GX_GDDR5 Frame A-2		
Size: A3	Document Number: MS-16J5	Rev: 0A
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N16P-GX(GDDR5 Frame B-1)

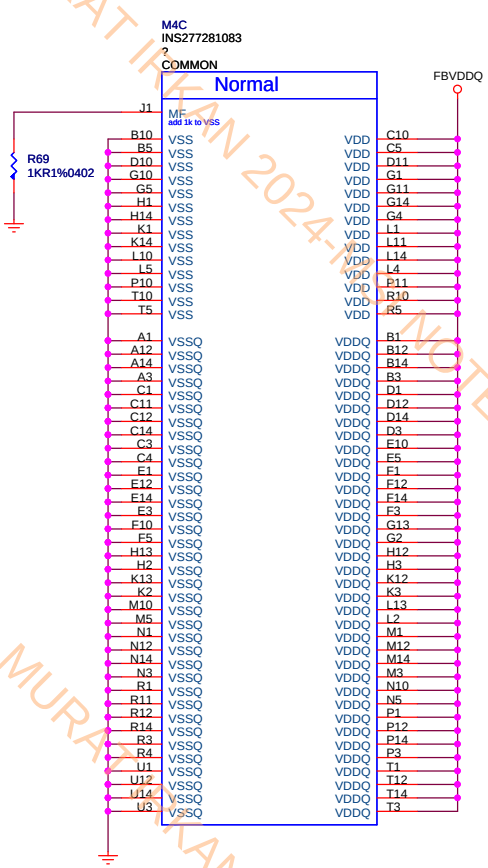


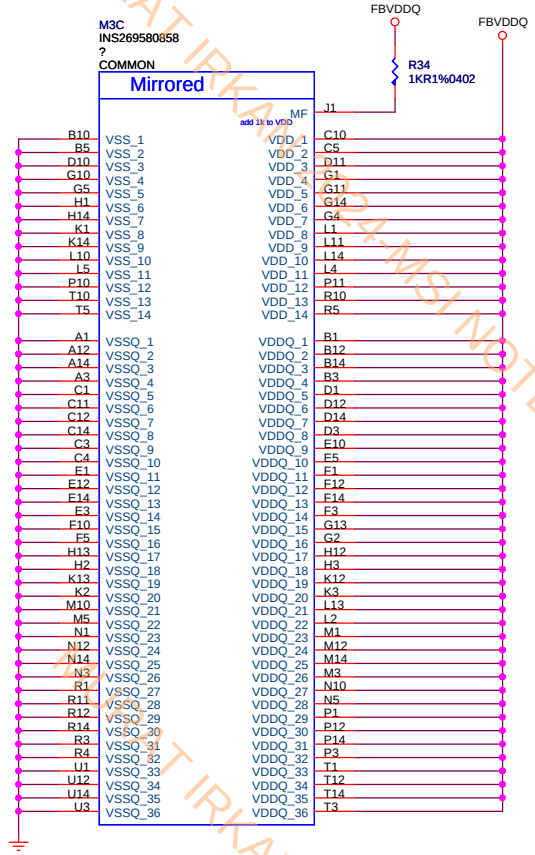
MURAT IRKAN 2024-MSI NOTEBOOK

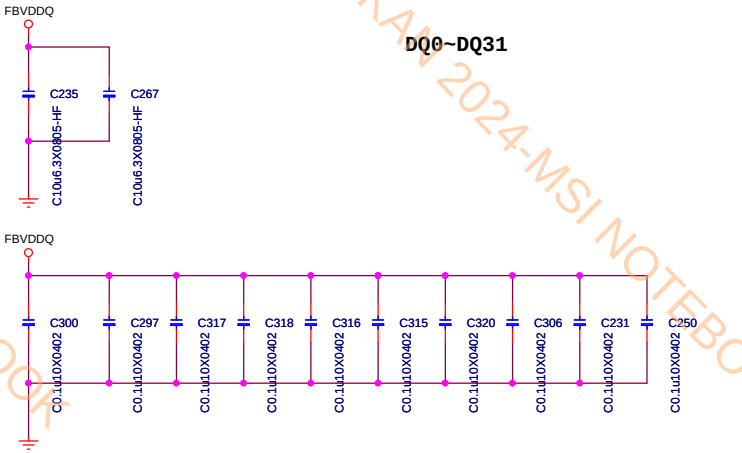
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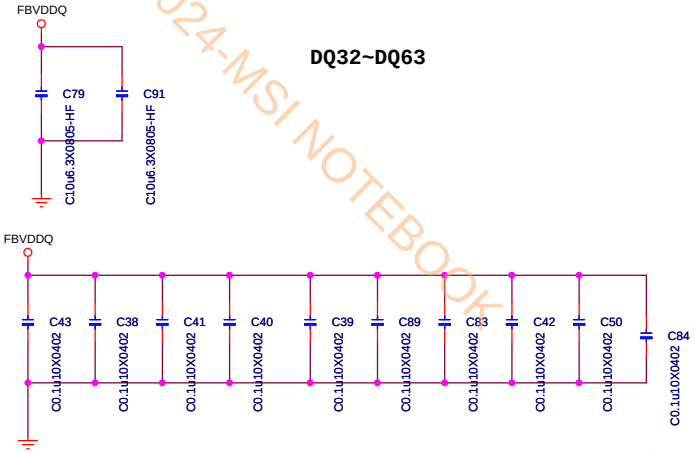
MICRO-STAR INT'L CO.,LTD.			
Title: N16P-GX_GDDR5 Frame B-1			
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 14 of 55		

N16P-GX(GDDR5 Frame B-2)







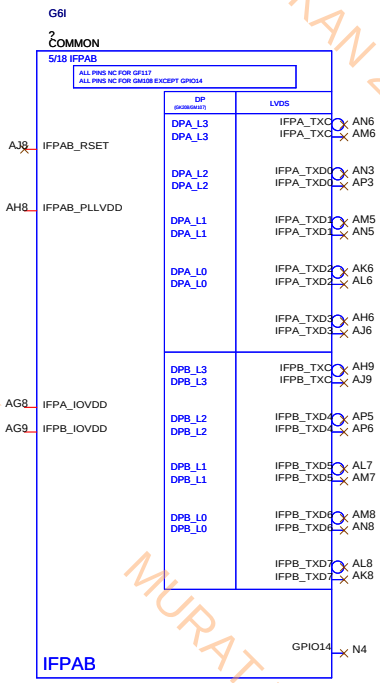


MICRO-STAR INT'L CO.,LTD.		
Title: N16P-GX_GDDR5 Frame B-2		
Size: A3	Document Number: MS-16J5	Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 15 of 55	

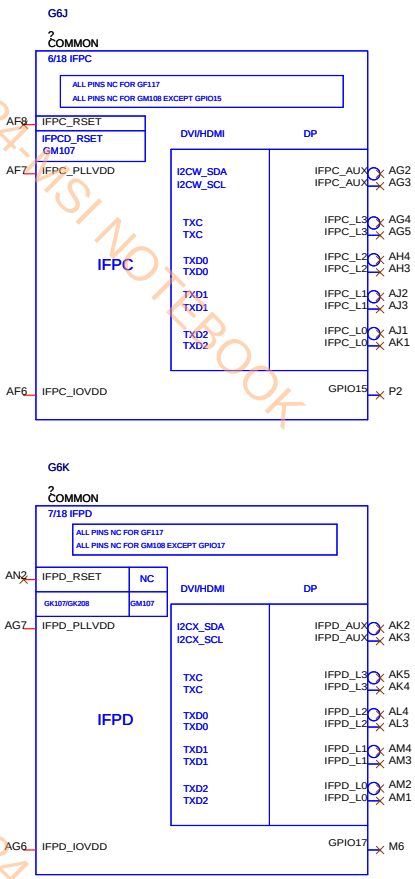
MURAT IRKAN 2024-MSI NOTEBOOK

N16P-GX(Display IF)

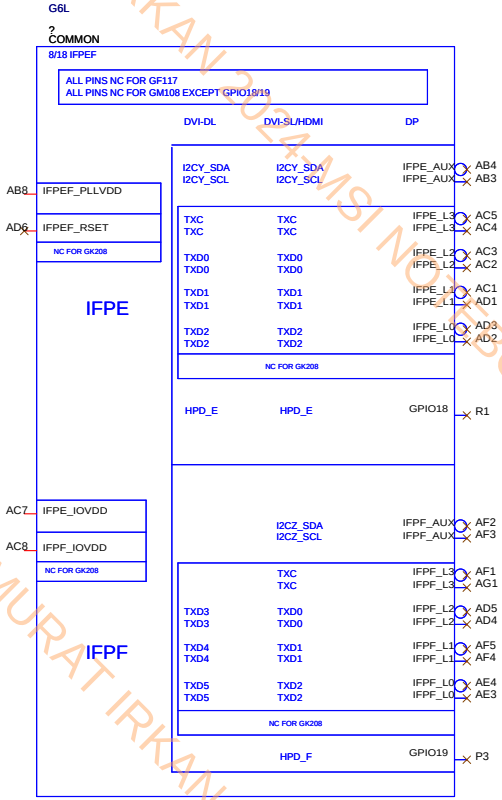
IFP A/B LVDSDual Link



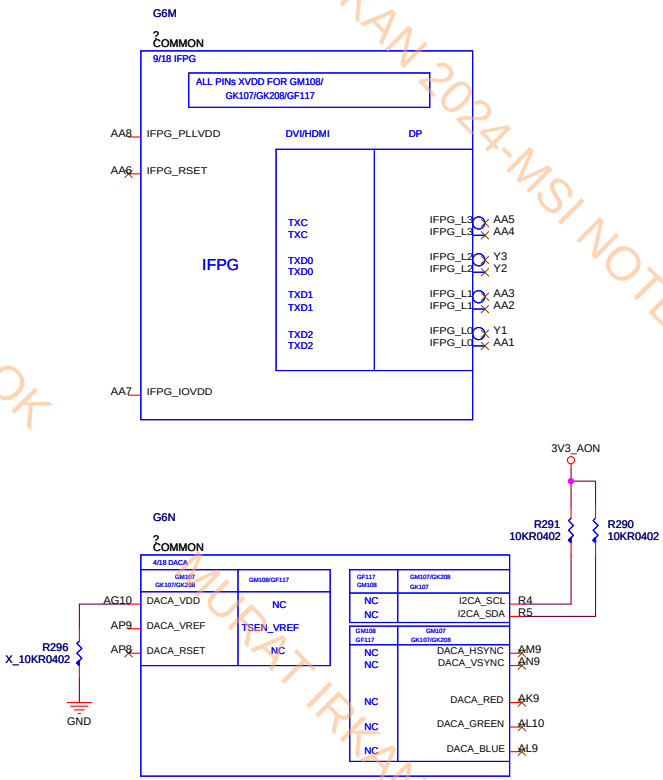
IFP C Native HDMI OR DP



IFP E/F Dual Link TMDS DVI-I



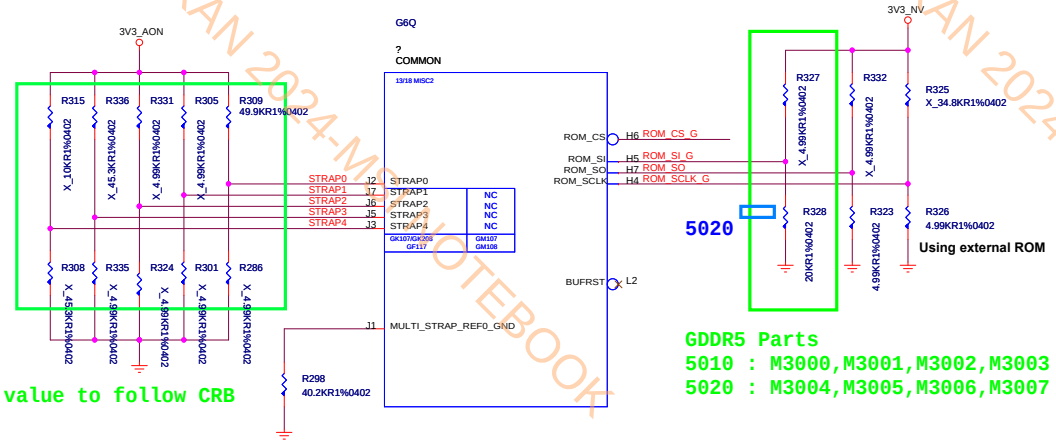
DAC A VGA



msi MICRO-STAR INT'L CO.,LTD.			
Title: N16P-GX_Display IF			
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015		Sheet 16 of 55	

MURAT IRKAN 2024-MSI NOTEBOOK
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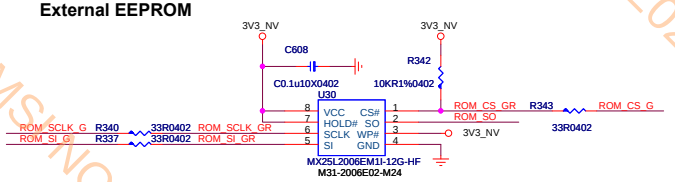
ROM, MULTI-LEVEL STRAPS



GDDR5 Parts
5010 : M3000,M3001,M3002,M3003
5020 : M3004,M3005,M3006,M3007

Modify R value to follow CRB

External EEPROM



M31-2006E02-M24

DEFAULT SETTING

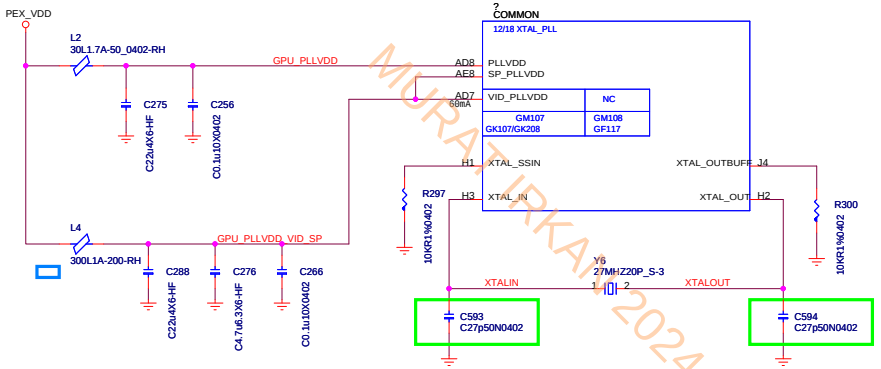
BOT 記得轉階層

ROM_S2 Samsung 128Mx32bit R11-0203T12-W08 X_20KR1%0402		V_BOT5 5020 M12-4132525-S02 X_K4G41325FC-HC03
ROM_H2 Hynix 128Mx32bit R11-0153T12-W08 X_15KR1%0402		V_BOT4 5020 M12-5GC4H05-H23 X_H5GC4H24MFR-T2C
ROM_M2 Micron 128Mx32bit R11-2492T12-W08 X_24.9KR1%0402		V_BOT6 5020 M12-4032B05-E59 X_EDW4032BABG-60-F-HF
ROM_H3 Hynix 128Mx32bit R11-3482T12-W08 X_34.8KR1%0402		V_BOT7 5020 M12-5GC4H65-H23 X_H5GC4H24AJR-T2C

ROM_SI RAM_CFG[3:0]

Micron GDDR5 : M12-4032B05-E59	0x4 25K PD	Micron 128Mx32bit
Samsung GDDR5 : M12-4132525-S02 V	0x3 20K PD	Samsung 128Mx32bit
Hynix GDDR5 : M12-5GC4H05-H23	0x2 15K PD	Hynix 128Mx32bit
Hynix GDDR5 : H5GC4H24AJR-T2C V	0x6 35K PD	

ROM_SO	DEVID_SEL PCIE_CFG SMB_ALT_ADDR VGADEVICE	5K PD
ROM_SCLK	SOR_EXposed[3:0]	5K PD
STRAP0		50K PU 3V3_AON
STRAP1		Reserved
STRAP2		Reserved
STRAP3		Reserved
STRAP4		Reserved



C3438 and C3439 change to 27pF for SA

	GND	3V3
5K	0000	1000
10K	0001	1001
15K	0010	1010
20K	0011	1011
25K	0100	1100
30K	0101	1101
35K	0110	1110
45K	0111	1111
	PD	PU

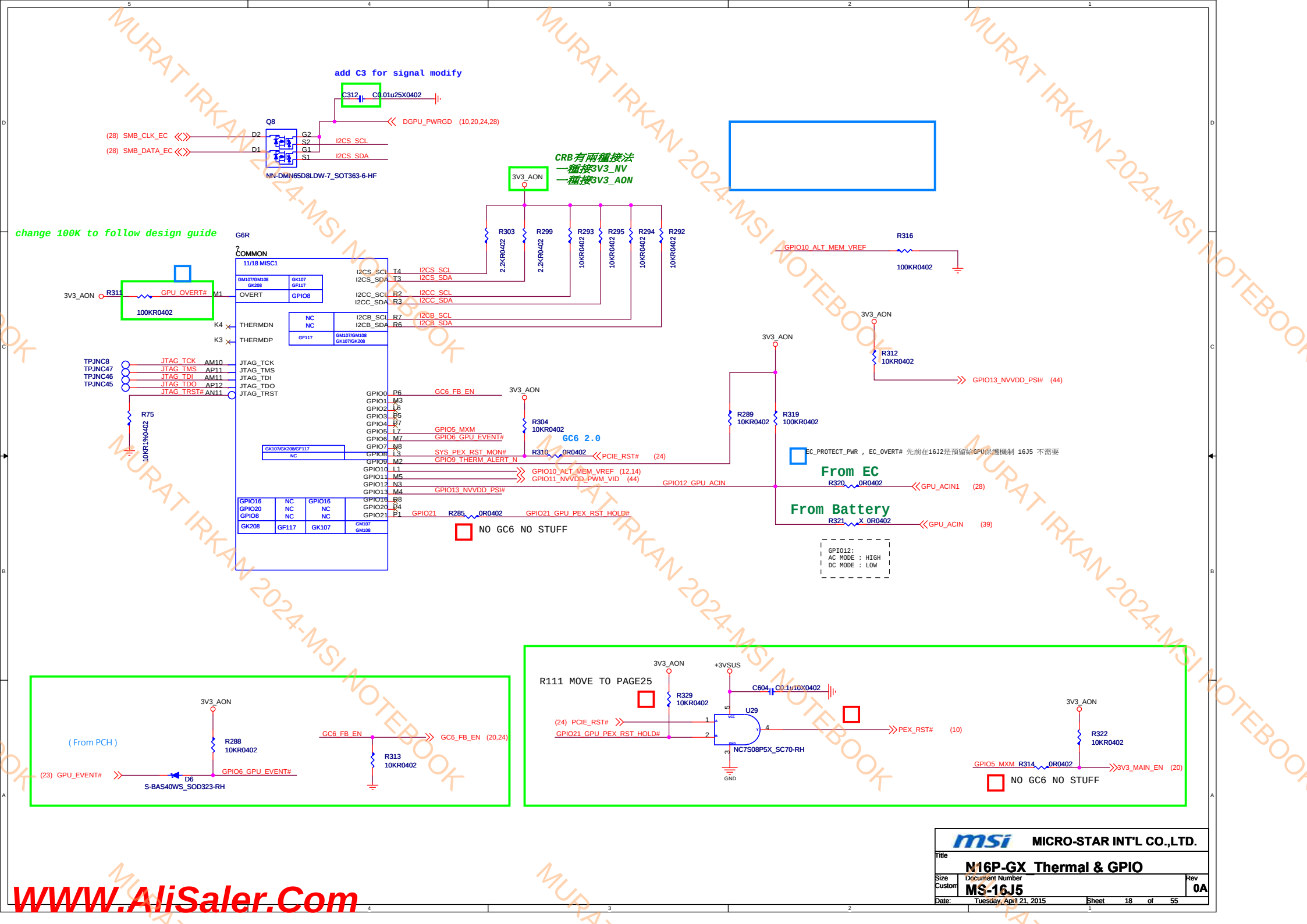
MICRO-STAR INT'L CO.,LTD.

Title: N16P-GX_ROM & HW Straps

Size: Custom | Document Number: MS-16J5 | Rev: 0A

Date: Tuesday, April 21, 2015 | Sheet 17 of 55

MURAT IRKAN 2024-MSI NOTEBOOK



MICRO-STAR INT'L CO.,LTD.

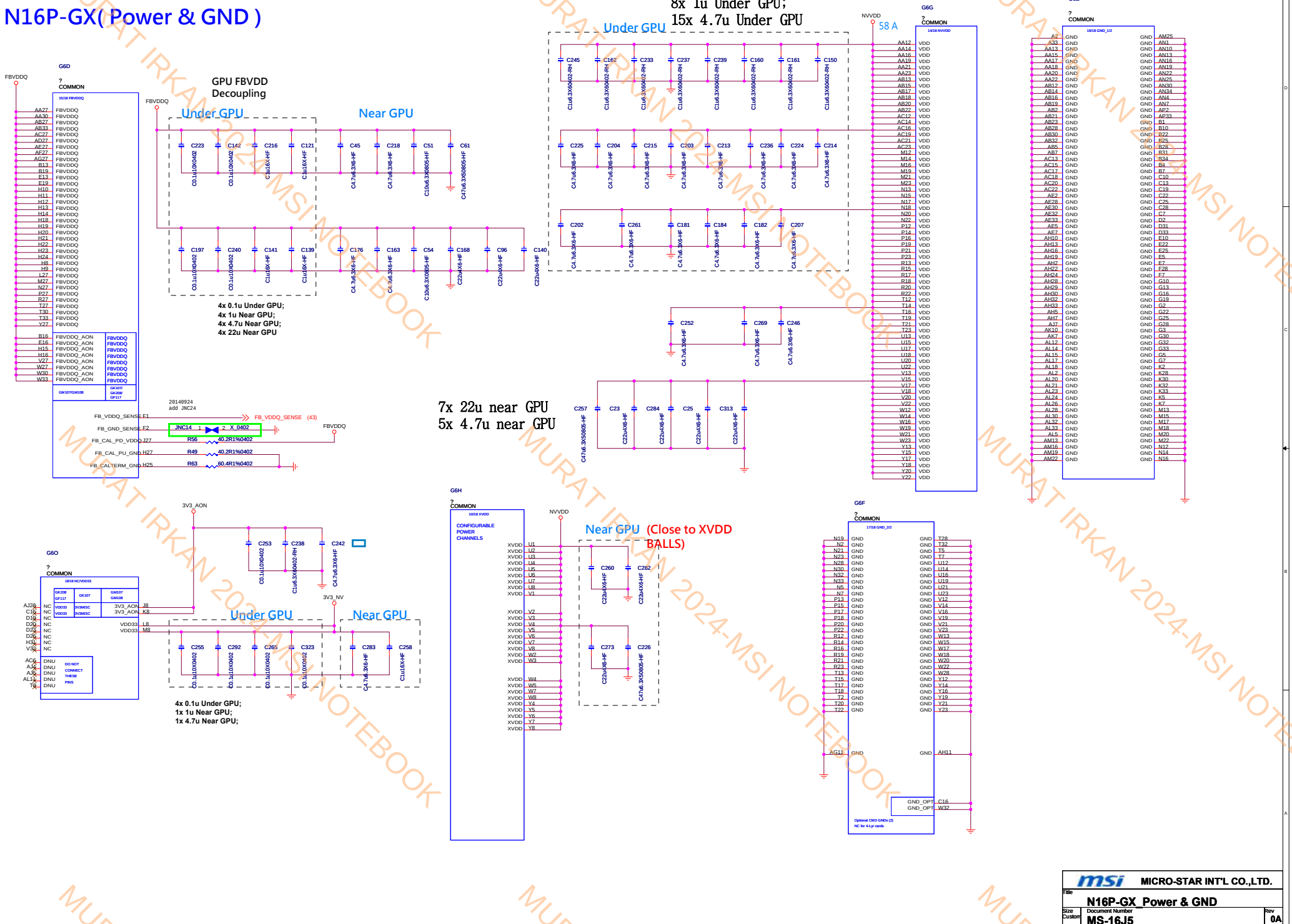
Title: N16P-GX_Thermal & GPIO

Size: Custom | Document Number: MS-16J5 | Rev: 0A

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MURAT IRKAN 2024-MSI NOTEBOOK
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N16P-GX(Power & GND)



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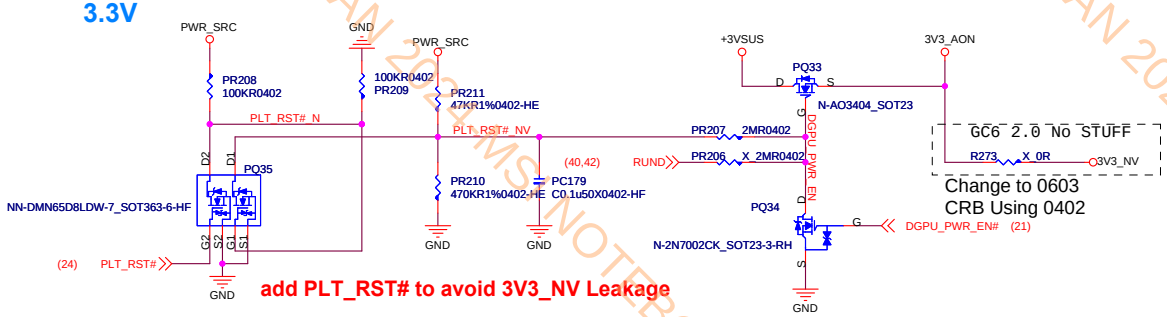
msi	MICRO-STAR INT'L CO.,LTD.
Title	N16P-GX_Power & GND
Size / Custom	Document Number: MS-16J5
Rev	0A
Date:	Tuesday, April 21, 2015
Sheet	19 of 55

N16P-GX(Power Control)

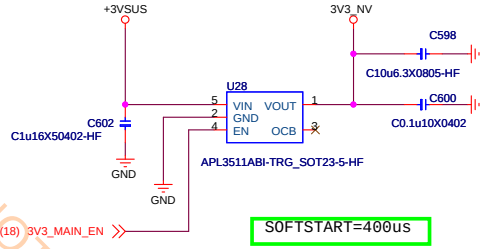
nVIDIA Power Sequence Control

3V3_AON -> 3V3_NV -> NVVDD ->PEX_VDD -> FBVDDQ -> DGPUPWRGD

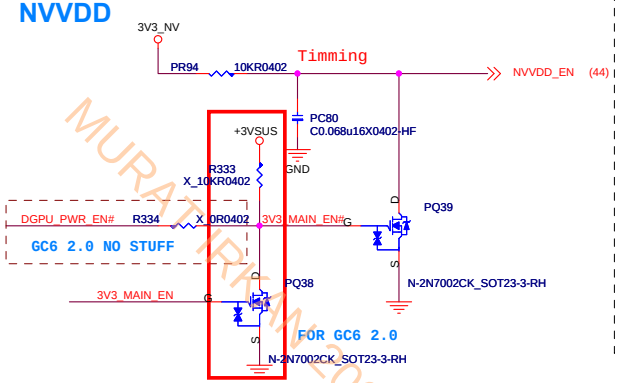
3.3V



GC6 2.0 STUFF



NVVDD

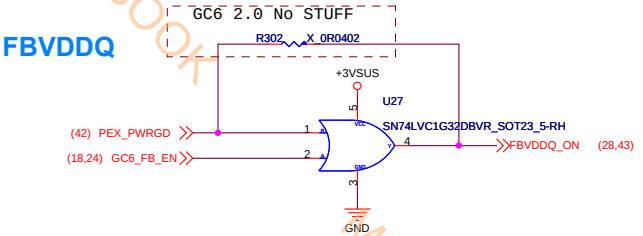


PEX_VDD

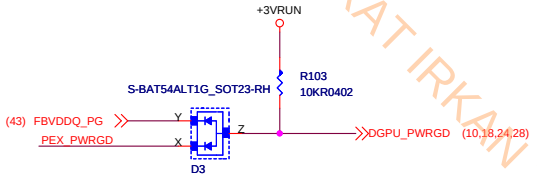
EDP Design Guide:
N16P-GX
NVVDD : 51.1A ; Peak 87A
3.3VRUN : 0.34A
PEX_VDD : 2.57A
FBVDDQ : 8.75A

EDP Design Guide:
N16P-GT
NVVDD : 30.2A ; Peak 53A
3.3VRUN : 0.34A
PEX_VDD : 2.57A
FBVDDQ : 8.75A

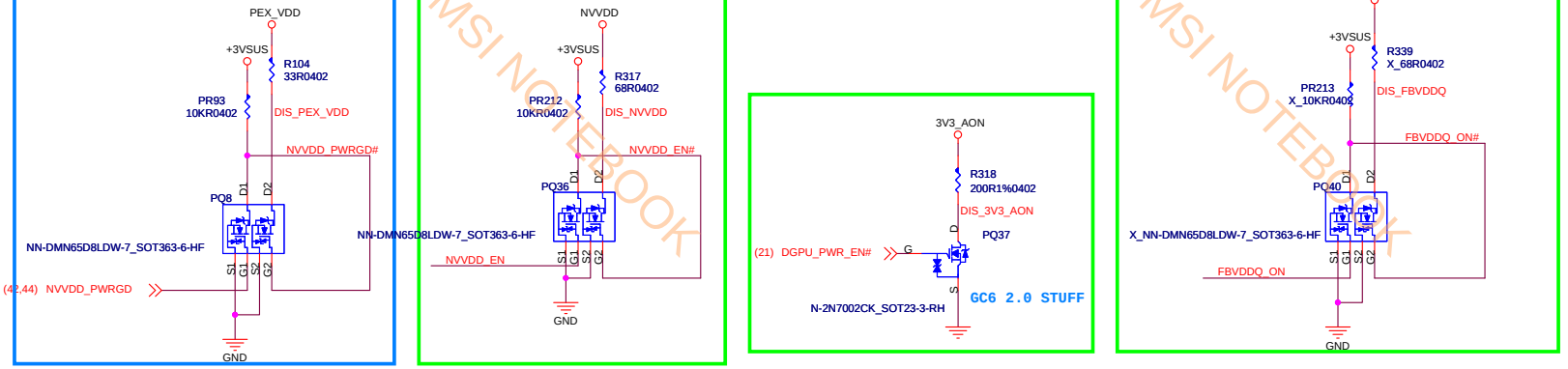
FBVDDQ



DGPU_PWRGD



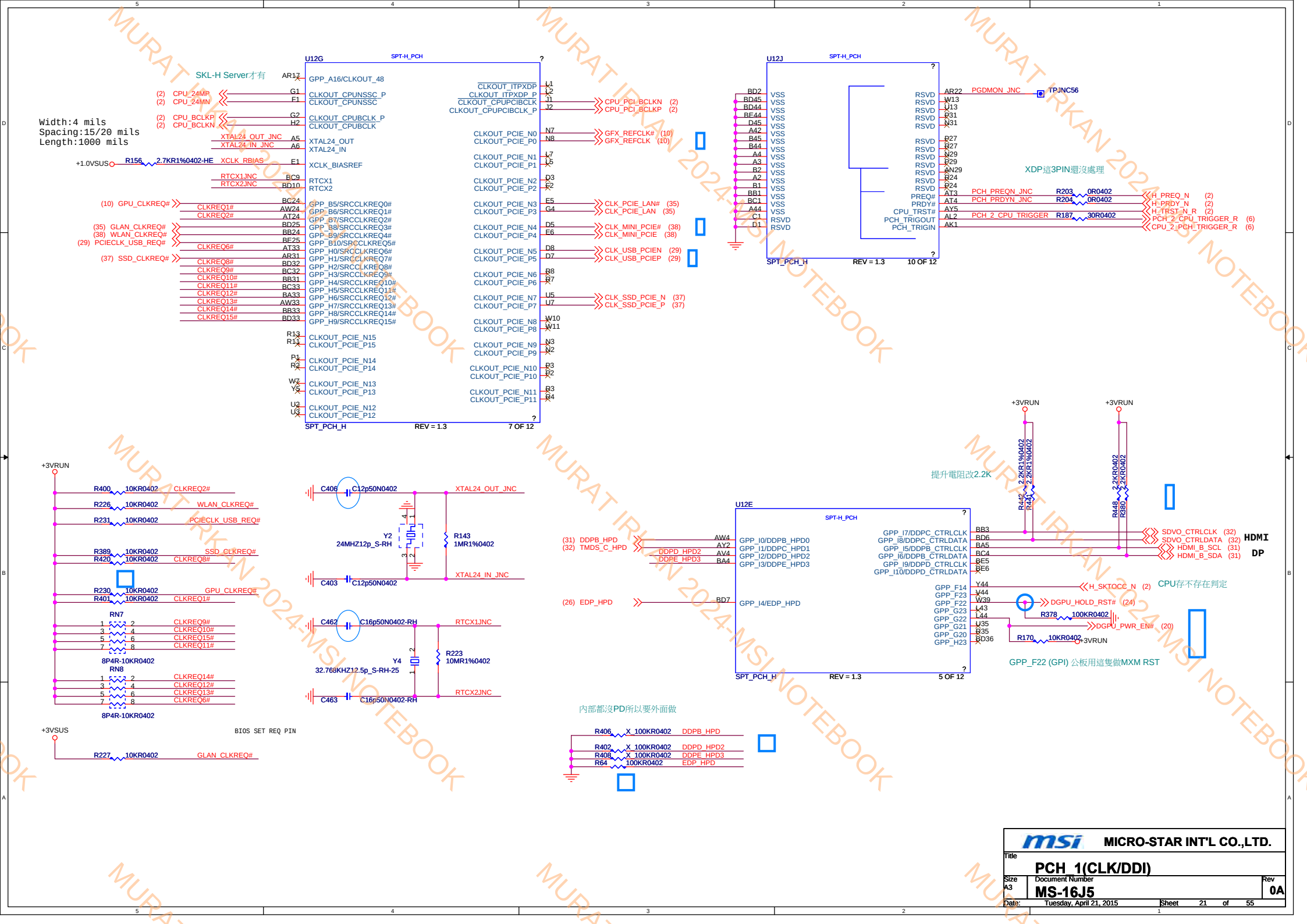
Discharge Circuit

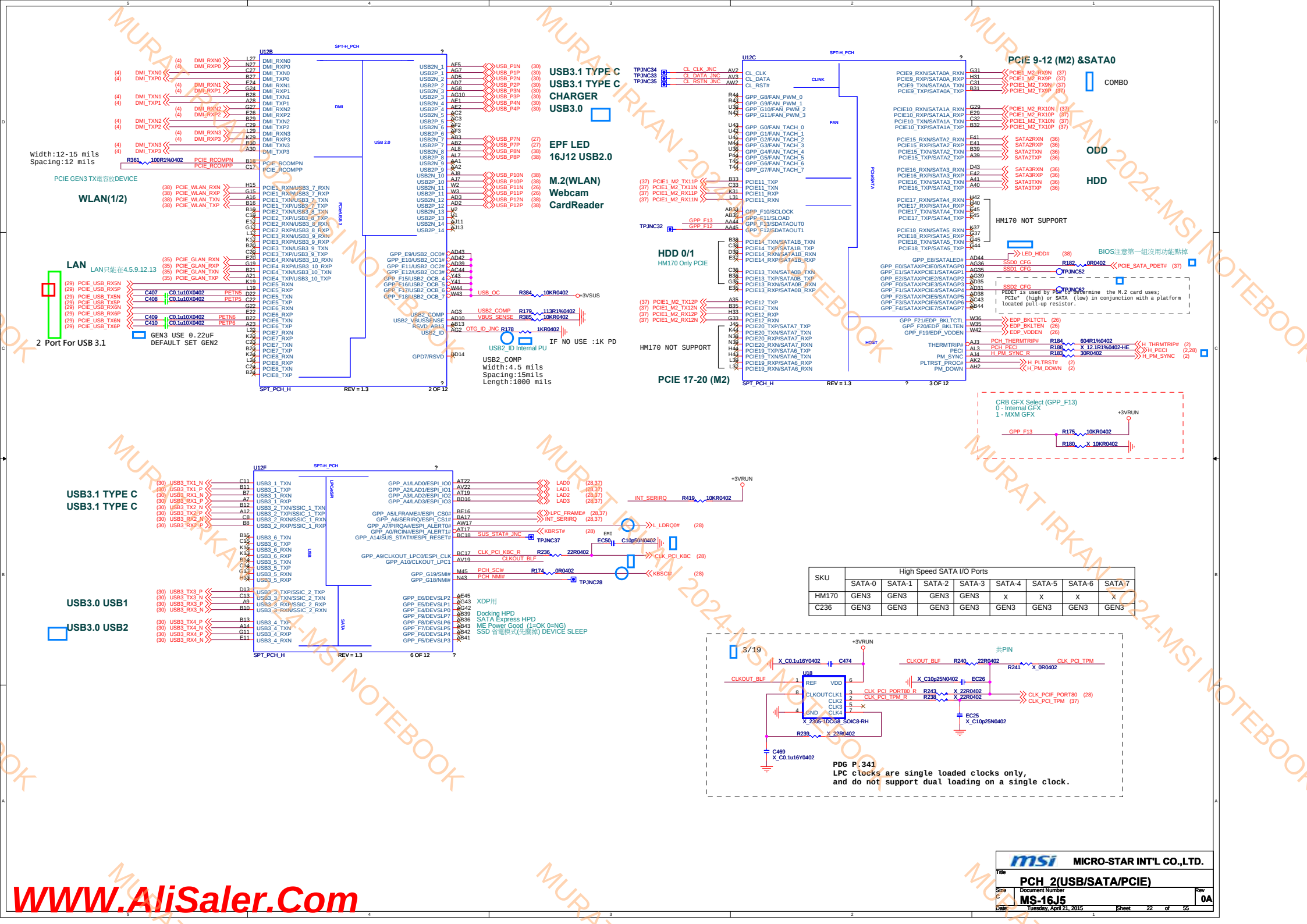


msi	MICRO-STAR INT'L CO.,LTD.		
Title	N16P-GX_Power Control		
Size: Custom	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 20 of 55		

4/13更換料號
D03-7002D10-D07 轉到 D03-65D8L09-D07

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Width:4 mils
Spacing:15/20 mils
Length:1000 mils
SKL-H Server才有
U12G
SPT-H_PCH
SPT_PCH_H
REV = 1.3
7 OF 12
U12J
10 OF 12
XDP這3PIN還沒處理
U12E
5 OF 12
提升電阻改2.2K
CPU存不存在判定
GPP_F22 (GPI) 公板用這隻做MXM RST
內部都沒PD所以要外面做
BIOS SET REQ PIN
HDMI
DP
MICRO-STAR INT'L CO.,LTD.
PCH_1(CLK/DDI)
MS-16J5
0A
Tuesday, April 21, 2015
Sheet 21 of 55



SKU	High Speed SATA I/O Ports							
	SATA-0	SATA-1	SATA-2	SATA-3	SATA-4	SATA-5	SATA-6	SATA-7
HM170	GEN3	GEN3	GEN3	GEN3	X	X	X	X
C236	GEN3	GEN3	GEN3	GEN3	GEN3	GEN3	GEN3	GEN3

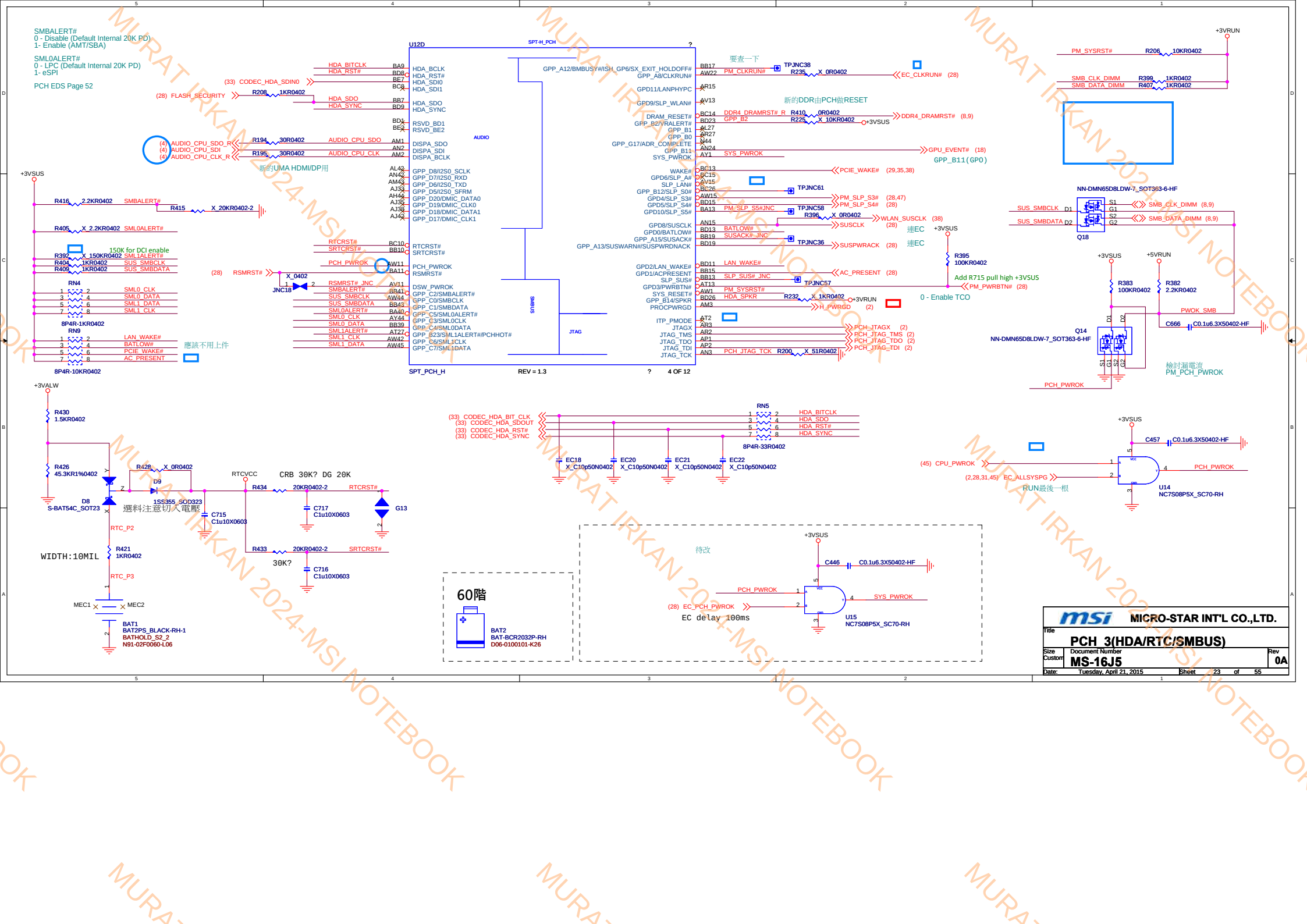
MICRO-STAR INT'L CO.,LTD.

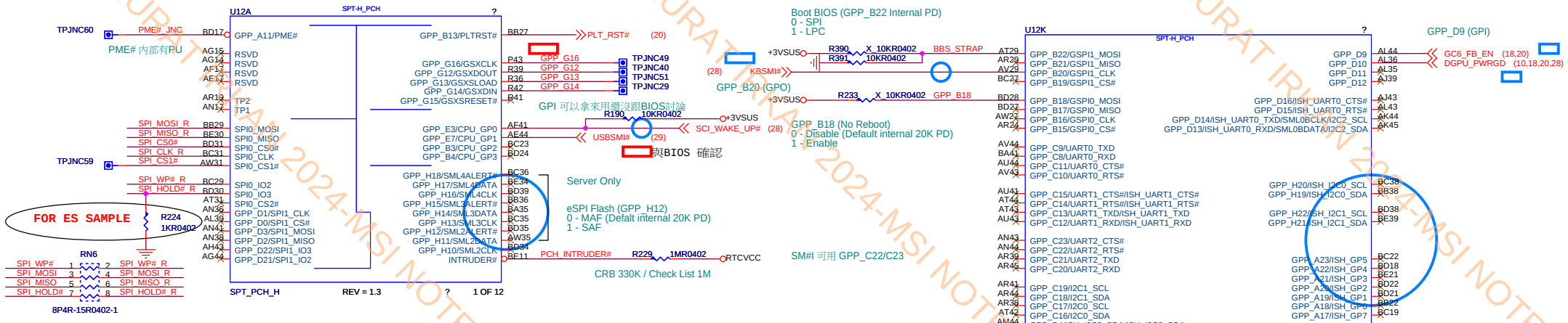
Title: PCH_2(USB/SATA/PCIE)

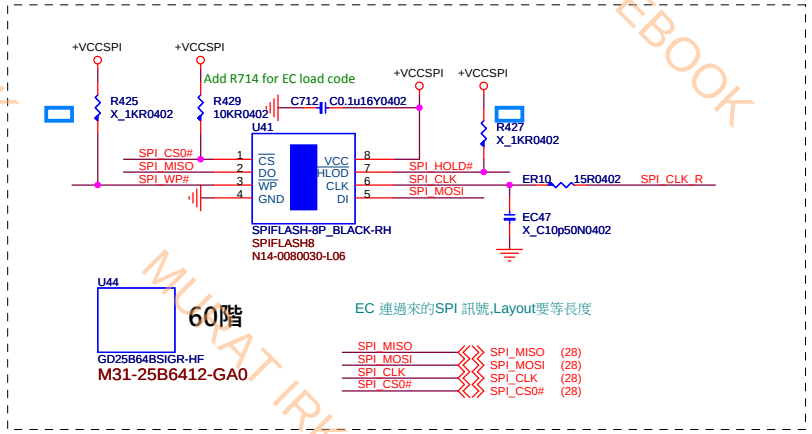
Size: C | Document Number: MS-16J5 | Rev: 0A

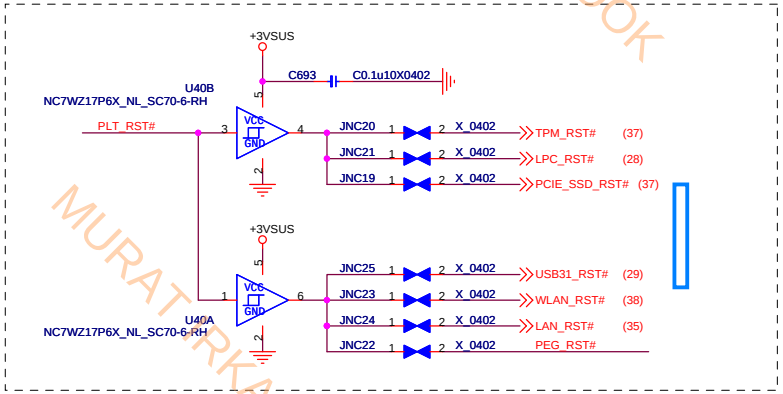
Date: Tuesday, April 21, 2015 | Sheet 22 of 55

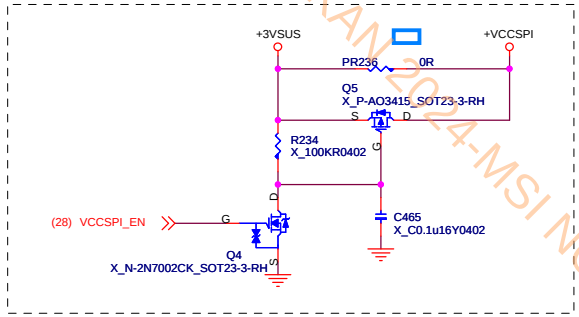
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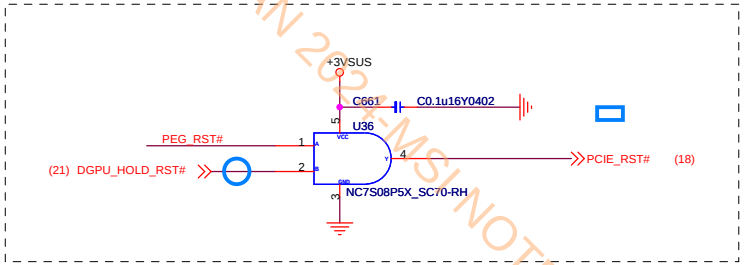

SMBALERT#
0 - Disable (Default Internal 20K PD)
1- Enable (AMT/SBA)
SML0ALERT#
0 - LPC (Default Internal 20K PD)
1- eSPI
PCH EDS Page 52
U12D
SPT-H_PCH
SPT_PCH_H
REV = 1.3
4 OF 12
AUDIO
SMBUS
JTAG
HDA_BITCLK
HDA_RST#
HDA_SDI0
HDA_SDI1
HDA_SDO
HDA_SYNC
RSVD_BD1
RSVD_BE2
DISPA_SDO
DISPA_SDI
DISPA_BCLK
GPP_D8/I2S0_SCLK
GPP_D7/I2S0_RXD
GPP_D6/I2S0_TXD
GPP_D5/I2S0_SFRM
GPP_D20/DMIC_DATA0
GPP_D19/DMIC_CLK0
GPP_D18/DMIC_DATA1
GPP_D17/DMIC_CLK1
RTCRST#
SRTCRST#
PCH_PWROK
RSMRST#
DSW_PWROK
GPP_C2/SMBALERT#
GPP_C0/SMBCLK
GPP_C1/SMBDATA
GPP_C5/SML0ALERT#
GPP_C3/SML0CLK
GPP_C4/SML0DATA
GPP_B23/SML1ALERT#/PCHHOT#
GPP_C6/SML1CLK
GPP_C7/SML1DATA
GPP_A12/BMBUSY#/ISH_GP6/SX_EXIT_HOLDOFF#
GPP_A8/CLKRUN#
GPD11/LANPHYPC
GPD9/SLP_WLAN#
DRAM_RESET#
GPP_B2/VRALERT#
GPP_B1
GPP_B0
GPP_G17/ADR_COMPLETE
GPP_B11
SYS_PWROK
WAKE#
GPD6/SLP_A#
SLP_LAN#
GPP_B12/SLP_S0#
GPD4/SLP_S3#
GPD5/SLP_S4#
GPD10/SLP_S5#
GPD8/SUSCLK
GPD0/BATLOW#
GPP_A15/SUSACK#
GPP_A13/SUSWARN#/SUSPWRDNACK
GPD2/LAN_WAKE#
GPD1/ACPRESENT
SLP_SUS#
GPD3/PWRBTN#
SYS_RESET#
GPP_B14/SPKR
PROCPWRGD
ITP_PMODE
JTAGX
JTAG_TMS
JTAG_TDO
JTAG_TDI
JTAG_TCK
要查一下
新的DDR由PCH做RESET
新的UMA HDMI/DP用
150K for DCI enable
應該不用上件
連EC
連EC
Add R715 pull high +3VSUS
0 - Enable TCO
檢討漏電流
PM_PCH_PWROK
RUN最後一根
待改
60階
遲料注意切入電壓
CRB 30K? DG 20K
30K?
WIDTH:10MIL
EC delay 100ms
MICRO-STAR INT'L CO.,LTD.
Title
PCH_3(HDA/RTC/SMBUS)
Size
Custom
Document Number
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Rev
0A
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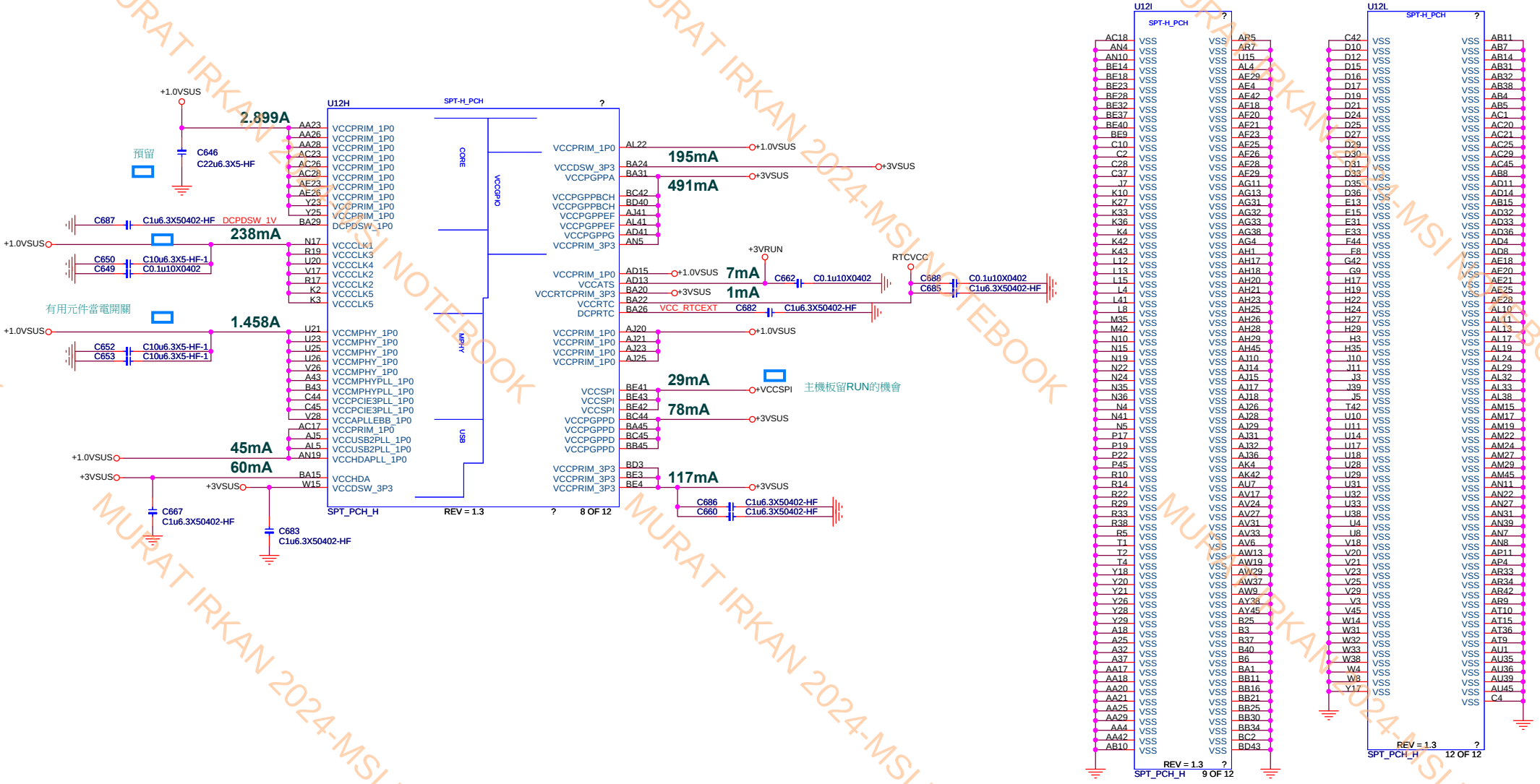
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Title: PCH_4(SPI/GPIO)

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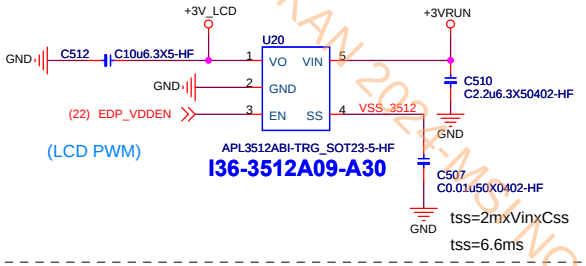


MICRO-STAR INT'L CO.,LTD.			
Title: PCH_5(Power1)			
Size: Custom	Document Number: MS-16J5		Rev: 0A
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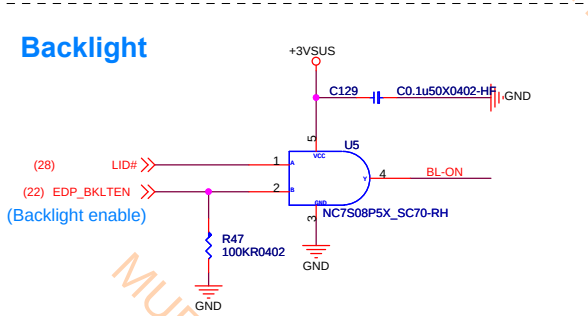
MURAT IRKAN 2024-MSI NOTEBOOK

eDP

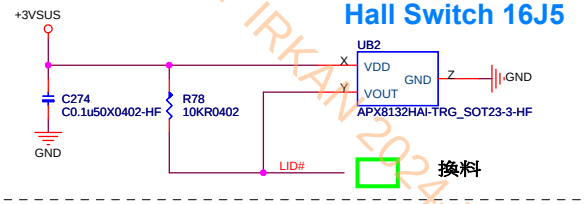
Pannel Device Logic Power



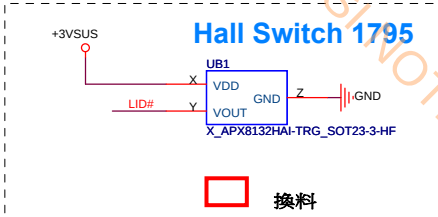
Backlight



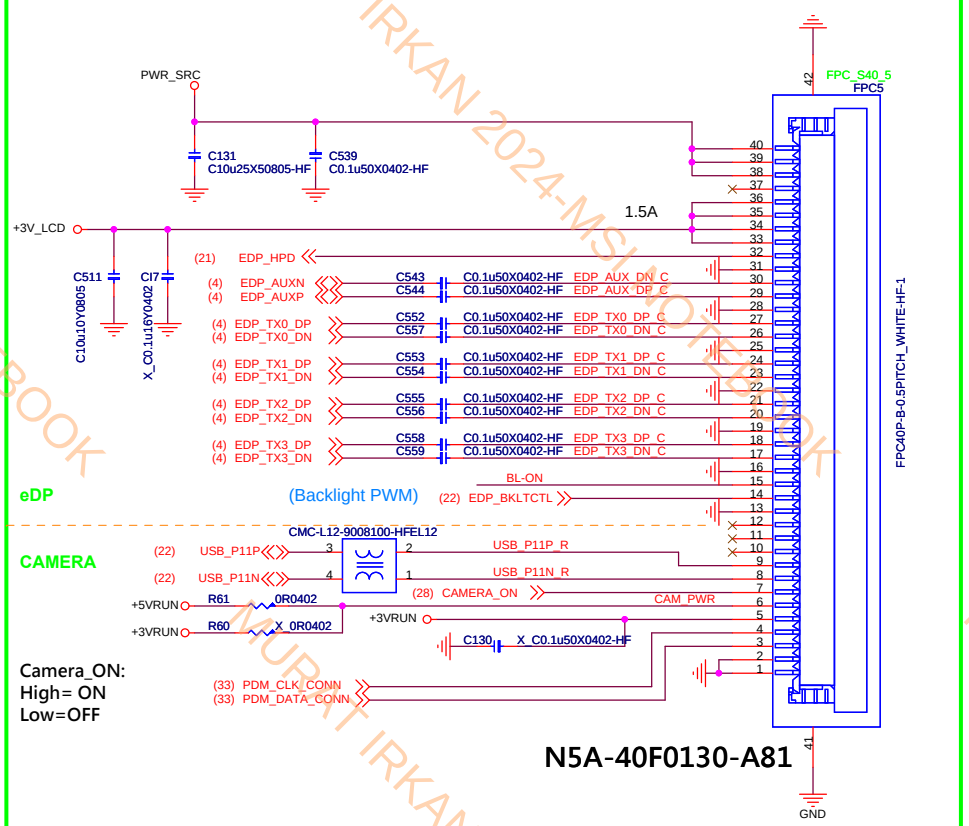
Hall Switch 16J5

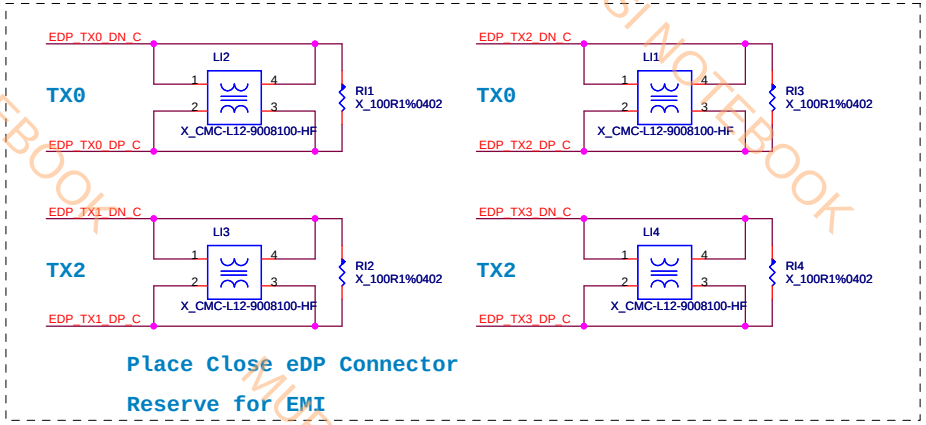


Hall Switch 1795



eDP CONN CAMERA





LCD Module Pin Define

Pin No	Symbol	Description
1	WP	EEPROM Write Protect(Keep open)
2	H_GND	High Speed Ground(0V)
3	eDP_Rx_3N	Complement Signal Link Lane 3
4	eDP_Rx_3P	True Signal Link Lane 3
5	H_GND	High Speed Ground(0V)
6	eDP_Rx_2N	Complement Signal Link Lane 2
7	eDP_Rx_2P	True Signal Link Lane 2
8	H_GND	H_GND
9	eDP_Rx_1N	Complement Signal Link Lane 1
10	eDP_Rx_1P	True Signal Link Lane 1
11	H_GND	H_GND
12	eDP_Rx_0N	Complement Signal Link Lane 0
13	eDP_Rx_0P	True Signal Link Lane 0
14	H_GND	H_GND
15	eDP_AUX_CH_P	True Signal Aux Channel
16	eDP_AUX_CH_N	Complement Signal Aux Channel
17	H_GND	H_GND
18	LCD_VCC	LCD logic and driver power
19	LCD_VCC	LCD logic and driver power
20	LCD_VCC	LCD logic and driver power
21	LCD_VCC	LCD logic and driver power
22	TEST	LCD Test Port
23	LCD_GND	LCD logic and driver ground(0V)
24	LCD_GND	LCD logic and driver ground(0V)
25	LCD_GND	LCD logic and driver ground(0V)
26	LCD_GND	LCD logic and driver ground(0V)
27	eDP_HPD	HPD signal pin
28	BL_GND	Backlight ground(0V)
29	BL_GND	Backlight ground(0V)
30	BL_GND	Backlight ground(0V)
31	BL_GND	Backlight ground(0V)
32	BL_ENABLE	Backlight enable
33	BL_PWM_DIM	System PWM signal input
34	SDA	I2C-bus Data
35	SCL	I2C-bus Clock
36	BL_PWR	Backlight power (5~21V)
37	BL_PWR	Backlight power (5~21V)
38	BL_PWR	Backlight power (5~21V)
39	BL_PWR	Backlight power (5~21V)
40	HSYNC	HSYNC output from Tcon

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Title: eDP /Camera

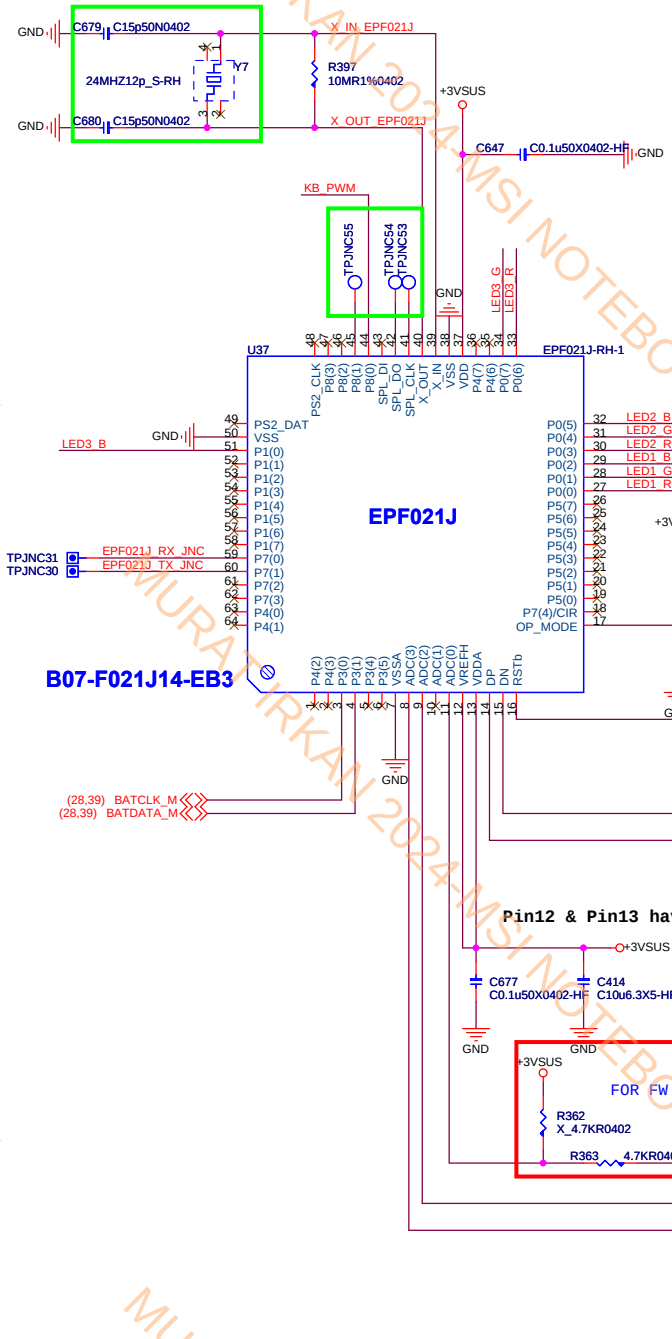
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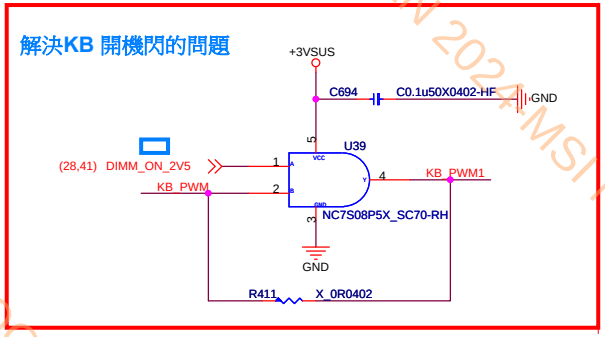
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LED 8051 Controller

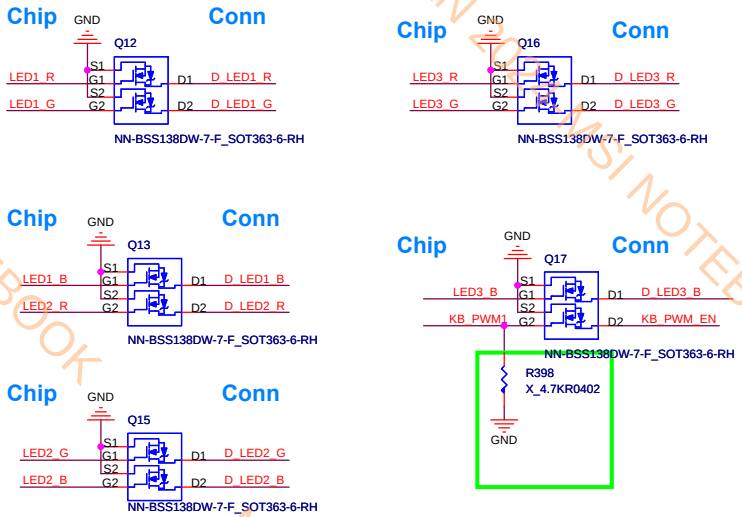
C749 and C750 change to 15pF for SA



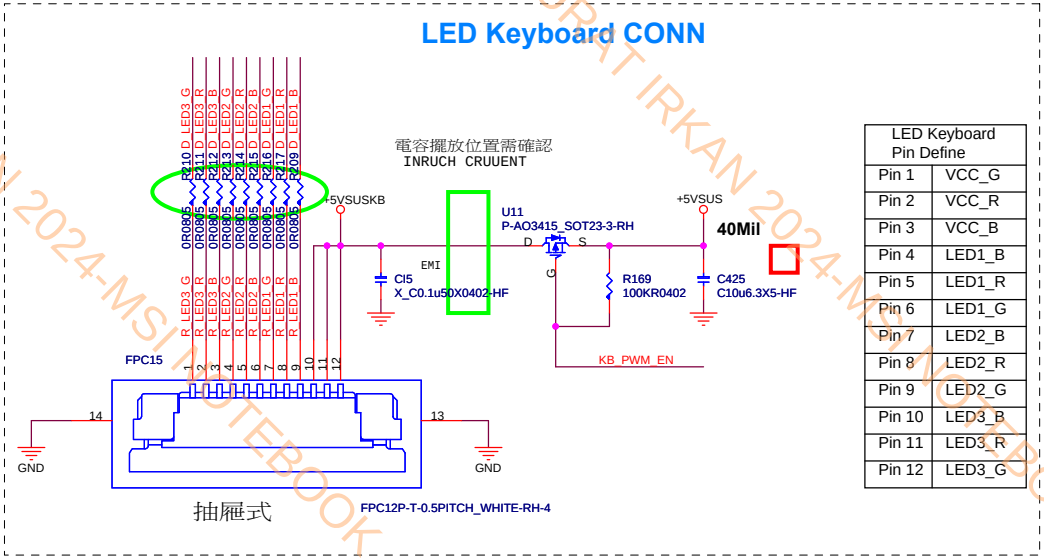
解決KB 開機閃的問題



EPF021J Sink current not enough, only using BSS138 (0.22A)



LED Keyboard CONN



LED Keyboard Pin Define	
Pin 1	VCC_G
Pin 2	VCC_R
Pin 3	VCC_B
Pin 4	LED1_B
Pin 5	LED1_R
Pin 6	LED1_G
Pin 7	LED2_B
Pin 8	LED2_R
Pin 9	LED2_G
Pin 10	LED3_B
Pin 11	LED3_R
Pin 12	LED3_G

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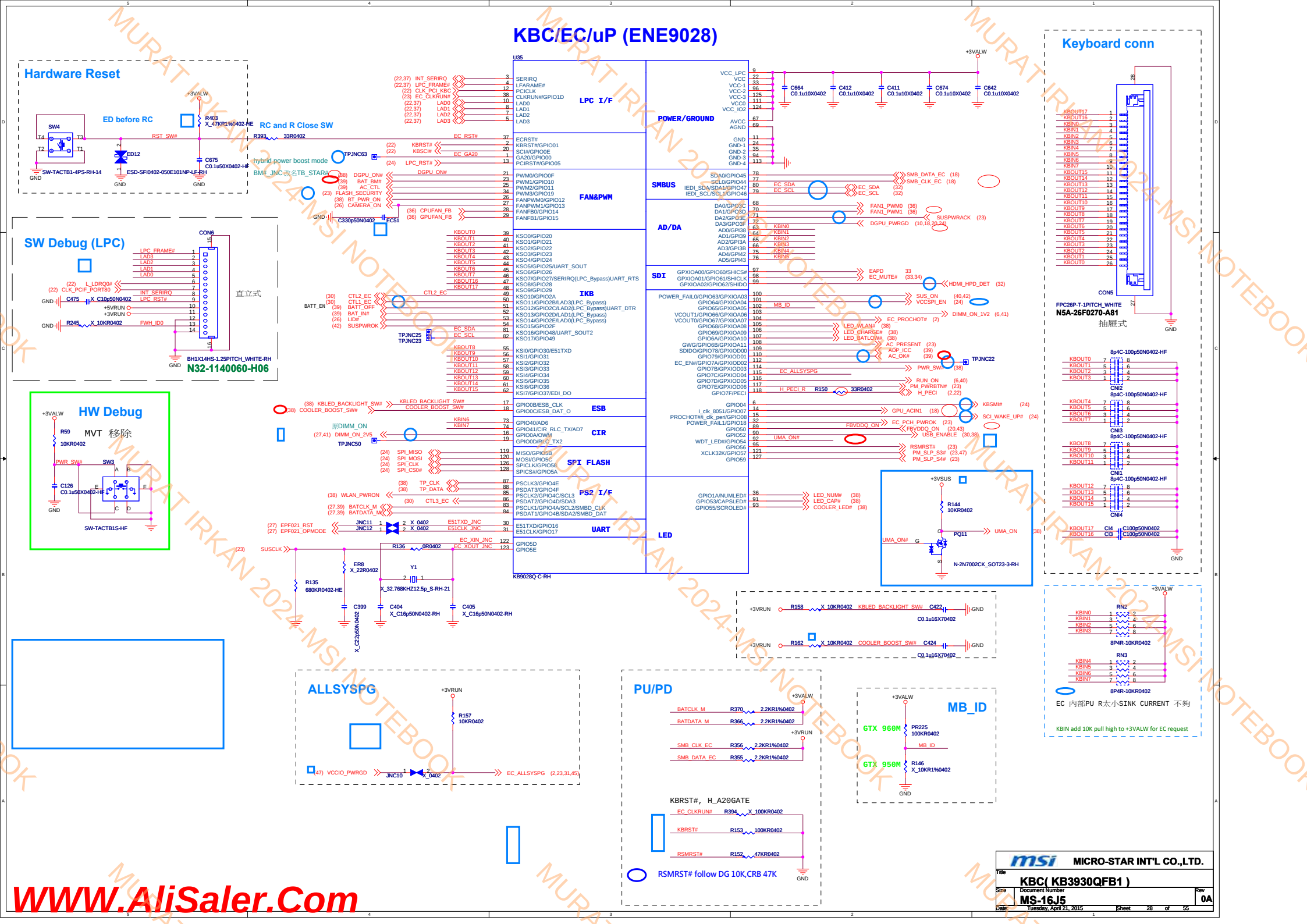
Title: LED_8051

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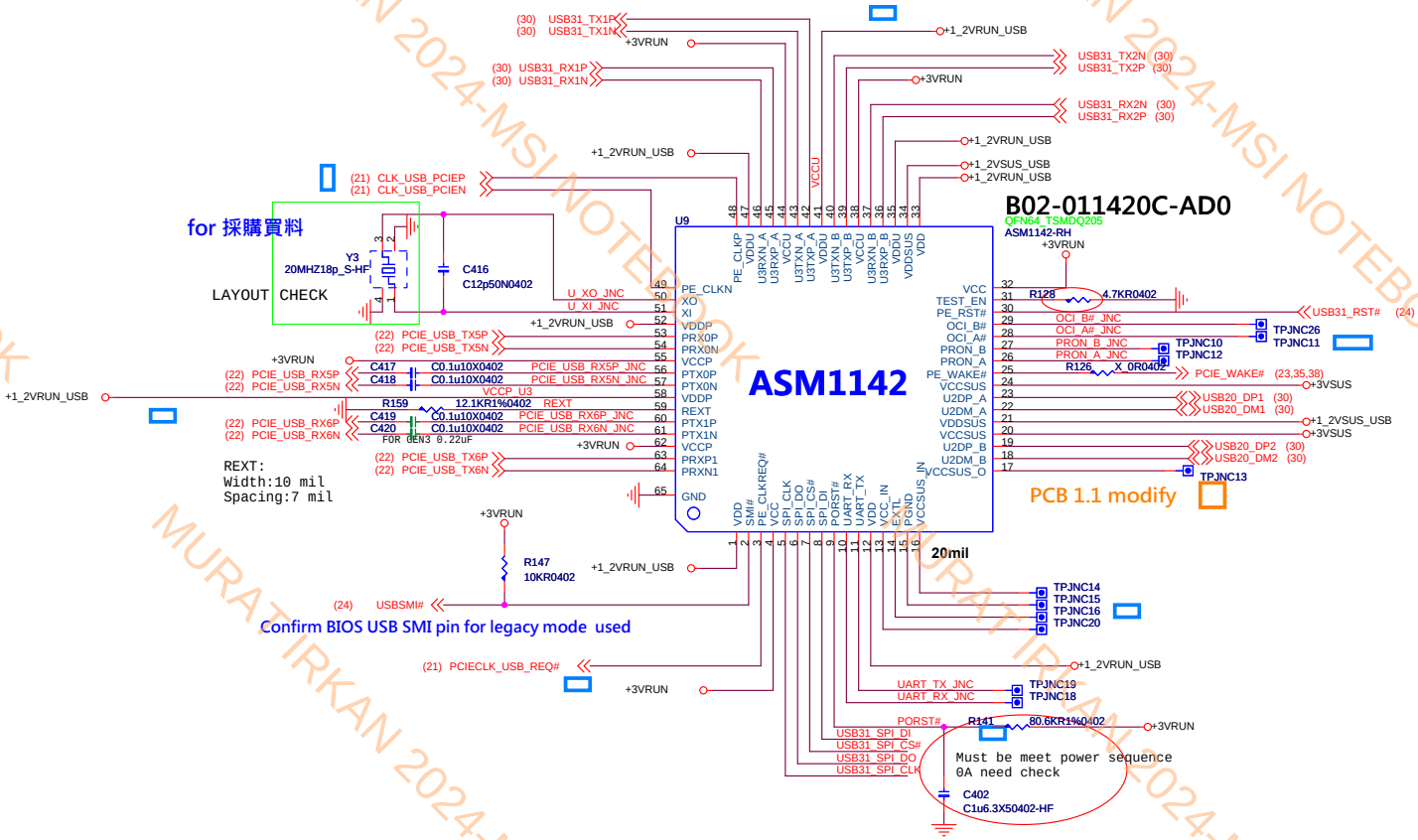
KBC/EC/uP (ENE9028)

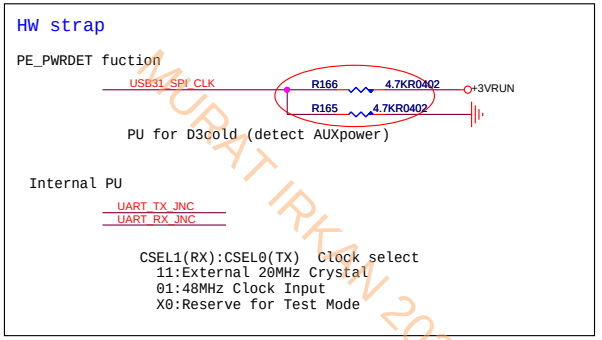


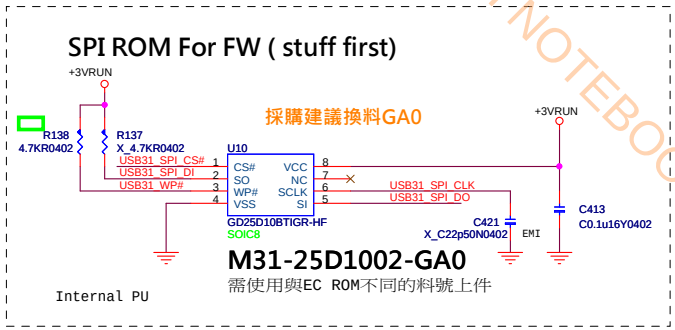
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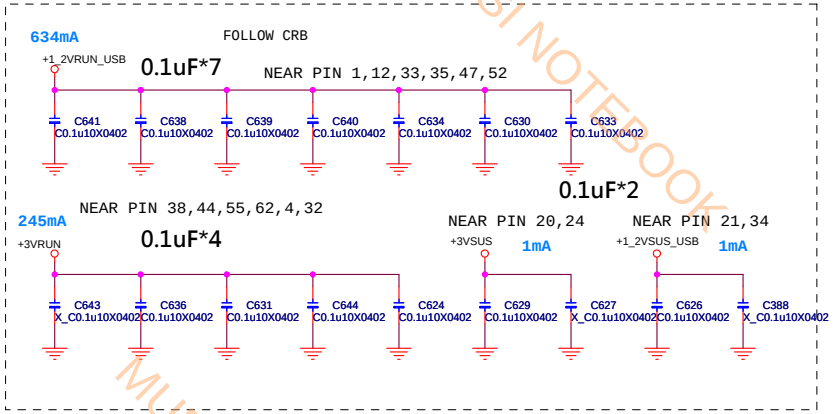
PCIE to USB 3.1

USB 3.0/ USB 3.1









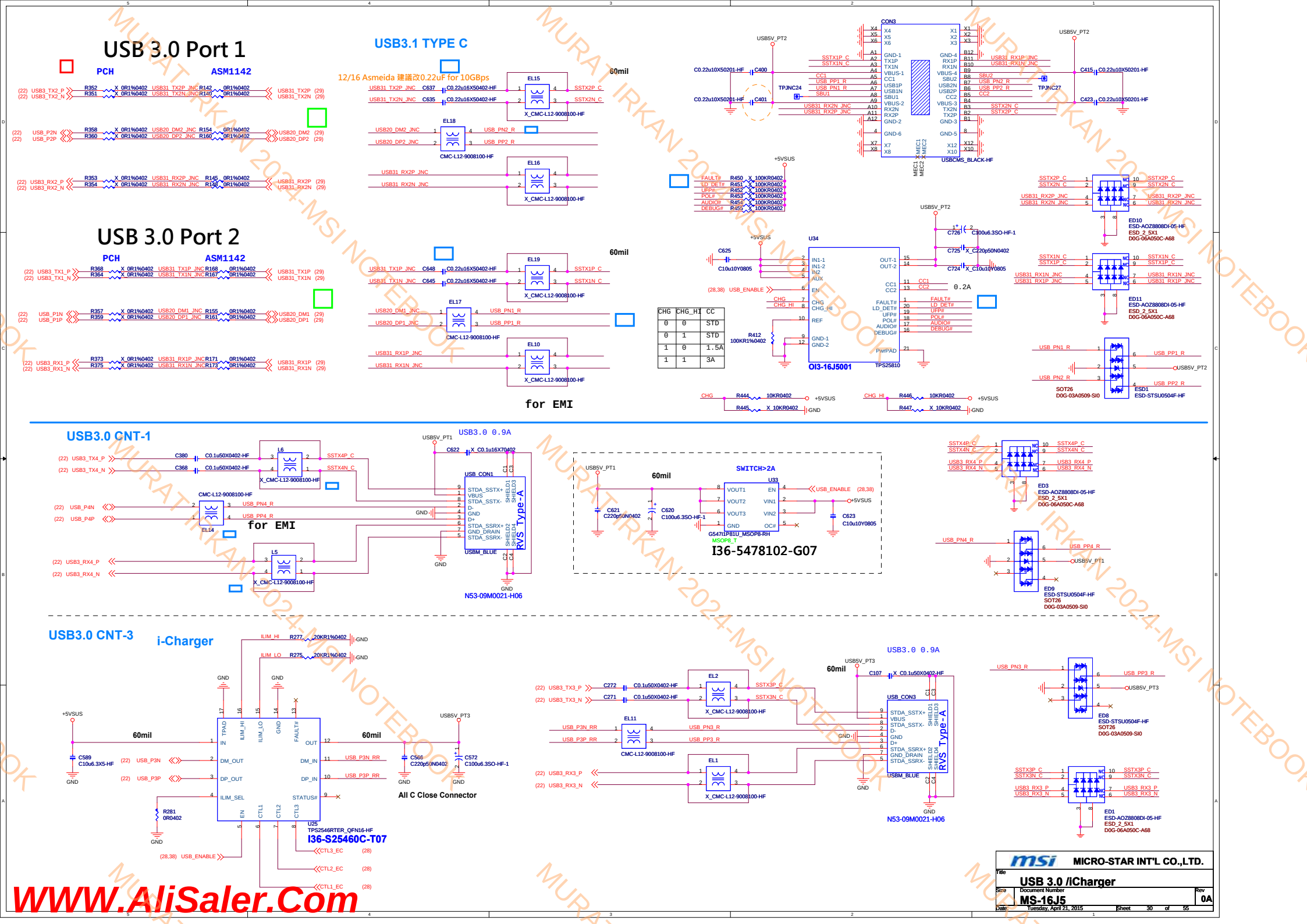
MICRO-STAR INT'L CO.,LTD.

Title: USB 3.0 / USB 3.1

Size | Document Number: MS-16J5 | Rev: 0A

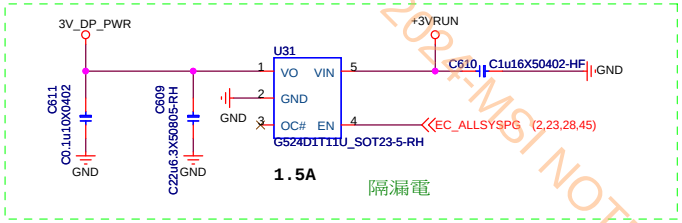
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USB 3.0 Port 1
PCH
ASM1142
USB3.1 TYPE C
12/16 Asmeida 建議改0.22uF for 10GBps
60mil
USB 3.0 Port 2
for EMI
CHG CHG_HI CC
0 0 STD
0 1 STD
1 0 1.5A
1 1 3A
USB3.0 CNT-1
USB3.0 0.9A
SWITCH>2A
OI3-16J5001
TPS25810
I36-5478102-G07
N53-09M0021-H06
USB3.0 CNT-3
i-Charger
All C Close Connector
I36-S25460C-T07
TPS2546RTER_QFN16-HF
USB3.0 0.9A
N53-09M0021-H06
MICRO-STAR INT'L CO.,LTD.
USB 3.0 /iCharger
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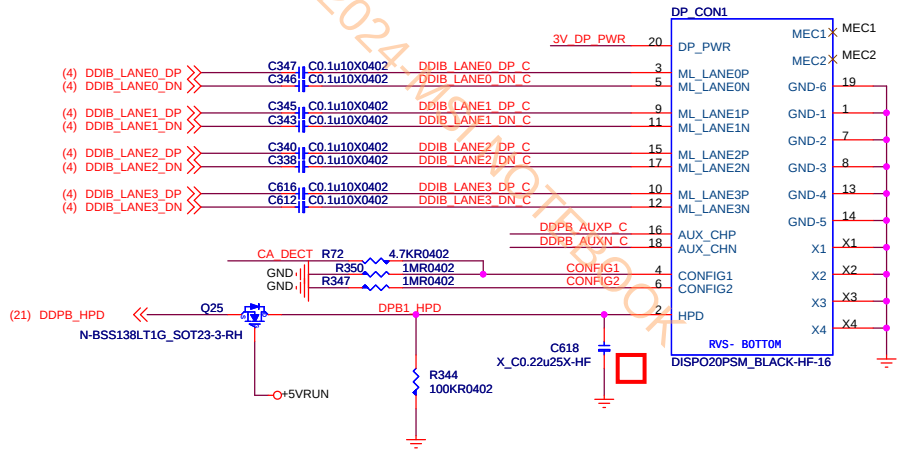
Display Port

The preset trip limit must not exceed 3A at the Upstream device connector DP_PWR pin and 1.5A at the Downstream device connector DP_PWR pin.

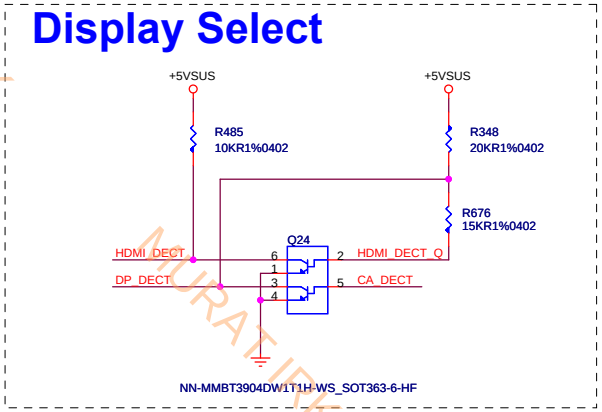


ESD Contact ± 5 KV & Air ± 15 KV

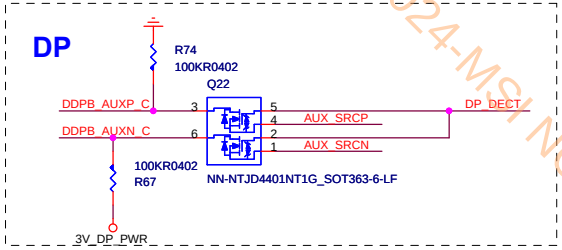
Display Port



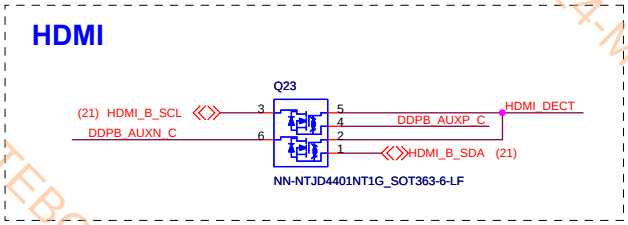
Display Select

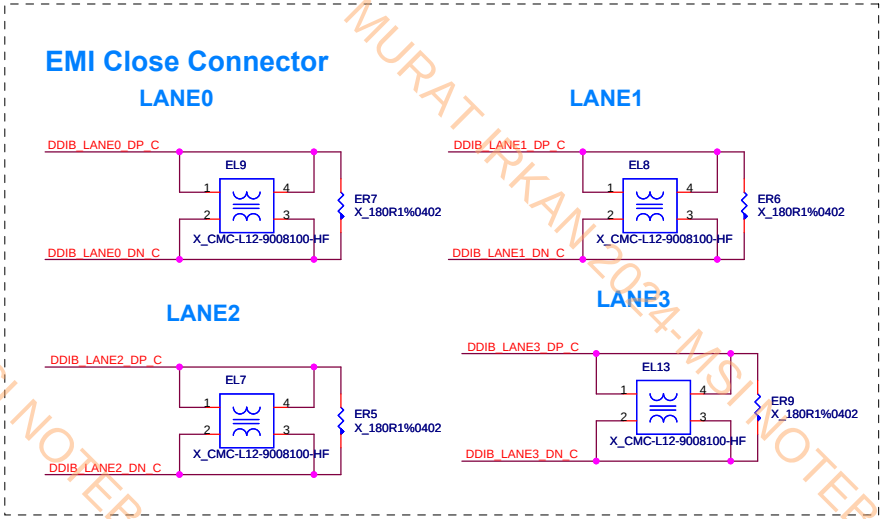


Dual Mode Switch









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Title: DP		
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HDMI Level Shifter

HPD_SNK Internal PD 130kohm ESD 10KV

HP_DET	
PLUGGED	5V
UNPLUGGED	0V

V1.0 reserved for AOC TV issue.

R100 2.2KR1%0402

U7 SN75DP139RSBR-RH

HPD Level Shift 5V to 3V

Q7 SN-DMN65D8LDW-7_SOT363-6-HF

FOR DEBUG PORT

HDMI Connector

HDMI1 HDMI19PM_BLACK-HF-7

D5 S-BAT54ALT1G_SOT23-RH

D4 X_BAV99LT1_SOT23

PF1 F-0805L075WR-HF

W>20mils

Place Under DUT

EMI Close Connector

TX0 / TX1 / TX2 / CLK

X_CMC-L12-9008100-HF

OE_N

Output Enable and power saving function for High Speed Differential level shifter path.
H: Power Saving OUT_Dx outputs = high impedance
L: Normal Mode OUT_Dx outputs = active

DDC I2C buffer offset select
H:Offset 1
L:Offset 2
Hi-Z:Offset 3

TMDS outputs rise and fall time select
H:Edge Rate:Slowest SRC =High adds ~60ps
L:Edge Rate: Slow SRC =Low adds ~30ps
Hi-Z:Edge Rate VCC/2

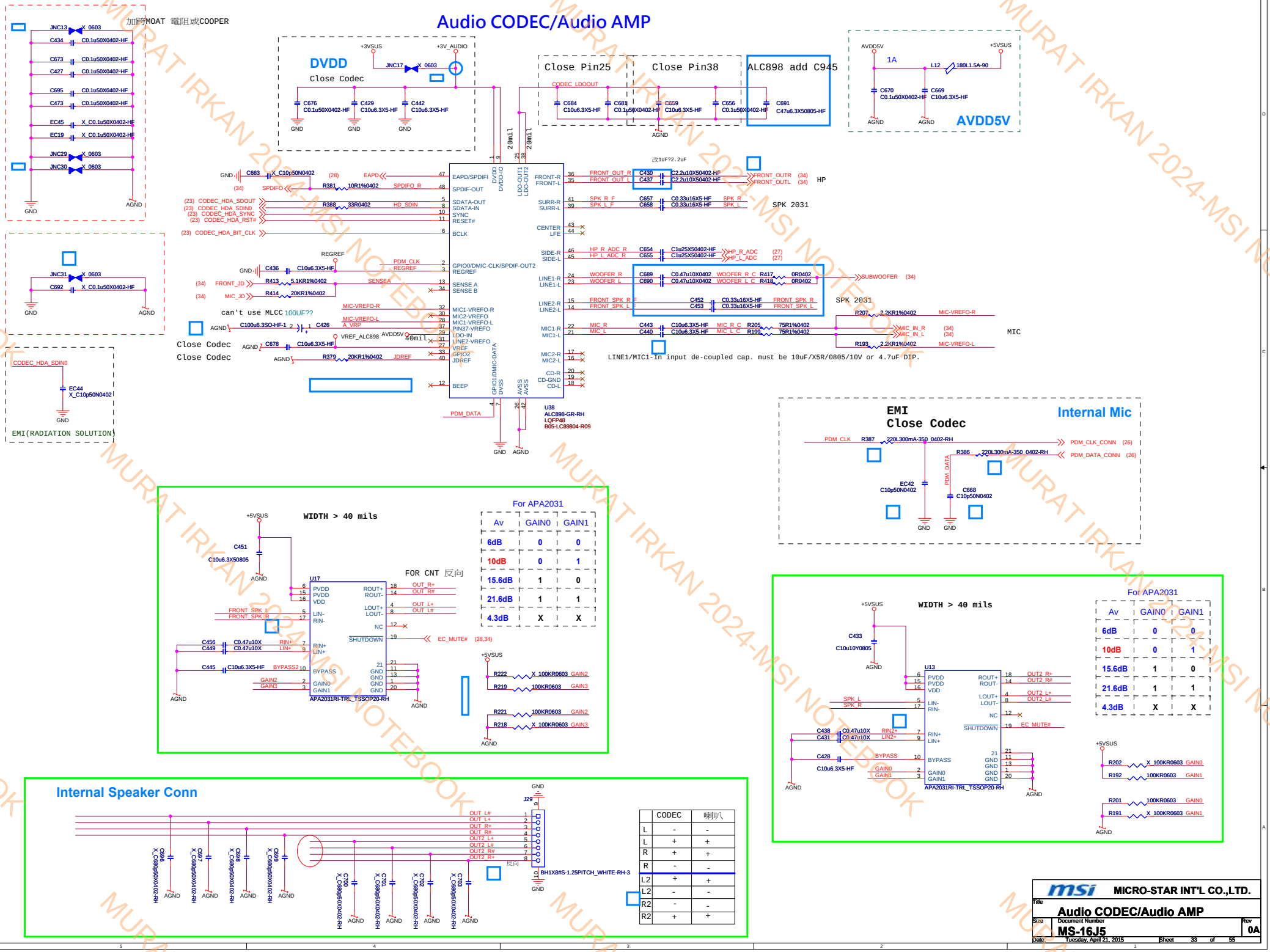
HPD_SOURCE Logic and Level Select
H:HPD Inversion HPD_SOURCE VOH =0.9V
L:HPD non-inversion HPD_SOURCE VOH =3.2V

Enables or Disables the DDC I2C buffer
H:DDC Buffer enabled
L:DDC Buffer disabled

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Audio CODEC/Audio AMP



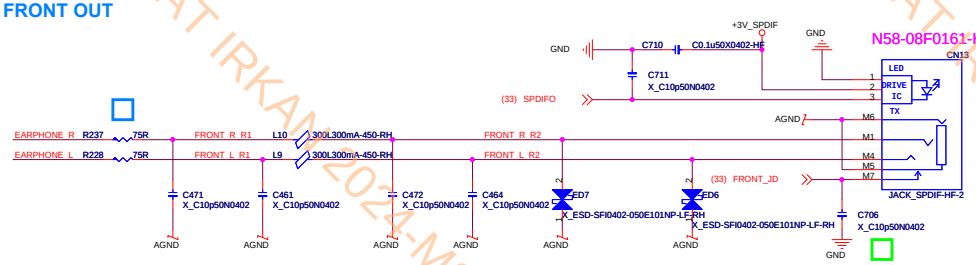
Title	Audio CODEC/Audio AMP
Size	Document Number: MS-16J5
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Date	Tuesday, April 21, 2015
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MICRO-STAR INT'L CO.,LTD.

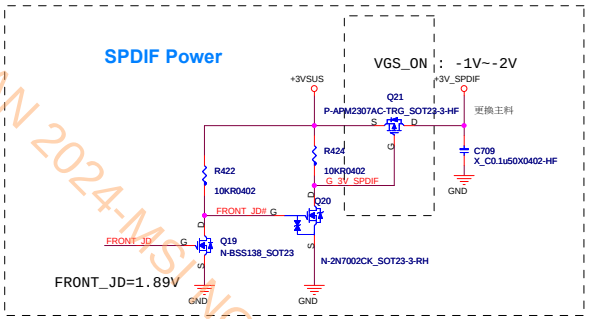
MURAT IRKAN 2024-MSI NOTEBOOK

Audio CONN /Woffer

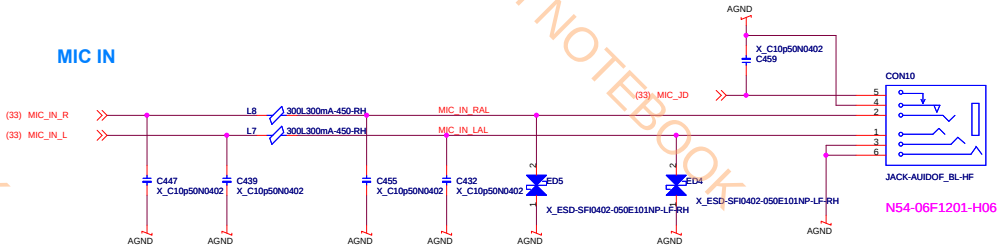
FRONT OUT



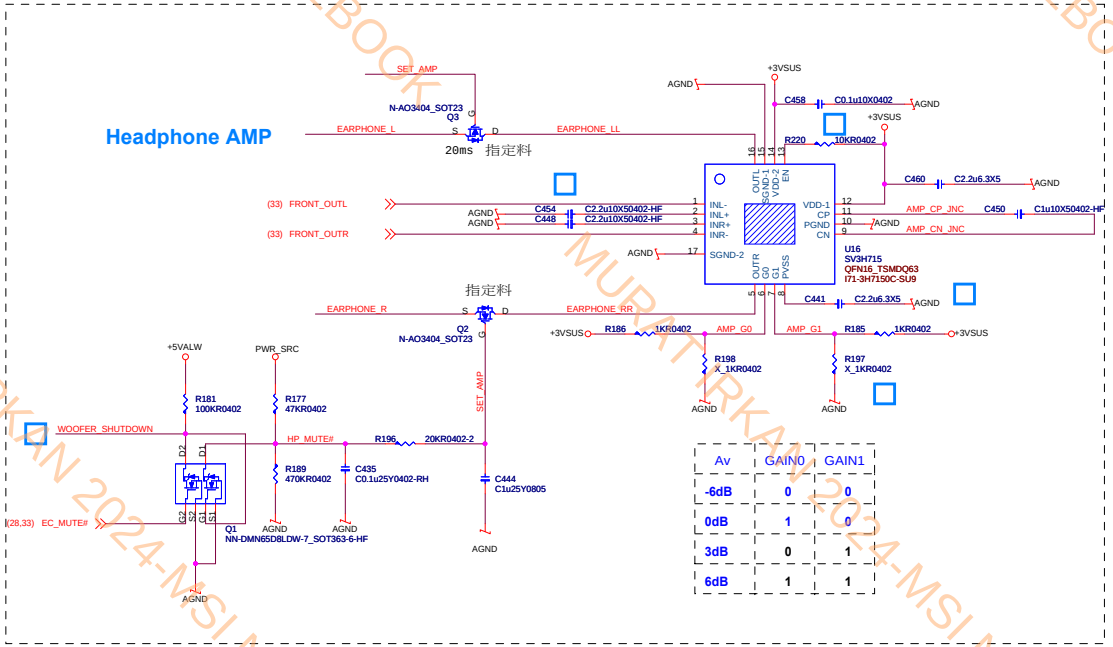
SPDIF Power



MIC IN



Headphone AMP

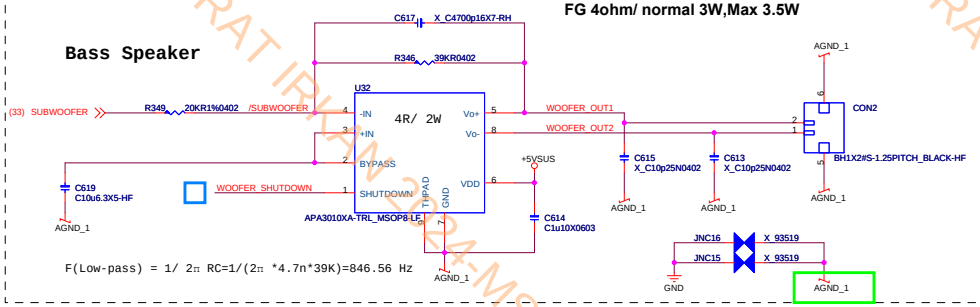


Av	GAIN0	GAIN1
-6dB	0	0
0dB	1	0
3dB	0	1
6dB	1	1

ALC892 SPC MAX 為 1.2Vrms
gain= -2*(R370/R371)= -2*(40K/20K) = -4
Vout= 0.58Vrms *4 = 2.32Vrms , Po=(2.32*2.32)/3.8=1.42W

Woofer SPEC
YG 3.8ohm / normal 3W,Max 3.5W
FG 4ohm/ normal 3W,Max 3.5W

Bass Speaker

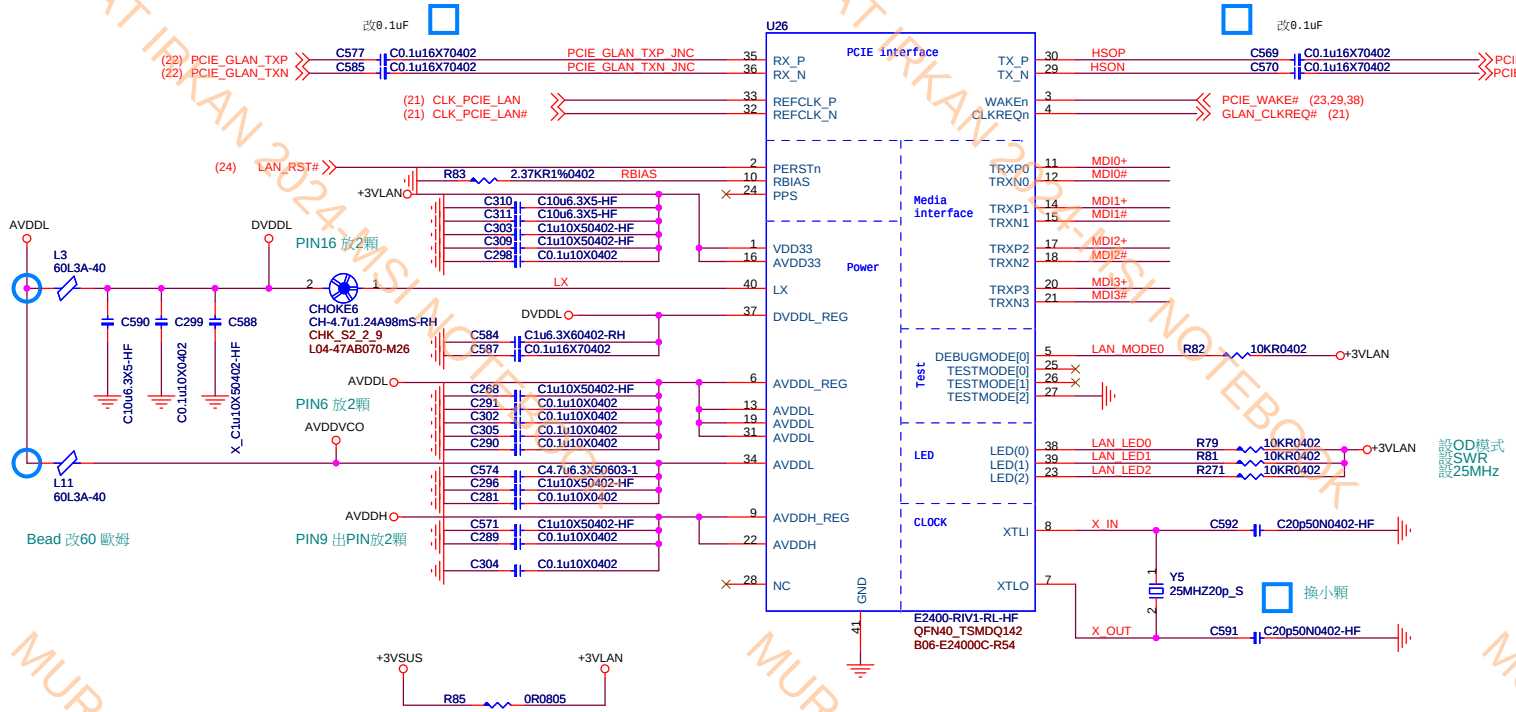


WOOFER AMP 擺太遠獨立切AGND_1

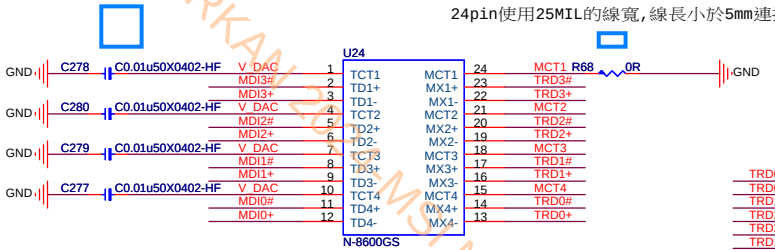
msi	MICRO-STAR INT'L CO.,LTD.	
Title	Audio / Woffer	
Size	Document Number MS-16J5	Rev 0A
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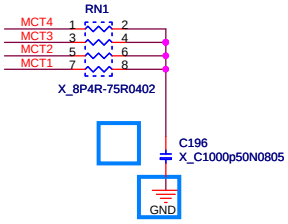
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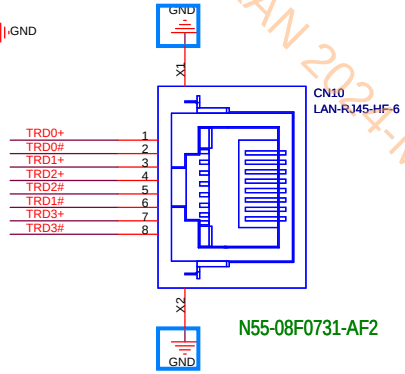
GIGA LAN(BigFoot BFN2400B)

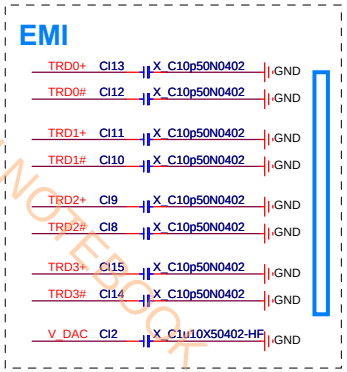


24pin使用25MIL的線寬,線長小於5mm連接GND









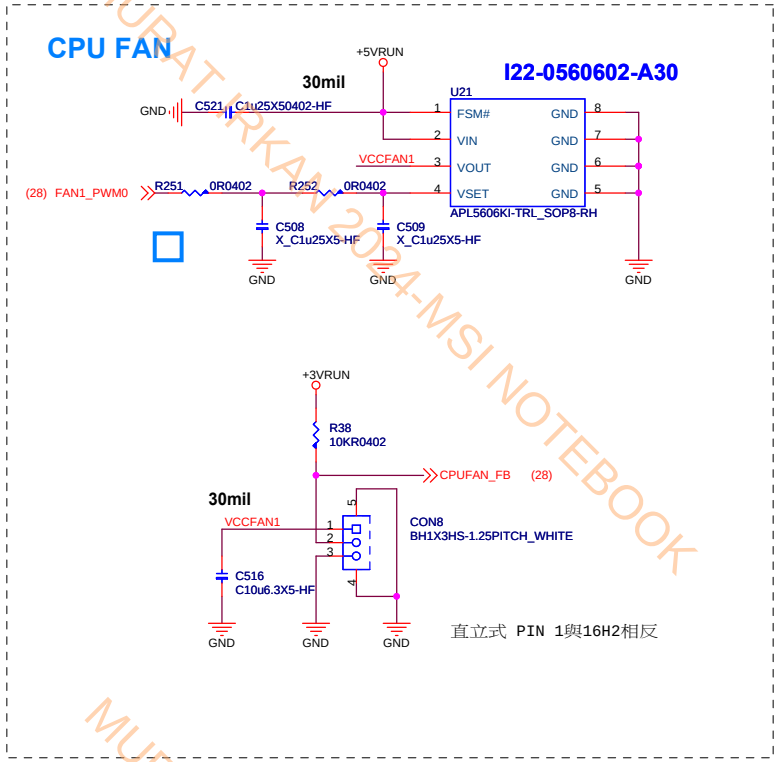
MICRO-STAR INT'L CO.,LTD.

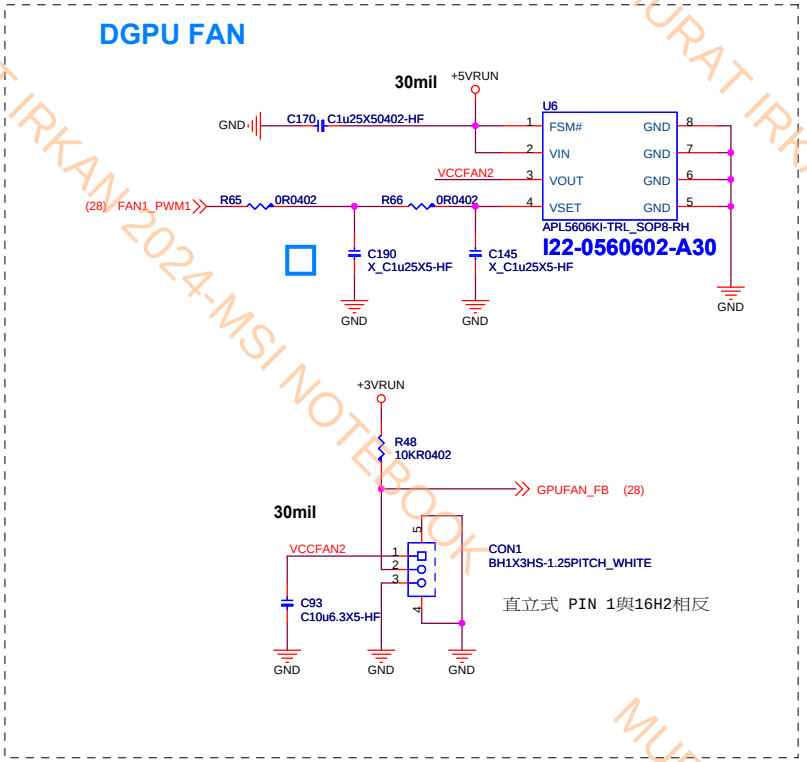
Title: GIGA LAN(BigFoot BFN2400/AR8161)

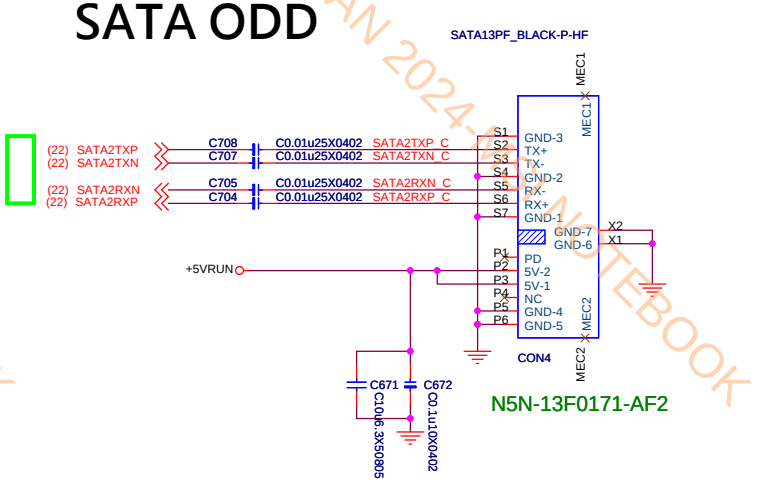
Size | Document Number: MS-16J5 | Rev: 0A

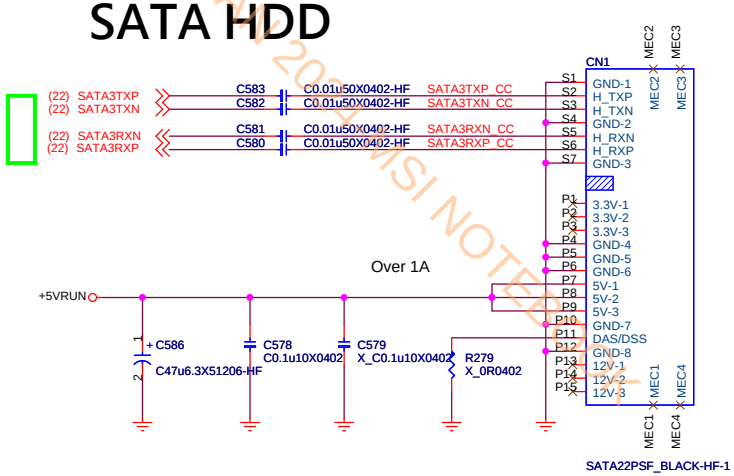
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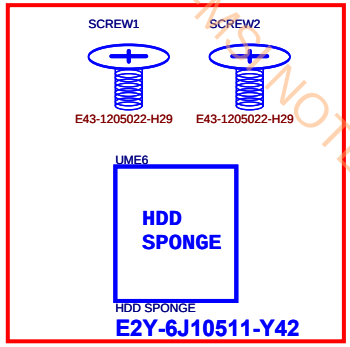






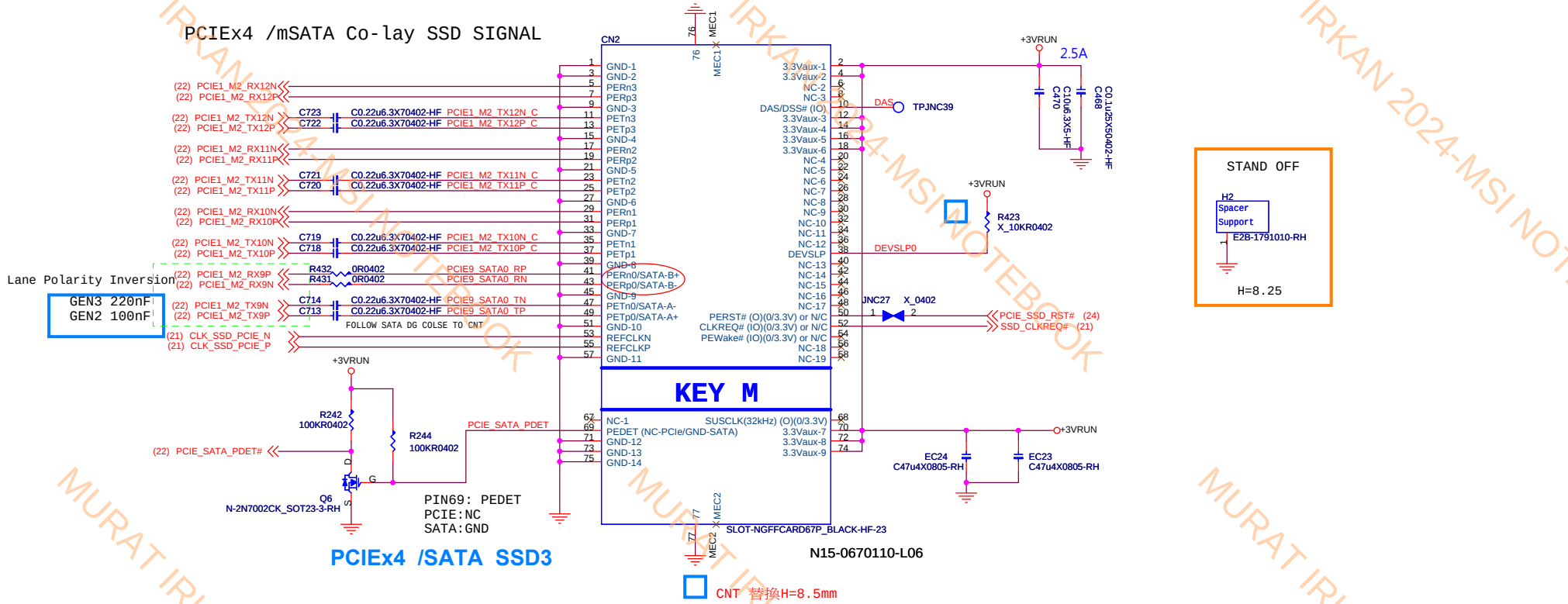
22.GND
21.TX
20.TX#
19.GND
18.RX#
17.RX
16.GND

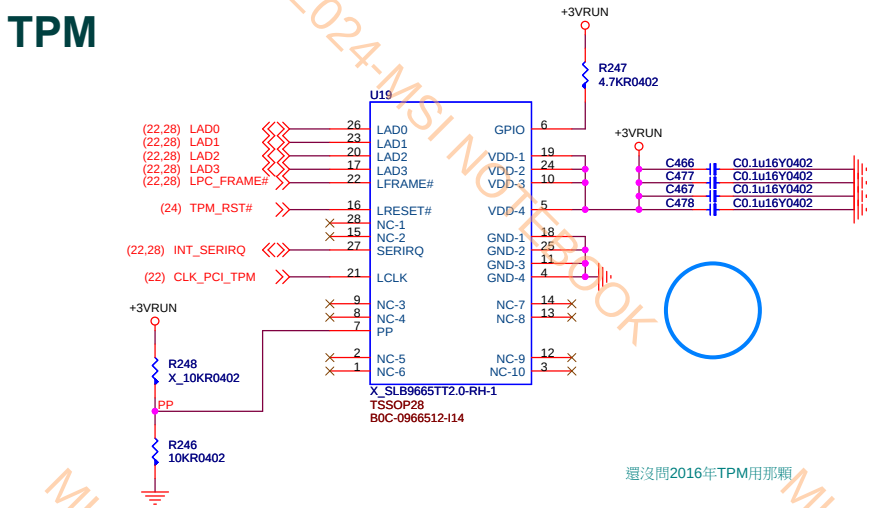
15.V33
14.V33
13.V33
12.GND
11.GND
10.GND
9.V5
8.V5
7.V5
6.GND
5.Reserved
4.GND
3.V12
2.V12
1.V12



Title	ODD/HDD/FAN	
Size A3	Document Number MS-16J5	Rev 0A
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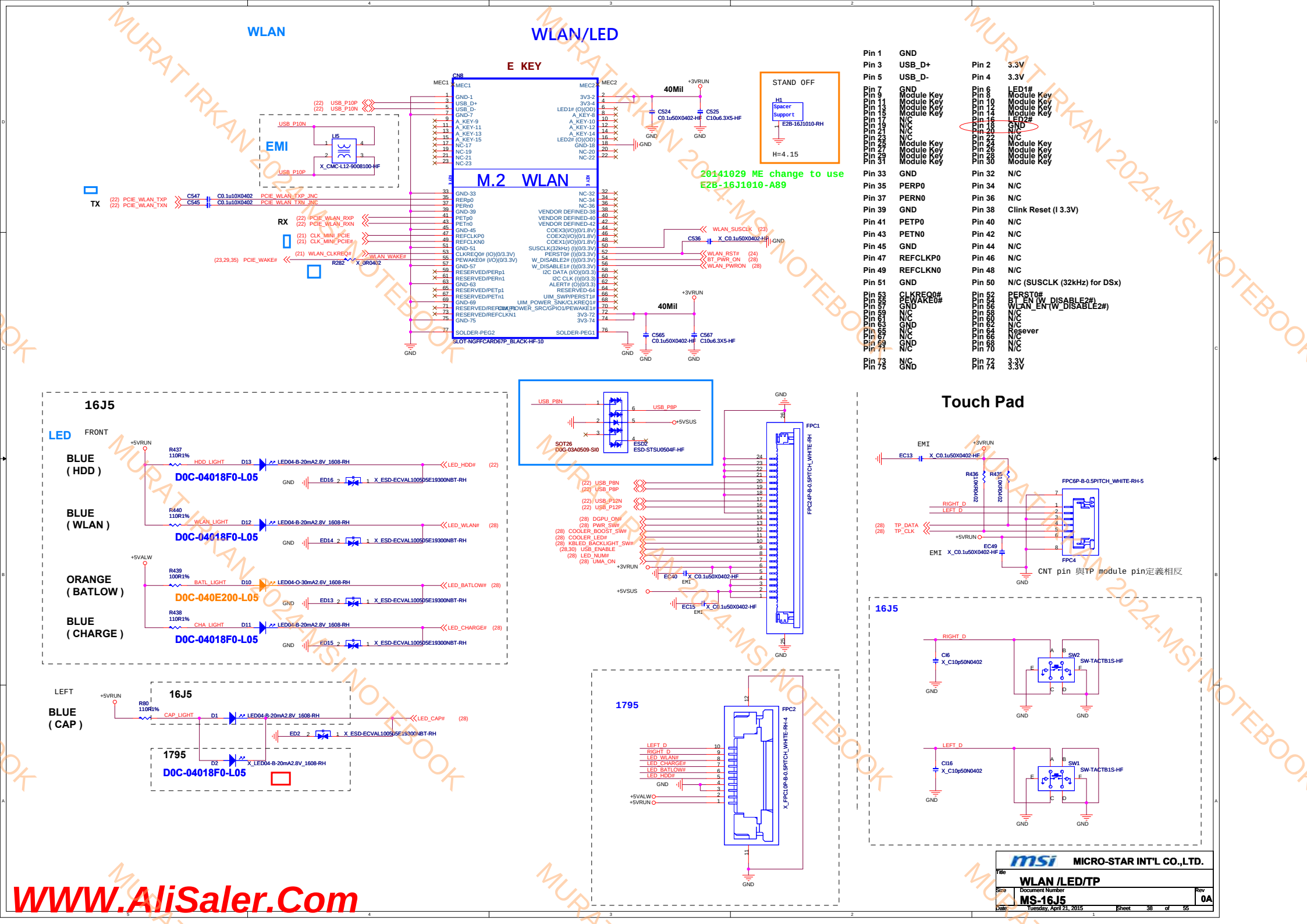
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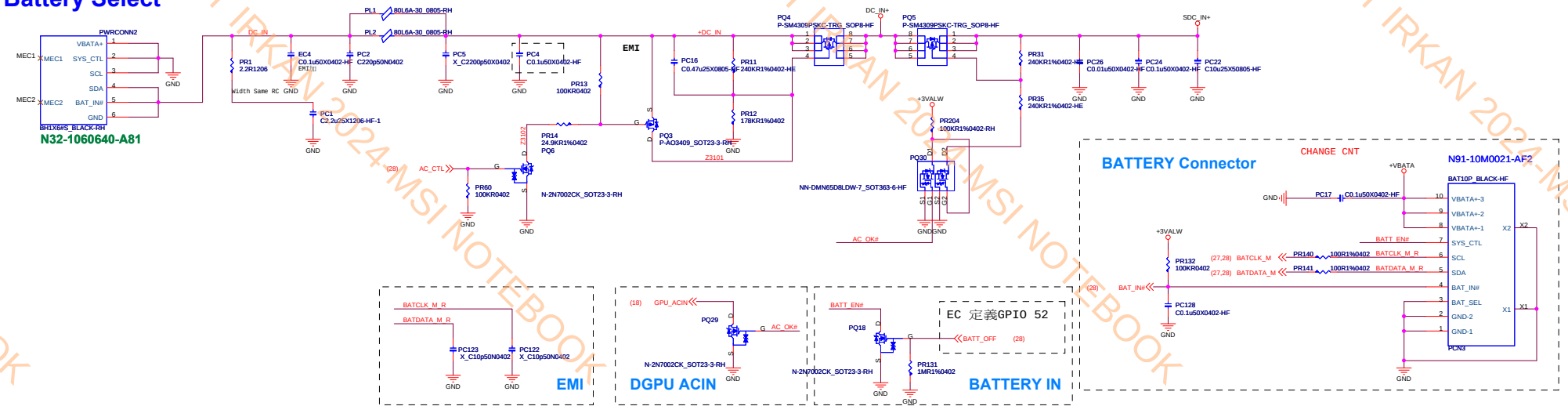
MICRO-STAR INT'L CO.,LTD.			
Title: SSD			
Size	Document Number: MS-16J5		Rev: 0A
Date: Tuesday, April 21, 2015	Sheet 37 of 55		

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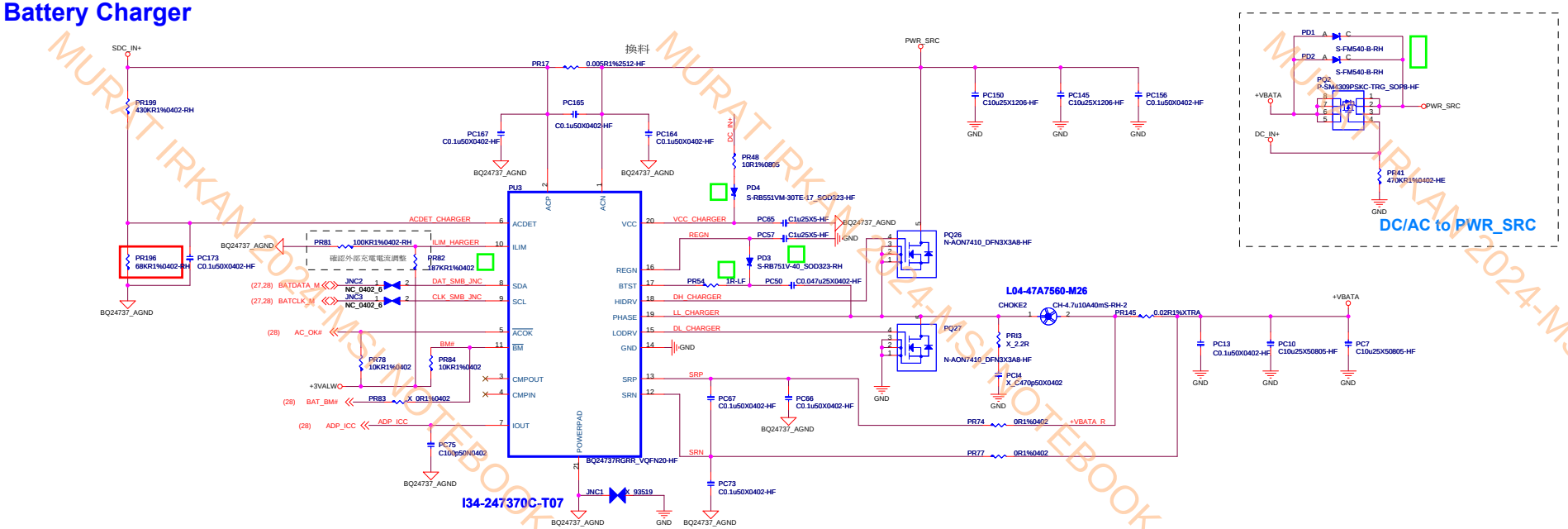

WLAN
WLAN/LED
E KEY
CN8
M.2 WLAN
SLOT-NGFFCARD67P_BLACK-HF-10
EMI
X_CMC-L12-9008100-HF
TX
RX
STAND OFF
H1
Spacer Support
E2B-16J1010-RH
H=4.15
40Mil
20141029 ME change to use E2B-16J1010-A89
Pin 1 GND
Pin 3 USB_D+
Pin 2 3.3V
Pin 5 USB_D-
Pin 4 3.3V
Pin 7 GND
Pin 6 LED1#
Pin 9 Module Key
Pin 8 Module Key
Pin 11 Module Key
Pin 10 Module Key
Pin 13 Module Key
Pin 12 Module Key
Pin 15 Module Key
Pin 14 Module Key
Pin 17 N/C
Pin 16 LED2#
Pin 19 N/C
Pin 18 GND
Pin 21 N/C
Pin 20 N/C
Pin 23 N/C
Pin 22 N/C
Pin 25 Module Key
Pin 24 Module Key
Pin 27 Module Key
Pin 26 Module Key
Pin 29 Module Key
Pin 28 Module Key
Pin 31 Module Key
Pin 30 Module Key
Pin 33 GND
Pin 32 N/C
Pin 35 PERP0
Pin 34 N/C
Pin 37 PERN0
Pin 36 N/C
Pin 39 GND
Pin 38 Clink Reset (I 3.3V)
Pin 41 PETP0
Pin 40 N/C
Pin 43 PETN0
Pin 42 N/C
Pin 45 GND
Pin 44 N/C
Pin 47 REFCLKP0
Pin 46 N/C
Pin 49 REFCLKN0
Pin 48 N/C
Pin 51 GND
Pin 50 N/C (SUSCLK (32kHz) for DSx)
Pin 53 CLKREQ0#
Pin 52 PERST0#
Pin 55 PEWAKE0#
Pin 54 BT_EN (W_DISABLE2#)
Pin 57 GND
Pin 56 WLAN_EN(W_DISABLE2#)
Pin 59 N/C
Pin 58 N/C
Pin 61 N/C
Pin 60 N/C
Pin 63 GND
Pin 62 N/C
Pin 65 N/C
Pin 64 Resever
Pin 67 N/C
Pin 66 N/C
Pin 69 GND
Pin 68 N/C
Pin 71 N/C
Pin 70 N/C
Pin 73 N/C
Pin 72 3.3V
Pin 75 GND
Pin 74 3.3V
16J5
LED
FRONT
BLUE (HDD)
BLUE (WLAN)
ORANGE (BATLOW)
BLUE (CHARGE)
D0C-04018F0-L05
D0C-040E200-L05
LEFT
BLUE (CAP)
1795
Touch Pad
CNT pin 與TP module pin定義相反
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WLAN /LED/TP
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Battery Select/Charger

Battery Select



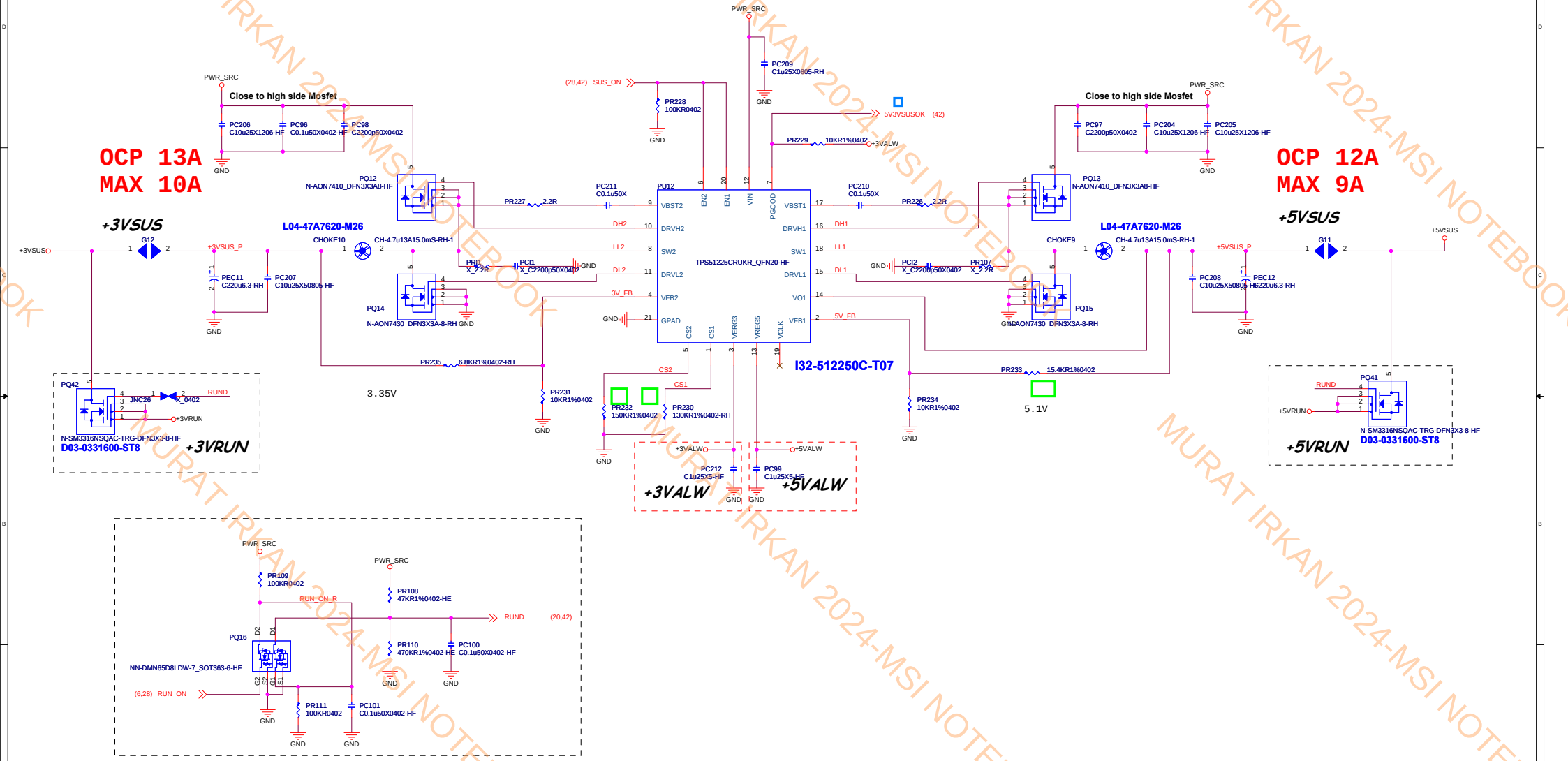
Battery Charger



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msi	MICRO-STAR INT'L CO.,LTD.	
Title	Battery Select/Charger	
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System Power

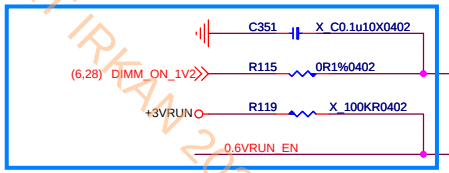


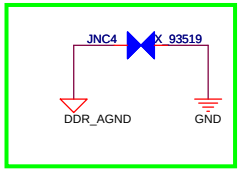
MICRO-STAR INT'L CO.,LTD.		
Title: System Power		
Size	Document Number: MS-16J5	Rev: 0A
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+1_2VDIMM/+0.6VRUN





MAX 2A

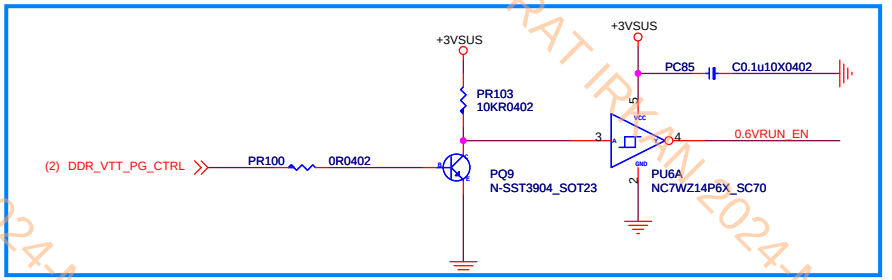
OCP 12A
MAX 9A

PU5
APW8819QAI-TRG_QFN20-HF
WQFN20_TSMDQ63
I32-8819Q0C-A30

Q10
N-AON7410_DFN3X3A8-HF

Q11
N-AON7430_DFN3X3A-8-RH

CHOKE7
CH-1.0u22A10mS-RH



+2.5V_MEM

IC MAX 3A
400mA

W>40mils

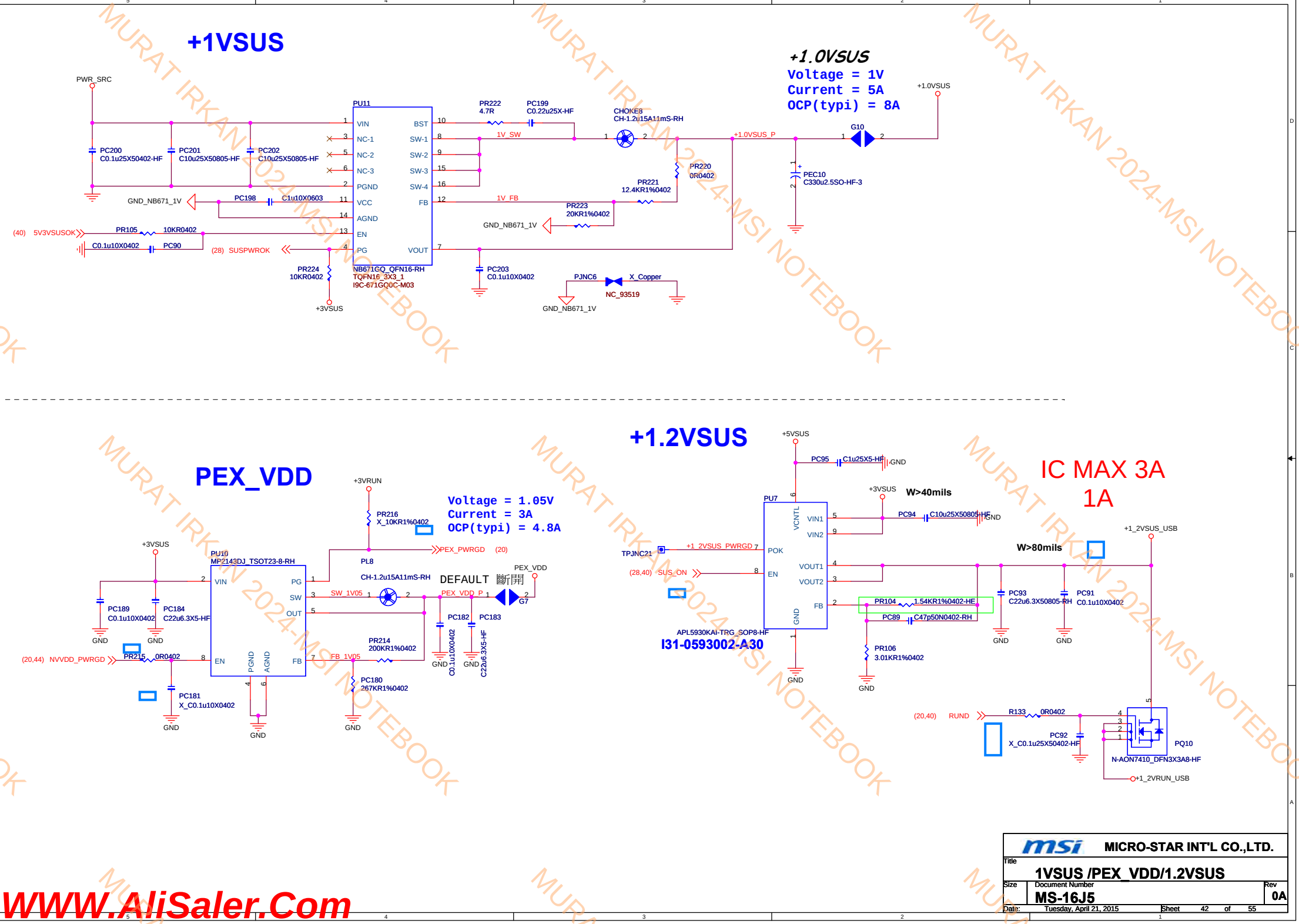
W>80mils

I31-0593002-A30
APL5930KAI-TRG_SOP8-HF



MICRO-STAR INT'L CO.,LTD.	
Title	+1.2VDIMM/+0.6VRUN/+2.5VMEM
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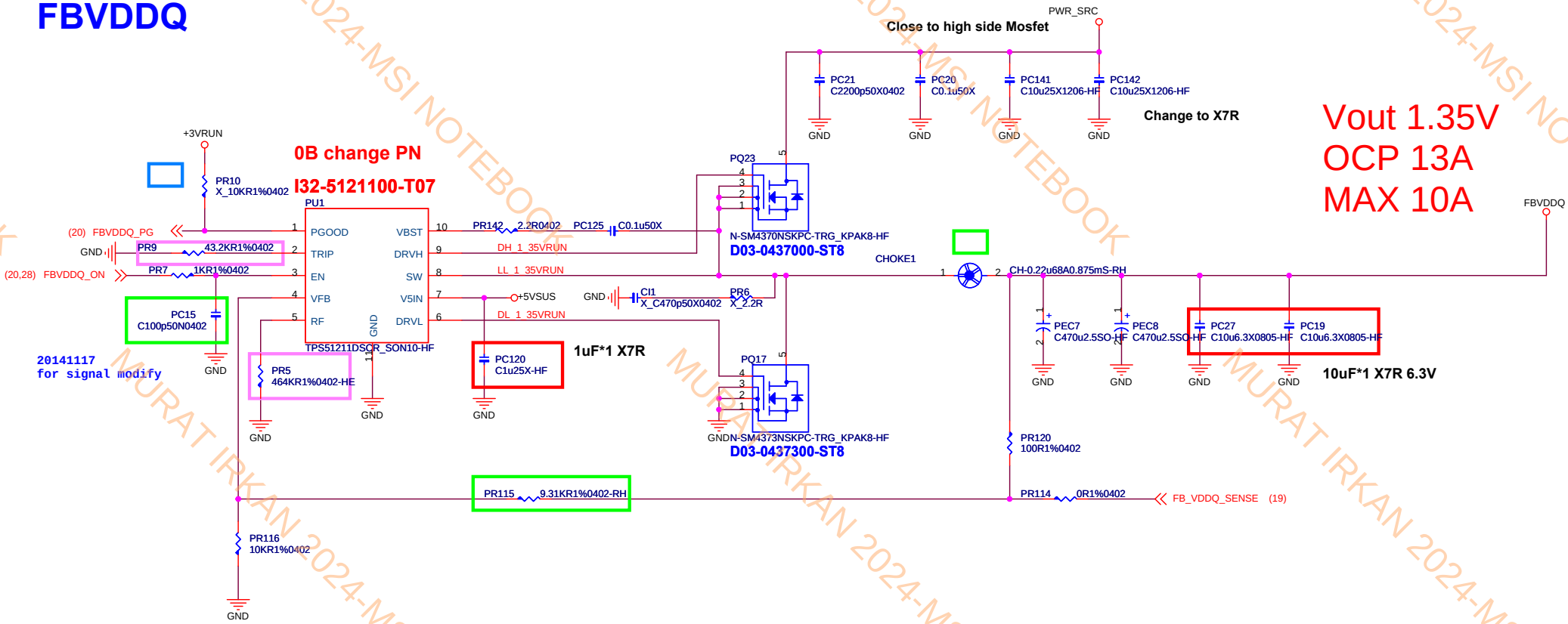
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DGPU POWER FBVDDQ

FBVDDQ



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Title: DGPU POWER FBVDDQ		
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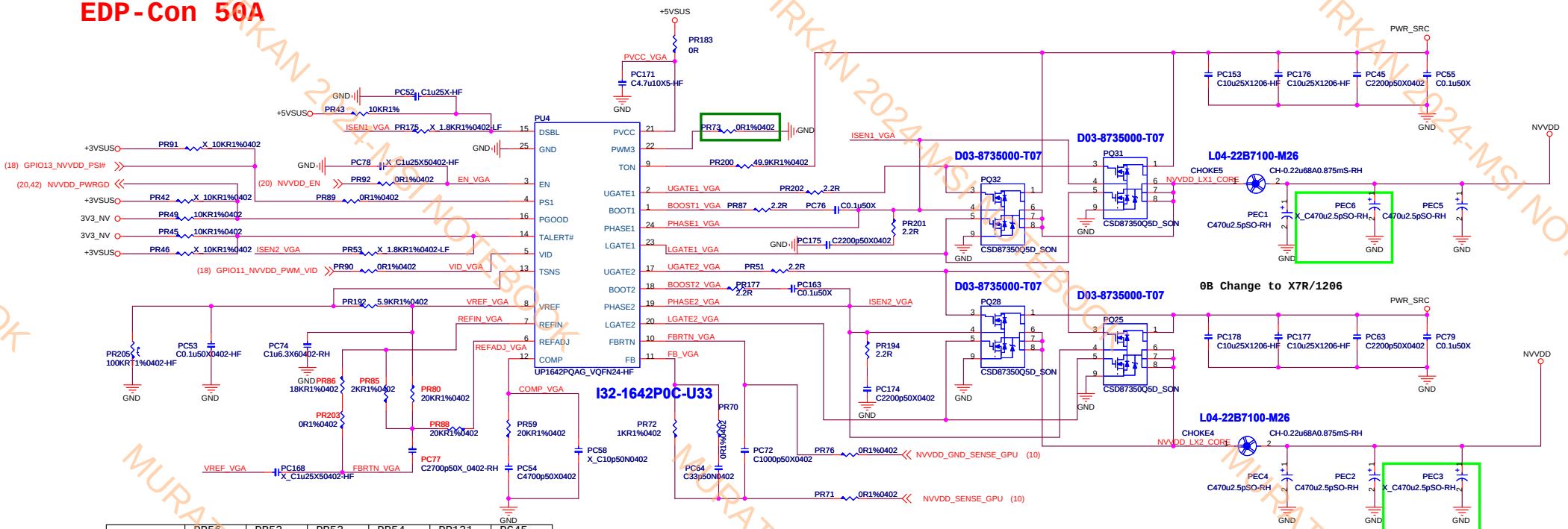
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DGPU POWER / UP1642PQAG

EDP-Peak 87A
EDP-Con 50A

DGPU POWER NVVDD

CONFIG A
VBoot:0.875V
Vmin:0.6V / Vmax:1.2V



	PR56	PR52	PR53	PR54	PR121	PC45
CONFIG	R1	R2	R3	R4	R5	C
N16E-GT N16P-GX-B	20K	20K	2K	18K	0	2.7nF

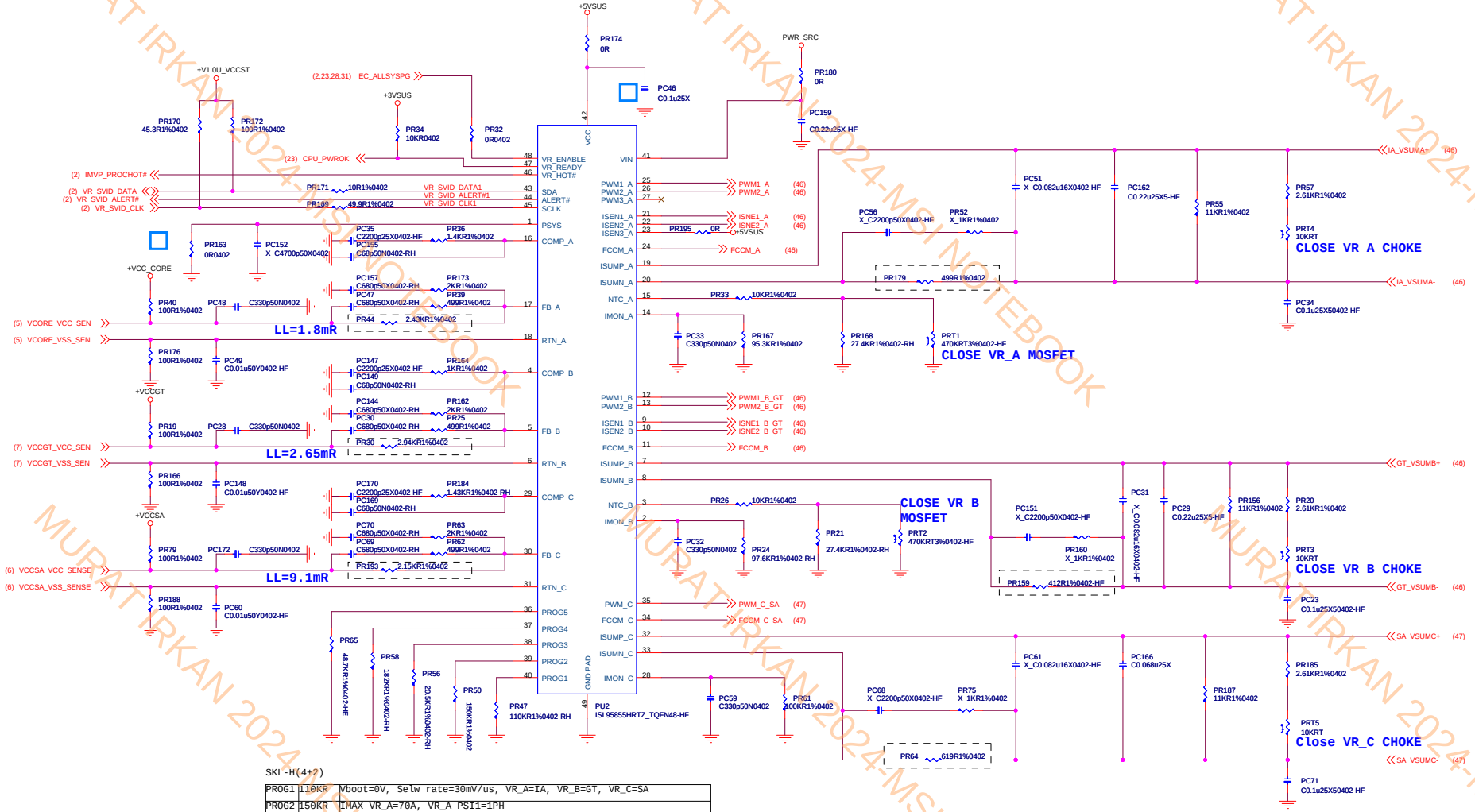
20141029 power modify for GPU power setting
PR52,PR56: 39K ohm to 20K ohm
PR53: 1.5K ohm to 2K ohm
PR54: 30K ohm to 18K ohm
PR121: 1.5K ohm to 0 ohm
PC45: 1.5nF to 2.7nF

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Title	DGPU POWER NVVDD	
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Skylake H-line 42 45W ISL95855



SKL-H(4+2)

PROG1	110KR	Vboot=0V, Selw rate=30mV/us, VR_A=IA, VR_B=GT, VR_C=SA
PROG2	150KR	IMAX VR_A=70A, VR_A PSI1=1PH
PROG3	20.5KR	IMAX VR_B=60A, DROOP VR_B Active
PROG4	182KR	DROOP VR_A Active, DROOP VR_C Active, VR_A VR_B Frequency=750kHz
PROG5	48.7KR	IMAX VR_C=12A, Frequency=450kHz

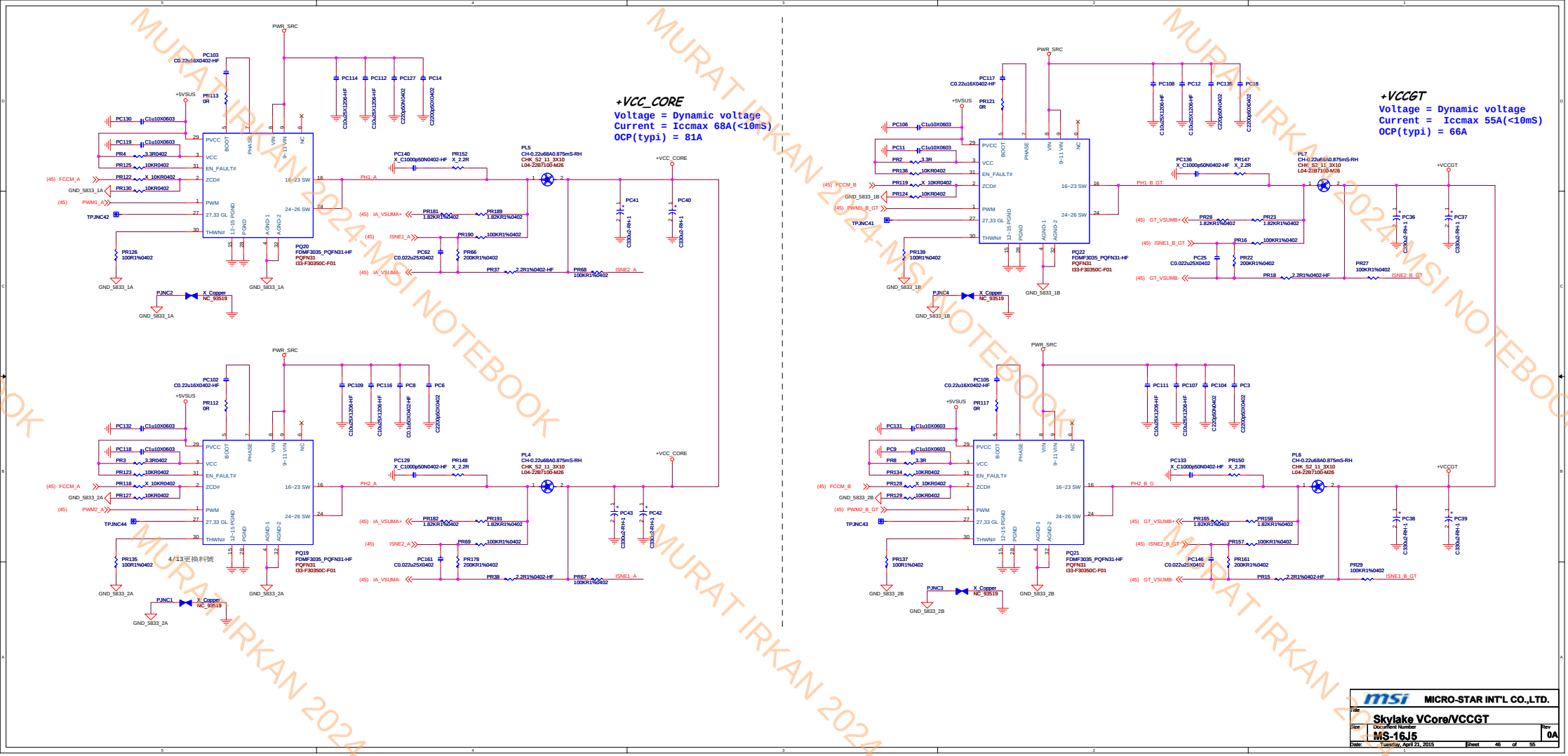
MICRO-STAR INT'L CO.,LTD.

Title: Skylake CPU Power

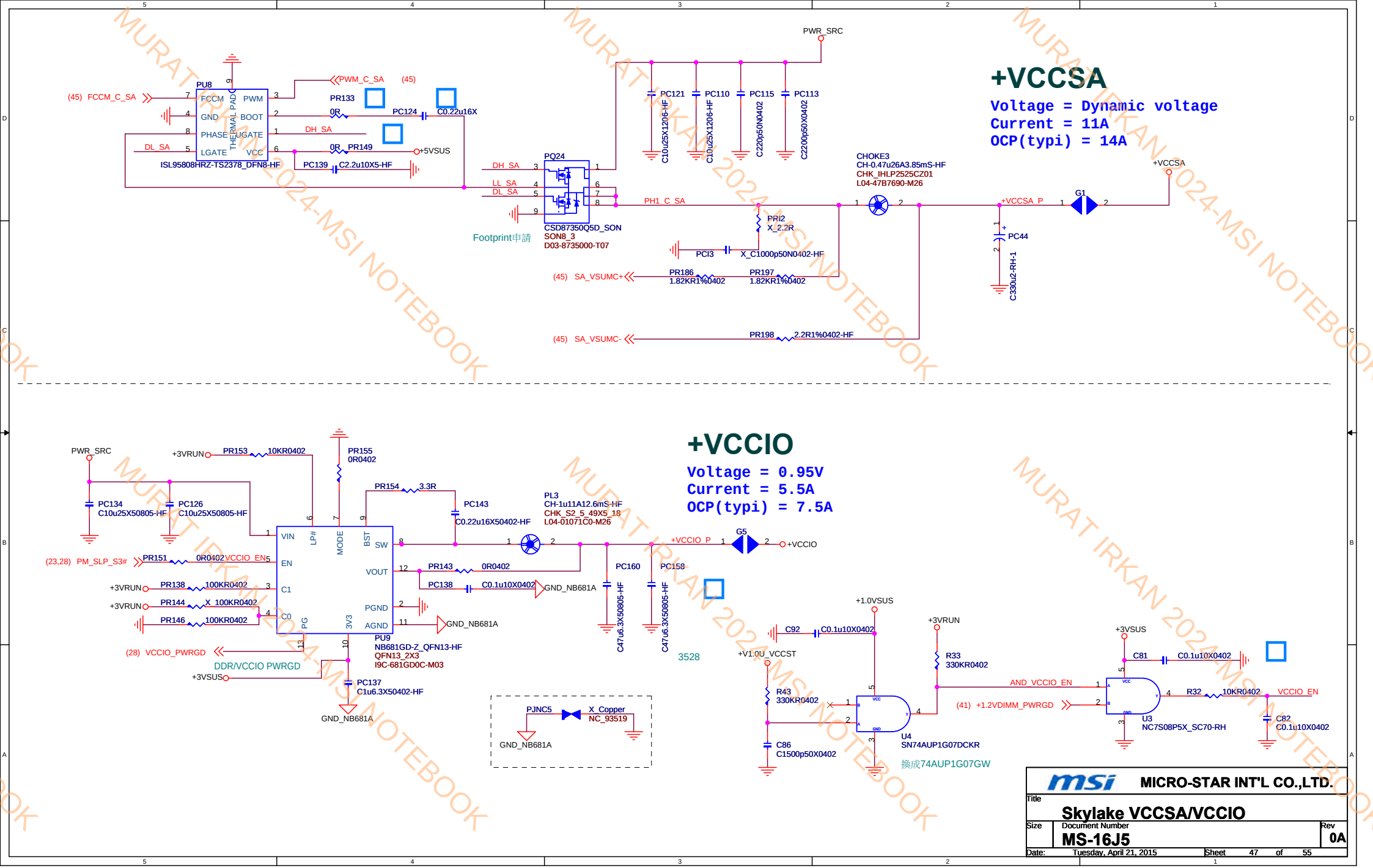
Size | Document Number: MS-16J5 | Rev: 0A

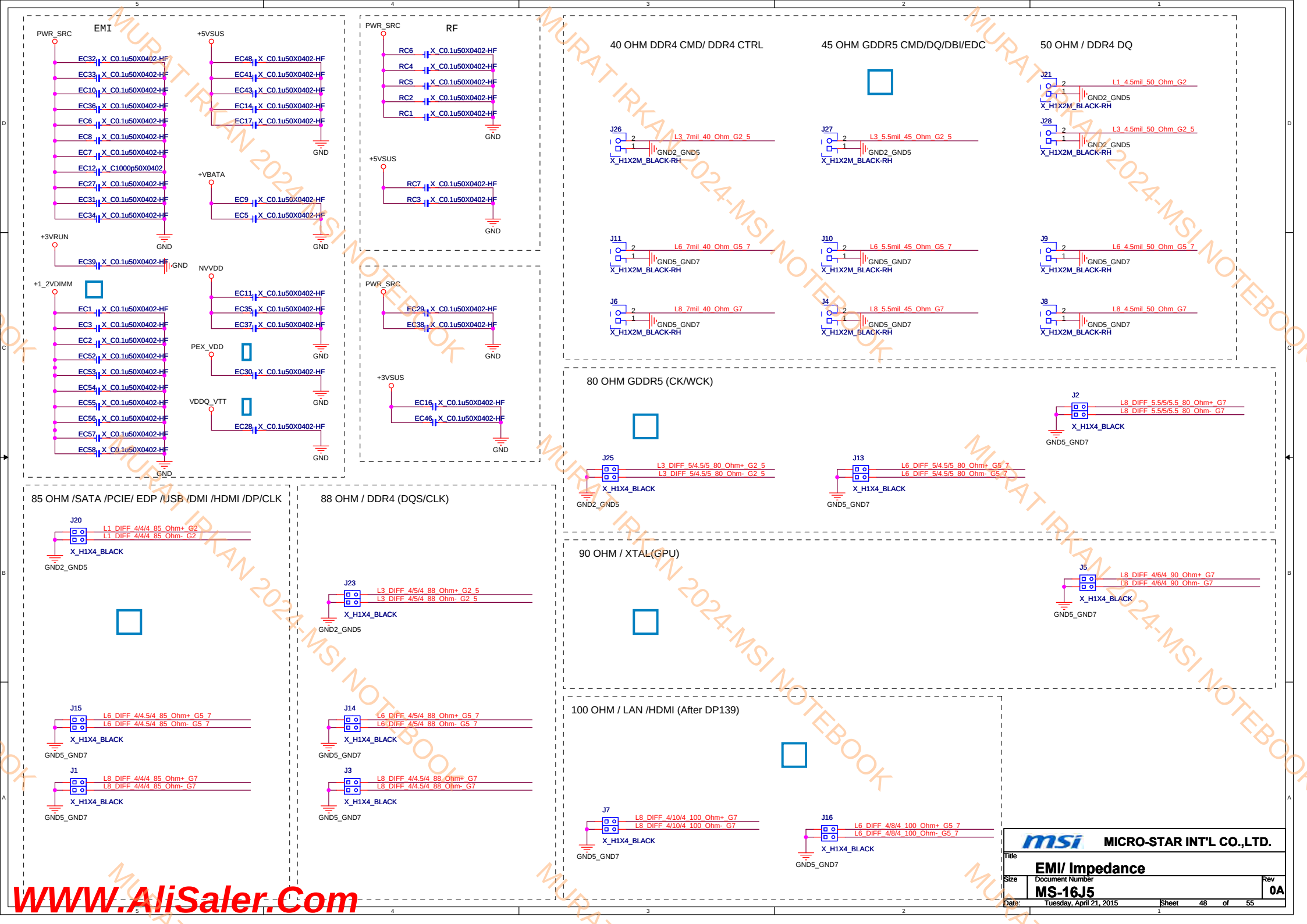
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+VCCSA
Voltage = Dynamic voltage
Current = 11A
OCP(typi) = 14A
PWR_SRC
PU8
ISL95808HRZ-TS2378_DFN8-HF
PC121
PC110
PC115
PC113
C10u25X1206-HF
C220p50N0402
C2200p50X0402
CHOKE3
CH-0.47u26A3.85mS-HF
CHK_IHLP2525CZ01
L04-47B7690-M26
PQ24
CSD87350Q5D_SON
SON8_3
D03-8735000-T07
Footprint申請
PC44
C330u2-RH-1
+VCCSA
+VCCIO
Voltage = 0.95V
Current = 5.5A
OCP(typi) = 7.5A
PL3
CH-1u11A12.6mS-HF
CHK_S2_5_49X5_18
L04-01071C0-M26
PU9
NB681GD-Z_QFN13-HF
QFN13_2X3
I9C-681GD0C-M03
DDR/VCCIO PWRGD
U4
SN74AUP1G07DCKR
換成74AUP1G07GW
U3
NC7S08P5X_SC70-RH
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Skylake VCCSA/VCCIO
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CPU/GPU Holes

MCPU4 H_R276D169_PB — GND X_CPU_HOLE
MCPU2 H_R276D169_PB — GND X_CPU_HOLE
MCPU3 H_R276D169_PB — GND X_CPU_HOLE
MCPU1 H_R276D169_PB — GND X_CPU_HOLE
MGPU2 H_R276D169_PB — GND X_CPU_HOLE
MGPU4 H_R276D169_PB — GND X_CPU_HOLE
MGPU1 H_R276D169_PB — GND X_CPU_HOLE
MGPU3 H_R276D169_PB — GND X_CPU_HOLE

OD 2.3

MH7 H_R197D118_PT X_ME_SCREW HOLE
MH4 H_R197D118_PT X_ME_SCREW HOLE
修改內徑3mm

MH9 H_R197D91 X_ME_SCREW HOLE
MH5 H_R197D91 X_ME_SCREW HOLE
MH6 H_R197D91 X_ME_SCREW HOLE

FM7 FM18 FM22 FM3 FM17 FM10 FM16 FM13
FM23 FM8 FM9 FM21 FM19 FM20 FM11 FM24
FM2 FM6 FM14 FM1 FM4 FM12 FM15 FM5
X_FM

FM25 FM26 FM27 FM31 FM32 FM33
FM28 FM29 FM30 FM34
X_FM
4/13 新增

Only 16J5

UME3 CPU BKT — CPU BKT — E2M-7810111-A89
UME4 GPU BKT — GPU BKT — 307-6H20211-C22
UME5 CPU MYLAR — CPU MYLAR — E2P-6J11211-Y42

UME7 LED SPONGE — LED SPONGE — E2Y-6J10611-Y42
UME8 LED SPONGE — LED SPONGE — E2Y-6J10611-Y42
UME17 ME_MYLAR — Hide Caps Lock LED_MYLAR — E2Y-6324111-Y42
UME11 HIDE_GAP_MYLAR — HIDE_GAP_MYLAR — E2P-6J11411-Y42

UME1 HDMI Lable — HDMI ROYALTY — Y01-RHDMI03-000
UME12 USB3_CON_MYLAR — MYLAR — E2M-3570611-Y42
UME14 HDMI/DP MYLAR — MYLAR — E2M-3570611-Y42

Only 1795

UME13 MB MYLAR — X_MB MYLAR — E2P-7910311-G40
UME15 SPONGE — X_SPONGE — E2Y-7910211-G40

UME2 BIOS Lable — X_BIOS_LABEL — G51-LA01678-A09
For MP

版邊 內3 / 外8

MH1 H_R197D118_PT X_ME_SCREW HOLE
HOLES_8X8_D3MM_VIA8 MH8 X_8x8
MH3 X_H_R197D118_PT_V3 H_R197D118_PT_V3 — GND
MH2 X_H_R197D118_PT_V3 H_R197D118_PT_V3 — GND

HOLE10 HOLE11 HOLE8 HOLE6 HOLE4 HOLE15 HOLE12 HOLE1
X_NPTH_80 NPTH_80 / X_npth_98 npth_98
HOLE14 HOLE3 HOLE2 HOLE7 HOLE5 HOLE13 HOLE9
X_NPTH_80 NPTH_80

PCB1 P30-16J510A-H73
P30-16J510A-H73
Hannstar: P30-16J510A-H73
TRIPOD: P30-16J510A-T53

Top Spring

PAD7 X_HS-MS1011-RH ATE_C006_106
E23-1011040-CA7
PAD3 X_HS-MS1011-RH ATE_C006_106
PAD2 X_HS-MS1011-RH ATE_C006_106
PAD8 X_HS-MS1011-RH ATE_C006_106
E23-1011040-CA7 E23-1011040-CA7 E23-1011040-CA7

BOT Spring

PAD4 X_HS-MS1011-RH ATE_C006_106
PAD6 X_HS-MS1011-RH ATE_C006_106
E23-1011040-CA7 E23-1011040-CA7

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